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Mobility and Transport Connectivity Series

Creating a Green Marine Fuel Market in South Africa

WORLD BANK GROUP

PROBLUE

PPIAF
Enabling Infrastructure Investment

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Acronyms

ADB	Asian Development Bank
AfDB	African Development Bank
AIS	Automatic Identification System
APGA	Australian Pipelines and Gas Association
ASU	Air Separation Unit
CAPEX	Capital Expenses
CBA	Critical Biodiversity Area
CBAM	Carbon Border Adjustment Mechanism
CCA	Customs Controlled Area
CCS	Carbon Capture and Sequestration
CEM Hubs	Clean Energy Marine Hubs
CfD	Contract for Difference
CMTP	Comprehensive Maritime Transport Policy
DAC	Direct Air Capture
DFFE	Department of Forestry, Fisheries, and the Environment
DFI	Development Finance Institution
DOT	Department of Transport
DRI	Direct Reduced Iron
DSI	Department of Science and Innovation
DTIC	Department of Trade, Industry and Competition
DWT	Deadweight Tons
EBRD	European Bank of Reconstruction and Development
ECA	Export Credit Agency
ECBA	Economic Cost Benefit Analysis
EIA	Environmental Impact Assessment
EMDC	Emerging Markets and Developing Countries
ENPV	Economic Net Present Value
EPC	Engineering, Procurement, and Construction
ESG	Environmental, Social, Governance
ETS	Emission Trading Scheme
EU	European Union

FEED	Frontend Engineering Design
GDP	Gross Domestic Product
GH2	Green Hydrogen
GHCS	Green Hydrogen Commercialization Strategy
GHG	Greenhouse Gas
GHNP	Green Hydrogen National Program
GMQ	General Maintenance Quay
HFO	Heavy Fuel Oil
HVAC	High-Voltage Alternating Current
HVDC	High-Voltage Direct Current
IAPH	International Association of Ports and Harbors
ICS	International Chamber of Shipping
IDB	Inter-American Development Bank
IDC	Industrial Development Corporation
IDZ	Industrial Development Zone
IFI	International Financial Institution
IMC	International Maritime Centre
IMO	International Maritime Organization
IPG	International Partners Group
IRA	Inflation Reduction Act
ISA	Infrastructure South Africa
JET-IP	Just Energy Transition Implementation Plan
ktons	Kilotons
LCOA	Levelized Cost of Ammonia
LCOT	Levelized Cost of Transport
Li-ion	Lithium-Ion
LNG	Liquefied Natural Gas
LPG	Liquefied Petroleum Gas
LR2	Long Range 2
LSIR	Location-Specific Individual Risk
MDB	Multilateral Development Bank
MGO	Marine Gas Oil
MIDTFN	Maritime Industry Development Task Force Network

MoU	Memorandum of Understanding
MWh	Megawatt hours
NEMA	National Environmental Management Act
NERSA	National Energy Regulator of South Africa
NPV	Net Present Value
OPEX	Operational Expenses
PEM	Proton-Exchange Membrane
PICC	Presidential Infrastructure Coordinating Commission
PPA	Power Purchase Agreement
PPP	Public-Private Partnership
PV	Photovoltaic
QRA	Quantitative Risk Assessment
RE	Renewable Energy
RED III	Renewable Energy Directive III
REDZ	Renewable Energy Development Zones
REEA	Renewable Energy Environmental Impact Assessment
REIPPP	Renewable Energy Independent Power Producer Program
SABS	South African Bureau of Standards
SAMSA	South African Maritime Safety Authority
SAT	Single Axis Tracking
SEZ	Special Economic Zone
SIP	Strategic Integrated Project
SMR	Steam Methane Reformation
SOE	State-owned Enterprise
SOEC	Solid Oxide Electrolyzer Cell
SPC	Special Purpose Company
tpa	tons per annum
TNPA	Transnet National Ports Authority
TWh	Terawatt hours
UNFCCC	United Nations Framework Convention on Climate Change
VAT	Value Added Tax
VLSFO	Very Low Sulfur Fuel Oil
WACC	Weighted Average Cost of Capital

Glossary

Blue hydrogen – hydrogen produced from fossil fuels coupled with carbon dioxide capture and storage (combustion based) or carbon storage (pyrolysis based), also known as low carbon hydrogen.

Clean hydrogen – comprises of both green hydrogen and blue hydrogen production pathways.

Conventional hydrogen – hydrogen produced from fossil fuels, mainly natural gas, or coal, without carbon dioxide capture and storage.

Green hydrogen – hydrogen produced from water electrolysis using renewable electricity or from biomass, also known as renewable hydrogen.

Green hydrogen (GH2) based marine fuels – hydrogen-based green shipping fuels, such as green ammonia and green methanol, produced from renewable electricity, also known as e-fuels.

Marine fuels – fuel source for ships, mainly for propulsion and auxiliary power, also known as bunker-, ship-, or shipping fuels.

Executive Summary

Background

The World Bank addresses decarbonization of the global maritime transport sector from two intertwined angles. On a global level, the Bank supports the policymaking process at the International Maritime Organization (IMO) through analytics and advocacy. On a country level, the Bank assists its member countries in identifying development opportunities. And South Africa has been identified as a country with high potential to become a future supplier of green shipping fuels to the global fleet.

As part of this engagement, the Bank supported the development of two case studies: one in the Port of Saldanha in the Western Cape Province, and another in the future port of Boegoebaai in the Northern Cape Province. The objective was to understand the requirements and challenges of developing green shipping fuel production and supply in South Africa, making this knowledge available to public and private stakeholders, and serving as one of many building blocks to advance this growth opportunity for the country. Since November 2022, one virtual and two in-person stakeholder workshops, with over 100 South African and international participants from science, private and public sector, significantly benefited this work.

This report presents the outcomes of the World Bank engagement, putting South Africa's opportunity to become a leading supplier of green marine fuels in the national and global context.

Green hydrogen is slated to be in high demand, as countries aim to reduce carbon emissions, especially in sectors like transport, chemicals, and heavy industries. And South Africa has the potential to become a leader in this space. The country is actively pursuing the hydrogen economy as a strategic development opportunity. The aim is to capture a significant share of the global hydrogen market, thereby generating economic growth and quality jobs. The hydrogen economy is projected to contribute 3.6 percent to South Africa's GDP and create 380,000 jobs by 2050. South Africa is attracting private sector interest in green hydrogen projects, solidifying its position as an investment destination for this industry.

The maritime sector can catalyze this new industry as both a consumer and enabler. According to World Bank analysis, green ammonia or methanol are the most promising hydrogen-derived fuels for ocean-going ships. Deep-sea shipping, being globally regulated, can thus provide stable demand for green hydrogen. Because by 2050, about 64 percent of the shipping fleet's fuel mix is expected to be hydrogen-based in order to cut its carbon footprint. For the sector to meet its 2030 target, which is to make zero- or near-zero greenhouse gas energy, fuels, and technologies 5-10 percent of shipping's energy mix, at least five million tons of green hydrogen would be needed for shipping alone. At the same time, ships are indispensable for the global hydrogen trade, moving derivatives to overseas markets.

This report estimates that in a base case, ships visiting South Africa's eight commercial ports could demand around 56,000 tons of hydrogen as early as 2030, increasing to about 530,000 tons by 2050. This includes the energy demand from vessels on international voyages, which fall under the global greenhouse gas reduction rules for international shipping. The largest demand will come from the country's largest ports – Durban, and Richards Bay in Kwazulu-Natal – followed by the ports of Cape Town and Saldanha Bay in the Western Cape. In addition, the country could supply ships, which are not bound for South African ports, but are passing by the Cape of Good Hope – a major maritime waypoint. By 2050, this by-pass traffic could add a substantial 1.3 million tons of green hydrogen demand.

As a major exporter of mining products, large bulk carriers are the main drivers for shipping fuel demand in South Africa. The quantification of hydrogen demand from shipping offers project developers an additional market to expand hydrogen sales and mitigate single-counterpart risk. As opposed to other hydrogen markets, for which South Africa would compete on price in distant economies, demand from ships is unique, as it fosters local value creation within its domestic maritime and fuel supply industry. Since ships come to South Africa to refuel, it can reduce a potential competitive disadvantage with countries, which may either have better renewable resources or access to lower financing cost.[1] For port authorities and government, the analysis offers an indication of common-user infrastructure (CUI) needs and can help to inform short-, mid- and long-term port planning, among other factors.

Ports of call for South Africa's hydrogen ambitions

The focus locations of the World Bank support to study green shipping fuel production were Boegoebaai and Saldanha. Initially, both ports show very different characteristics to support the country's hydrogen ambition. The Boegoebaai greenfield project can boost the country's position as a high-volume exporter for green molecules. Meanwhile, Saldanha, the largest natural deepwater port in the Southern Hemisphere, can aggregate hydrogen demand from nearby industries and ship traffic in the short-term. Pre-feasibility studies for both ports prove technical suitability for green shipping fuel production, supply, and export.

The case study for the port of Saldanha (presented in this report) explores the production of green ammonia as a candidate marine fuel. It describes the technical development and scale of a possible project, including an indicative siting of process components. To produce approximately 50,000 tons of green hydrogen, converted into 280,000 tons of green ammonia, the estimated capital expenditure is around $2 billion. Different scenarios are analyzed to determine the financial viability of the project. The study emphasizes the opportunity for a phased approach to mitigate risk and target specific markets as they mature. The study also highlights the potential for South African ports to become hydrogen hubs.

Roadblocks to achieving South Africa's green hydrogen ambition

In South Africa, developing green hydrogen investment projects at scale faces two major challenges. Firstly, an electricity crisis. The country is facing severe constraints in terms of generation capacity to meet electricity demand. However, the inevitable excess electricity production from green hydrogen projects can add net generation capacity to the grid. Hydrogen as an energy carrier can address intermittency issues inherent to renewable power. It is necessary to simultaneously expand renewable power for electricity and green hydrogen. This will ensure that the electricity crisis is not neglected while pursuing the green hydrogen opportunity. A robust expansion of transmission capacity must also keep pace with proposed projects.

Secondly, the country's commercial seaports are facing severe operational challenges. Recently, they have seen hampered im- and exports that caused supply chain bottlenecks. For new commodities like hydrogen-derivatives, the port system will be confronted by new operational requirements. The protection of infrastructure and the restoring of operational efficiency is therefore key for a successful hydrogen economy.

[1] The higher cost of capital in emerging economies is a challenge for hydrogen projects since it significantly influences the production cost of hydrogen.

Purpose and approach of this report

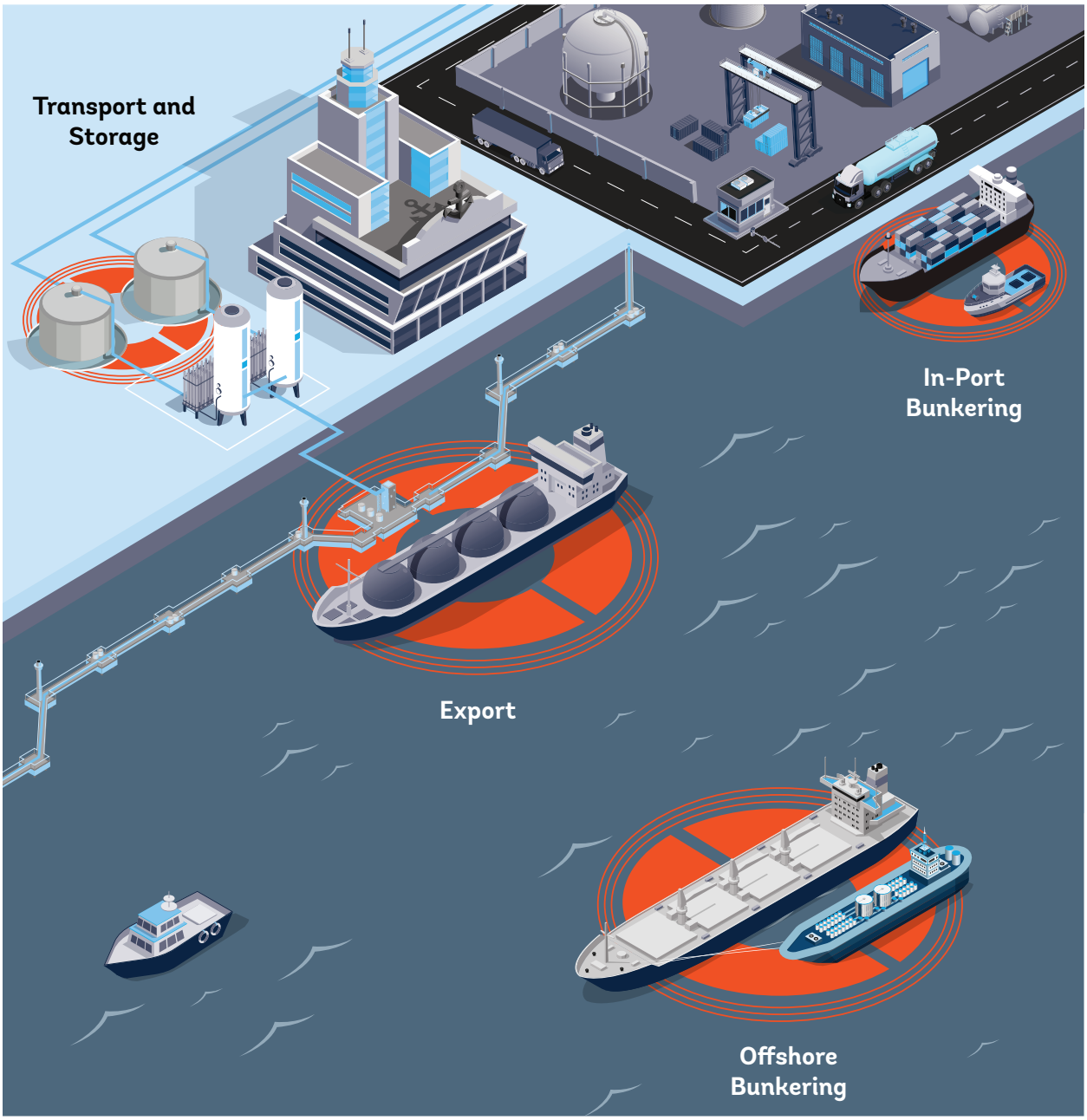
This report aims to answer how the maritime sector can help unlock the potential of the country's hydrogen economy. The nascent hydrogen economy and associated production of green shipping fuels promises economic and social development opportunities. The analysis approaches this growth opportunity with a risk lens and identifies priorities for action and further exploration. Proper risk management is key both for private investors and government advancing green hydrogen as a strategic sector. This can support to realize co-benefits while safeguarding social, fiscal and environmental integrity and avoid competition with other development priorities. Throughout the report, the analysis offers four risk spotlights specific to South Africa: Electricity crisis, maritime sector challenges, safety & environment, and fiscal considerations.

The findings can be summarized in five key messages and policy considerations:

1. **South Africa strategically advances a national green hydrogen economy, and the maritime sector plays a key role.** The government and the private sector are strategically positioning the country as an investment destination for green hydrogen. Through the Green Hydrogen Commercialization Strategy (GHCS), the country is moving towards implementation. The national port operator, Transnet National Ports Authority (TNPA), has identified ports as a key enabler and made initial steps to explore this growth opportunity. The relevance of the port and shipping sector to the successful implementation of the green hydrogen economy shall not be underestimated. It presents the bridging between the South African energy and transport system, where the ultimate success of the green hydrogen economy hinges on an able port system to adapt to new needs.

   Policy context: The government shall continue to expedite the strategic objectives set out in the GHCS. Facilitating investments in the expansion of transmission infrastructure as well as observing additionality for renewable generation dedicated to green hydrogen can help address the electricity crisis and avoid competing objectives. In this context, critical reforms in the power sector are already under way and show promise to facilitate investments into green hydrogen (Chapter 2.3). The GHCS identifies marine fuels as a focal downstream opportunity for green hydrogen. However, to ensure the safe and environmentally sound handling of green hydrogen derivatives, especially green hydrogen-based marine fuels in South African ports, it is necessary for the government, line-agencies, and relevant state-owned-enterprises to start developing an end-to-end regulatory framework today. Identifying regulatory touchpoints in the extended port area is an important first step (Figure E.1.), which involves key agencies like the Energy Regulator of South Africa (NERSA), the South African Bureau of Standards (SABS), Transnet National Ports Authority (TNPA), the South African Maritime Safety Authority (SAMSA), and the South African Revenue Service (SARS). This will ensure that the port system is in sync with the significant pipeline of private sector investments and activities within or outside port limits. It is also critical to maintain the social acceptance of these new activities.

Figure E.1. Regulatory touchpoints in the wider port system



Source: World Bank.

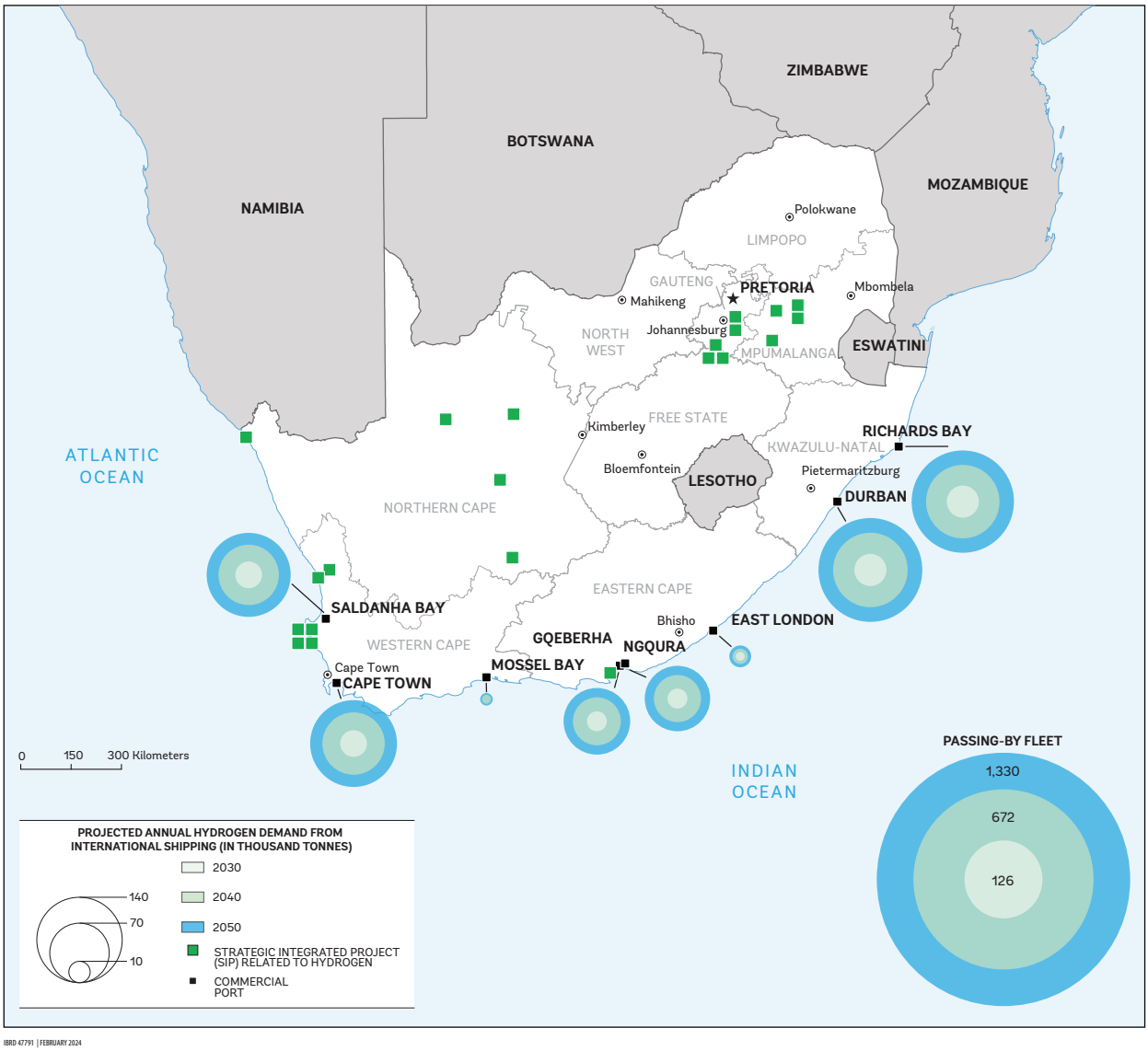
2. **International shipping can be a catalyst to commercialize South Africa's green hydrogen economy.** In a base case, up to 56,000 tons of annual hydrogen demand could come from the supply of marine fuel, as early as 2030. By 2050, this demand could rise to over 0.5 million tons per year. If capturing the demand from ships passing by the Cape, South Africa could supply international fleets with up to 182,000 tons by 2030 and a substantial 2 million tons of green hydrogen by 2050 (Figure E.2.). As opposed to other hydrogen markets – for which South Africa would compete on price in distant economies – the demand

from ships is unique. When "the customer comes to South Africa", the country could reduce competition over countries which may have better renewables resources or access to lower financing cost.

By 2030, South Africa can unlock around 200k tons of hydrogen production demand from ships, sharply rising to up to 2 million tons by 2050.

Policy context: To unlock the demand from this special market, policymakers can promote the adoption of global rules at the IMO which would mandate the uptake of green hydrogen-based marine fuels. This will be aligned with the ambition set out by the IMO 2023 Greenhouse Gas Strategy. Policy options include a goal-based greenhouse gas fuel standard combined with a global emissions pricing mechanism, both stringent enough to provide a bankable demand signal for green hydrogen-based marine fuels by 2030. Global rulesets made at the IMO will also prevent unilateral action to result in a competitive disadvantage. Additionally, to support the large-scale commercial supply of green hydrogen-based fuels, policymakers can create a clear and transparent licensing system for bunker fuel suppliers. In view of recent issues faced in the South African bunkering market, such processes should be transparent, and inclusive of social, environmental, and commercial concerns. Such a system can reduce market entry barriers and foster a favorable business climate for commercial bunkering. Pilot initiatives, for example as part of green shipping corridor initiatives, can help to build private sector confidence in this new marine fuel market.

Figure E.2. Hydrogen demand from international shipping in South Africa's commercial ports



Source: World Bank. For accessibility purposes, the locations of ports and projects are approximate only.

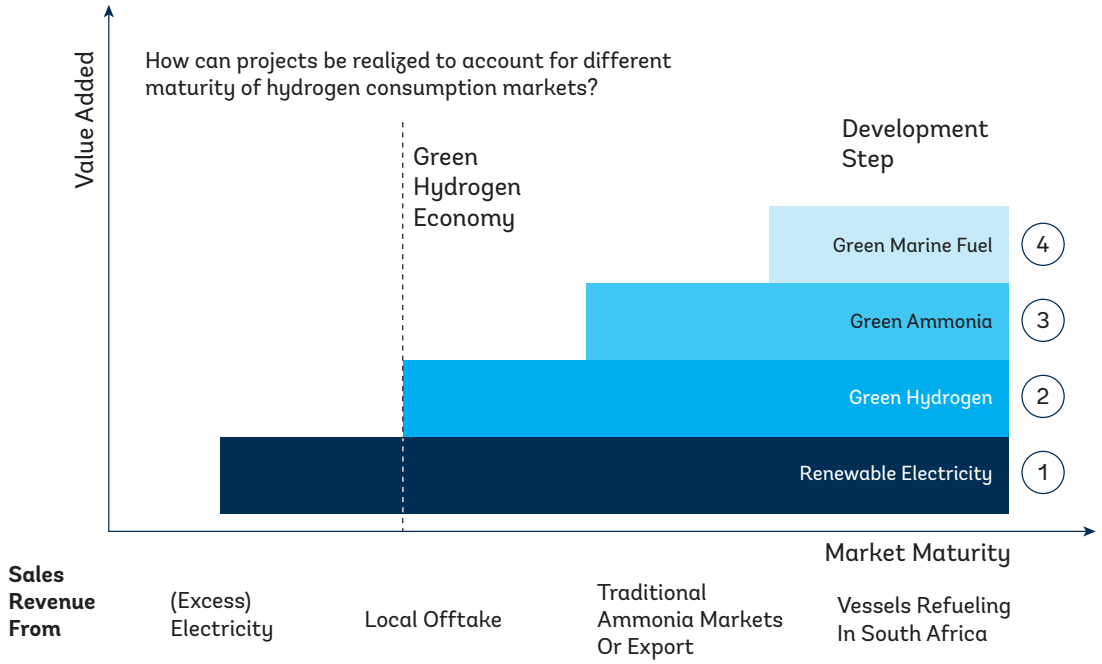
3. **Green shipping fuel production in South Africa is technically feasible but is financially challenged, not only in South Africa.** The case study unveils that less mature markets, such as the sale of green hydrogen-based marine fuel, turn out a less competitive business case today. Green hydrogen-based marine fuels could be produced at a price range between ≈1.4 to 4.3 times the cost of conventional marine fuels, depending on the financing conditions.[2] However due to recent increases in prices, such as those seen in conventional ammonia markets, green ammonia might become cost-competitive in Africa under less aggressive financing terms. Projects could be de-risked through the gradual phasing in of value-adding processes and hydrogen beneficiation, as different markets mature (Figure E.3.). Investment projects to

[2] Based on a weighted average cost of capital (WACC) range of 4-12 percent and a reduction in capital expenditure of between 0-40 percent, compared against a three-year max average price of VLSFO, and considering a delivered price (fuel supplied).

produce green hydrogen-based marine fuels do not differ much from projects like export-oriented green ammonia production and come with largely the same economic challenges. Accessing lower cost capital can reduce the competitiveness gap and investment risk for first-mover projects. For the global maritime sector, successful project delivery is critical to decarbonize in line with its sectoral climate targets and transition equitably for countries dependent on maritime transport, like South Africa.

Policy context: Given South Africa's limited fiscal space, public spending must be considered cautiously. Large-scale publicly funded subsidy programs do not appear to be adequate in the South African context. Against this backdrop, a reduction of the price gap for marine fuels must be pursued by other means. The marine fuel market deserves special attention though. As opposed to hydrogen markets for fertilizer or industrial feedstock, the use and type of marine fuel is regulated globally. Here, the price differential of green hydrogen-based marine fuels over conventional fuels becomes less relevant since international maritime regulation historically works through goal-based mandates. Policymakers can promote the global adoption of fuel standards at the IMO, which prescribe the use of low carbon fuels, unlocking this market for South Africa and ensuring a level playing field. Alongside this, contracts for difference in regions like the European Union, for instance, can subsidize the production price of green hydrogen derivatives for export. Where public and private involvement overlap, the public sector can support the private sector in the development of a sound project pipeline to successfully participate in, for example, double-auction schemes abroad.

Figure E.3. Stepwise development of a green shipping fuel project



Source: World Bank.

4. **Mobilizing private capital is key to successfully develop green shipping fuel projects.** The capital-intensive nature of large-scale green hydrogen projects exceeds the capacity of government and individual financial investors, such as banks. Platform approaches, where multiple (development) banks cooperate, can help to blend capital from commercial and concessional sources accommodating different risk profiles. Funding mechanisms in overseas markets, such as contracts for difference, can lock in production prices, which are financially more attractive. To further mobilize private capital, novel funding sources, such as revenues from a global emissions pricing mechanism can possibly provide capital injections in marine fuel production projects.

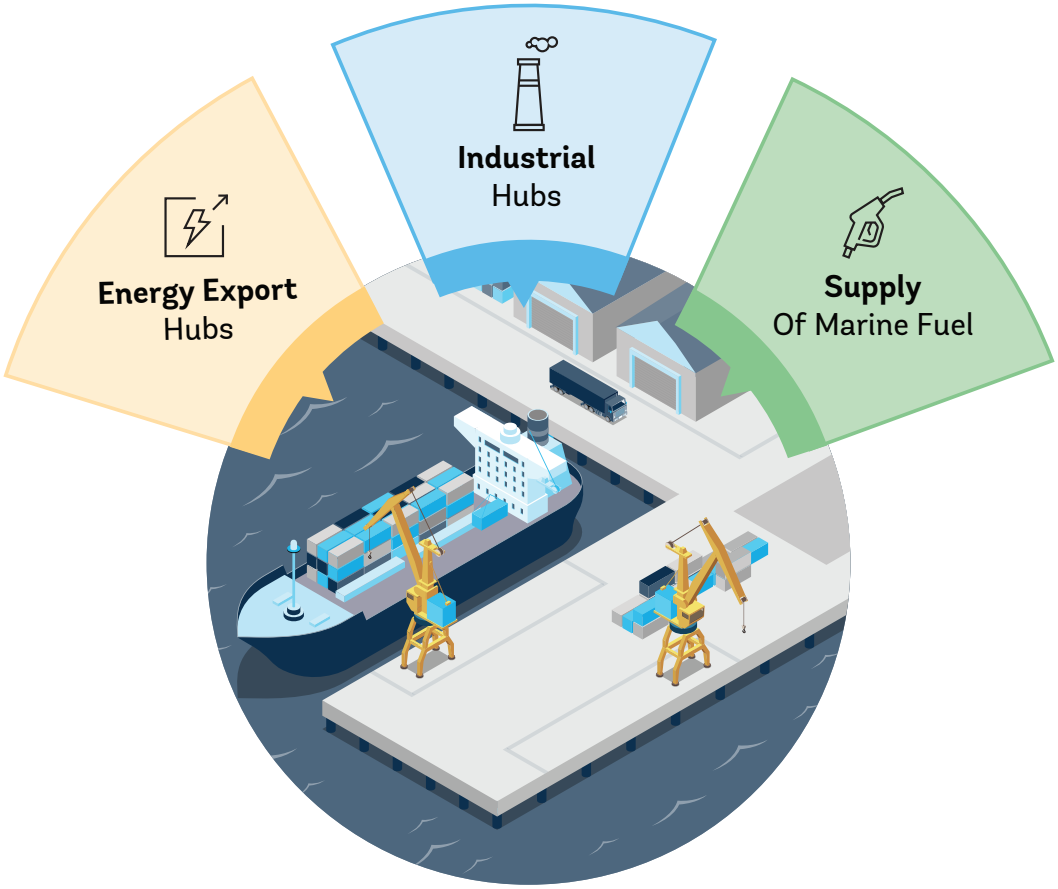
   Policy context: Policymakers can further strengthen partnerships with international partners and multilateral development banks to tap into non-commercial finance and risk mitigation instruments, which can bring down the production cost of green hydrogen derivatives. This will enable first-mover projects to reach financial closing, build confidence in the country's green hydrogen opportunity, and reduce the private sector's overall investment risk. IMO member states, including South Africa, have agreed to develop a global pricing mechanism for ship emissions. The government can work with IMO member states to ensure that revenues from such a mechanism can be used to support capital projects which produce green hydrogen-based marine fuels for the international fleet.[3]

5. **Ports in South Africa can develop into hydrogen hubs.** Ports will facilitate transport and storage for hydrogen export, supply green hydrogen-based fuel to ships, and act as production hubs, aggregating demand from various end-users. Port and maritime authorities can attract investments through the provision of land, common-user infrastructure, prepare licensing, and transparent rulesets. The country's special economic zones near or at seaports can offer fiscal incentives. The aggregation of several hydrogen demand centers within the port ecosystem can mitigate investment risk for various parties and reduce production cost.

[3] Also see World Bank publication on how revenues from a global emissions pricing scheme could be used (Dominioni, et al. 2023)

Figure E.4. Ports can develop into green hydrogen hubs

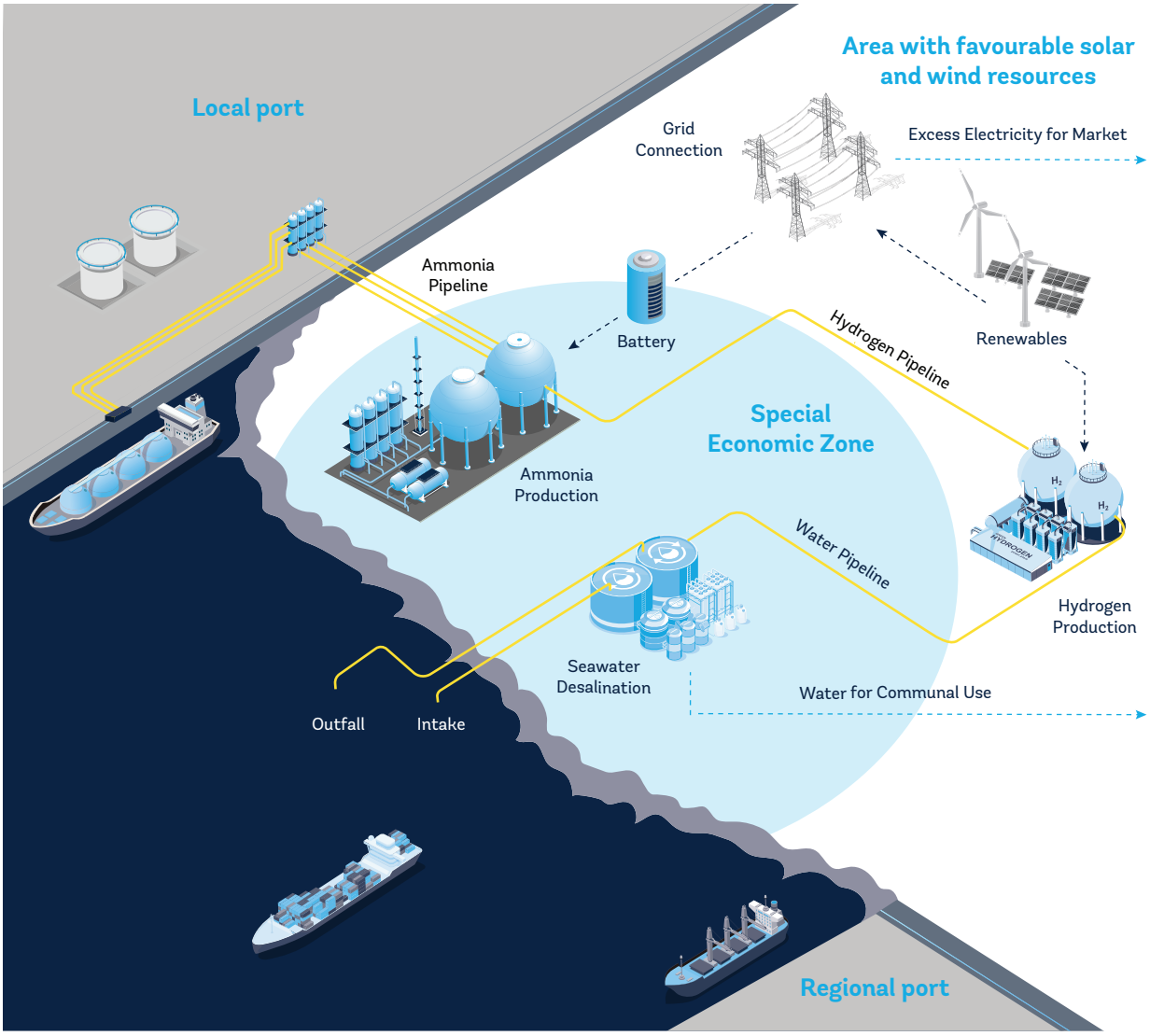
Different roles for ports



Source: World Bank.

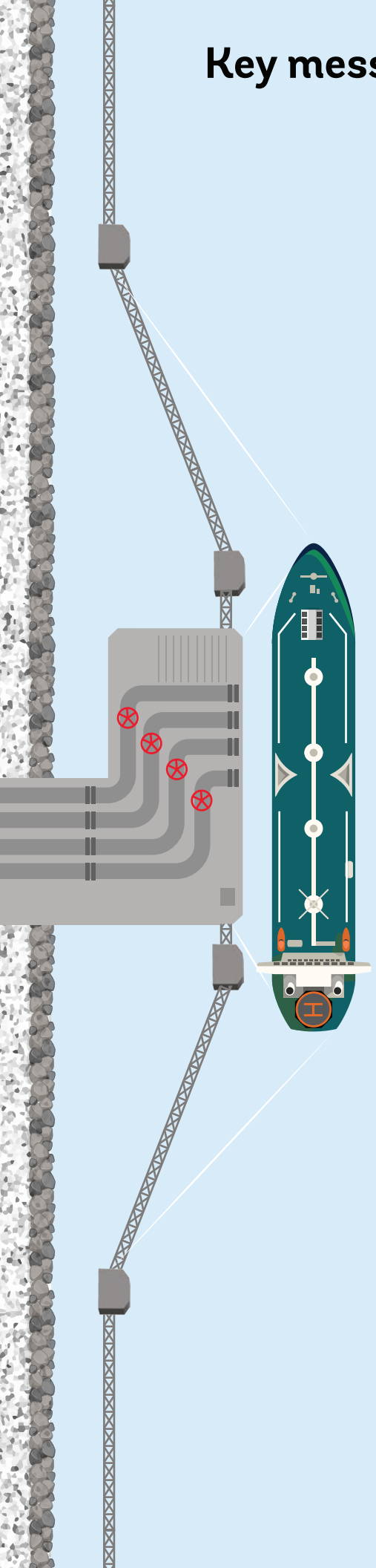
Policy context: On the port level, a review of required public common-user infrastructure, land-leases to the private sector, and the zoning of SEZs can be prioritized to reap the full benefits of hydrogen hubs in South Africa. Common-user infrastructure can be assessed in the municipal and regional context to realize co-benefits in areas like water resilience and transmission capacity (Figure E.5.). An efficient port system is key to developing hydrogen hubs and maritime gateways. Resolving the inefficiencies in the port sector should therefore be pursued in parallel.

Figure E.5. Green marine fuel production can create co-benefits



Source: World Bank.

Key messages and policy recommendations



1 South Africa strives towards a green hydrogen economy for which the maritime sector is crucial

- Advance objectives of the Green Hydrogen Commercialization Strategy.
- Facilitate investment in additional renewable power and grid capacity.
- Develop a regulatory framework to handle hydrogen derivatives in ports.

2 The marine fuel market can commercialize up to 2 million tons in hydrogen demand in South Africa

- Promote policy for uptake of hydrogen-based marine fuels at the IMO.
- Develop a robust licensing system for green marine fuel suppliers.
- Identify test cases to pilot the transfer of green marine fuel.

3 Producing hydrogen-based fuel is technologically feasible but the business case faces challenges

- Consider public spending carefully to protect fiscal space.
- Enable the marine fuel market through global rules.
- Develop investment pipelines for external support.

4 Mobilizing private capital is key to successfully develop green marine fuel projects in South Africa

- Identify risk mitigation instruments for first-mover projects.
- Partner with (development) finance institutions to scale private capital.
- Promote novel funding sources like global ship emissions pricing to support investments.

5 Ports in South Africa can develop into enabling hydrogen hubs

- Assess common-user infrastructure options at the port-, municipal- and regional levels.
- Attract greater private sector participation in port areas.
- Address operational inefficiencies affecting the port sector.

Introduction

South Africa is positioning itself as a major player in the production of green hydrogen, with ambitions to capture a significant share of the global hydrogen market via the maritime industry. Initiatives to facilitate the production and export of green hydrogen and related fuels are underway in several ports. These are supported by private sector interest and international partnerships. Projections predict both economic growth and job creation, and this report sums up how policymakers can tap into this potential.

01

South Africa has the potential to become a major producer of green hydrogen. This clean alternative is set to be in high demand as countries aim to reduce carbon emissions, especially in sectors like transport and heavy industries. The country is actively pursuing the hydrogen economy as a strategic development opportunity, aiming to capture a significant share of the global hydrogen market, thereby generating economic growth and quality jobs. The hydrogen economy is projected to contribute 3.6 percent to South Africa's GDP and create 380,000 jobs by 2050 (The Presidency Republic of South Africa 2023). The country is attracting private sector interest in green hydrogen projects, solidifying its position as an investment destination for renewable energy sources.

The South African port system features eight main commercial seaports which are important to the country's economy. As global shipping reduces greenhouse gas emissions, the country's ports will face new challenges and opportunities alike. As the demand for green shipping fuel increases globally, ports will reconfigure their offering to become a supplier of those fuels. Similar to their role in exporting minerals, ports will also become an important node in hydrogen supply chains. A whole emerging industry may benefit from the port ecosystem and vice versa.

Currently, around 21 hydrogen-related projects are registered with the South African government, spanning a broad range of hydrogen use-cases and project scales. Export-scale green hydrogen derivatives projects are developed mainly in the Eastern, Northern and Western Cape provinces.

The World Bank provided support to conduct case studies (at pre-feasibility level) to explore the potential of producing green shipping fuels in two South African ports: the future port of Boegoebaai in the Northern Cape Province and the port of Saldanha in the Western Cape Province. Both locations have attracted investor interest to develop GH2 production and come with notably different features. This report presents the outcomes of the World Bank engagement, putting South Africa's opportunity to become a leading supplier of green shipping fuels in the national and global context. A macro-assessment of hydrogen demand from international shipping complements the insights from the location specific study in this report.

A two-hour drive from Cape Town, the port of Saldanha is ideally located to aggregate demand from multiple hydrogen users. The plans to convert a local steel plant to run on green hydrogen is one of them. Such a project has the potential to become one of the largest first-mover projects in Sub-Saharan Africa, producing green direct reduced iron (DRI) for export markets and bringing back over 8,000 direct and indirect jobs in the next few years.

The Boegoebaai green hydrogen project, in contrast, is one of several projects which has the potential to boost the country's position as a large-scale exporter of green molecules. The Northern Cape provincial government has set a long-term ambition to develop 40 GW electrolysis capacity, producing around four million tons of hydrogen annually. A private anchor investor has confirmed this potential and is looking to develop an investment project. Meeting this goal will require around 80 GW of renewable energy. Boegoebaai, located 20 kilometers south of the Namibian border, offers ample available land for the gradual expansion of solar parks and wind farms. The site provides favorable conditions for developing a vertical hydrogen value chain, including domestic production of photovoltaic modules and electrolyzers. Besides the export of molecules, the port will also be able to capitalize on the Northern Cape's mining industry, which contributes 28 percent to the province's GDP. The new port will enable cost-effective export of minerals like manganese, critical for the energy transition.

Overall, as we will see later in the report, hydrogen projects are expected to be developed mainly by the private sector. The report, however, discusses how the public and private sector can create a market for marine fuel made from green hydrogen, including but not limited to infrastructure, enabling environment and finance.

Chapter 2	Introduces the national context of South Africa's green hydrogen ambition and the universal barriers of the global hydrogen economy
Chapter 3	Outlines the connection between green hydrogen and the maritime sector
Chapter 4	Analyzes the hydrogen demand coming from international shipping in all eight commercial South African seaports
Chapter 5	Presents a case study for the port of Saldanha to produce and supply green ammonia to ships, including a step-by-step analysis of design drivers, technical options as well as a financial and economic analysis at pre-feasibility stage
Chapter 6	Connects the findings from Chapter 4 and 5 with universal economic challenges of hydrogen investment projects. Also discusses a selection of financing and funding sources and the South African context
Chapter 7	Discusses the special role of ports in supporting the hydrogen economy
Chapter 8	Concludes and translates the findings into key findings for policymakers and industry
Risk spotlights	Throughout the report, concise reflections offer the reader a synopsis on current issues, which constitute possible implementation risks of green hydrogen in South Africa. Many of which are in the process of being addressed but will require continuous effort and monitoring. These are: The electricity crisis, maritime sector challenges, safety & environment and fiscal considerations.

The future port of Boegoebaai combines two of South Africa's major strengths: Minerals and green hydrogen.

South Africa's green hydrogen ambition

As countries across the globe race to meet decarbonization targets set by the Paris Agreement, South Africa is playing to win. Their aim: to capture 25 percent of the world's green hydrogen market. The target: Sectors which demand hydrogen, like shipping. Despite challenges, the government hopes to align with the private sector to drive demand, supply, and growth for the green hydrogen sector in the country.

02

2.1. Fueling decarbonization

As many countries and sectors aim to reduce carbon emissions in line with the temperature targets set by the Paris Agreement, green hydrogen is expected to be in high demand going forward. Hydrogen is particularly important for sectors that are difficult to decarbonize, such as transport (maritime, aviation), chemical production (ammonia, methanol), and heavy industries (iron, steel). For countries like South Africa, that have abundant renewable energy resources, the production of green hydrogen can stimulate the growth of related industries, contribute to domestic decarbonization efforts, and generate revenue through exports.

Planning for a green future

The government of South Africa is pursuing the green hydrogen economy as a strategic development opportunity. The country's goal to capture 25 percent of the global hydrogen and fuel cell market originated from the Hydrogen South Africa program. Led by the Department of Science and Technology (now Department of Science and Innovation, DSI), it was approved by the cabinet in 2007 (University of the Western Cape 2014). The Green Hydrogen Society Roadmap developed by the DSI in 2021, further emphasized the importance of green hydrogen in the government's economic growth and greening objectives (Department of Science and Innovation 2021).

Most recently, the implementation of the Green Hydrogen Commercialization Strategy (GHCS), developed by the Department of Trade, Industry and Competition (DTIC), was approved by the cabinet in October 2023 (DTIC 2023). Therein, the green hydrogen economy is projected to contribute 3.6 percent to the country's GDP and generate 380,000 jobs by 2050. Green hydrogen is also a key sector in the country's Just Energy Transition Plan (JET-IP) (The Presidency, Republic of South Africa 2023), which requires approximately ZAR 319 billion ($16.8 billion) in funding for GH2 in the five-year period from 2023 to 2027.

In the global comparison, South Africa is projected to be able to compete with other future green hydrogen producing nations on production cost. Since primary export markets are expected to be the European Union and the United Kingdom, countries that are geographically closer – Morocco, in particular – will be significant competitors. However, as countries seek diversification of energy imports, the geographic distance only plays a minor role. There are already ongoing EU initiatives on hydrogen imports from South Africa. An increased collaboration with neighboring Namibia has the potential to solidify the country's role as a prime export destination, including to other markets such as East Asia (DTIC 2023).

By 2050, South Africa's green hydrogen economy is projected to contribute 3.6 percent to GDP and generate 380,000 jobs.

Private sector interests, both domestic and foreign, are increasingly considering South Africa as an attractive investment destination for hydrogen. Concrete projects are emerging, and the Presidential Infrastructure Coordinating Commission (PICC) is identifying a growing number of Strategic Integrated Projects (SIPs) through the Green Hydrogen National Program (GHNP). SIPs are projects that hold significant economic or social importance, contribute to national strategies or policies, or have a certain monetary value. These projects benefit from an expedited approval process and shorter timeframes for delivery, as outlined in the Infrastructure Development Act of 2014 (The Presidency, Republic of South Africa 2014). Currently, there are 21 hydrogen-related projects registered with the Department of Public Works and Infrastructure South Africa (Infrastructure South Africa 2022).[4]

[4] Includes projects which are registered (and gazetted) and pending final registration.

2.2. Leveraging opportunities, confronting challenges

There are multiple opportunities for South Africa's government to leverage when it comes to hydrogen. The transport sector plays a crucial role in the hydrogen economy as both a consumer and enabler. Firstly, transport is expected to present the largest user of hydrogen by 2050 (Figure 2.1.) (IEA, 2023). Sectors like maritime shipping, aviation, and heavy-duty vehicles require hydrogen and hydrogen-based fuels like ammonia and methanol to reduce emissions. Secondly, transport is necessary to move hydrogen molecules into consumer markets and develop a hydrogen transport network globally. Seaports in particular play a crucial role in enabling the export and handling of green hydrogen derivatives, while pipeline infrastructure is essential for transporting molecules.

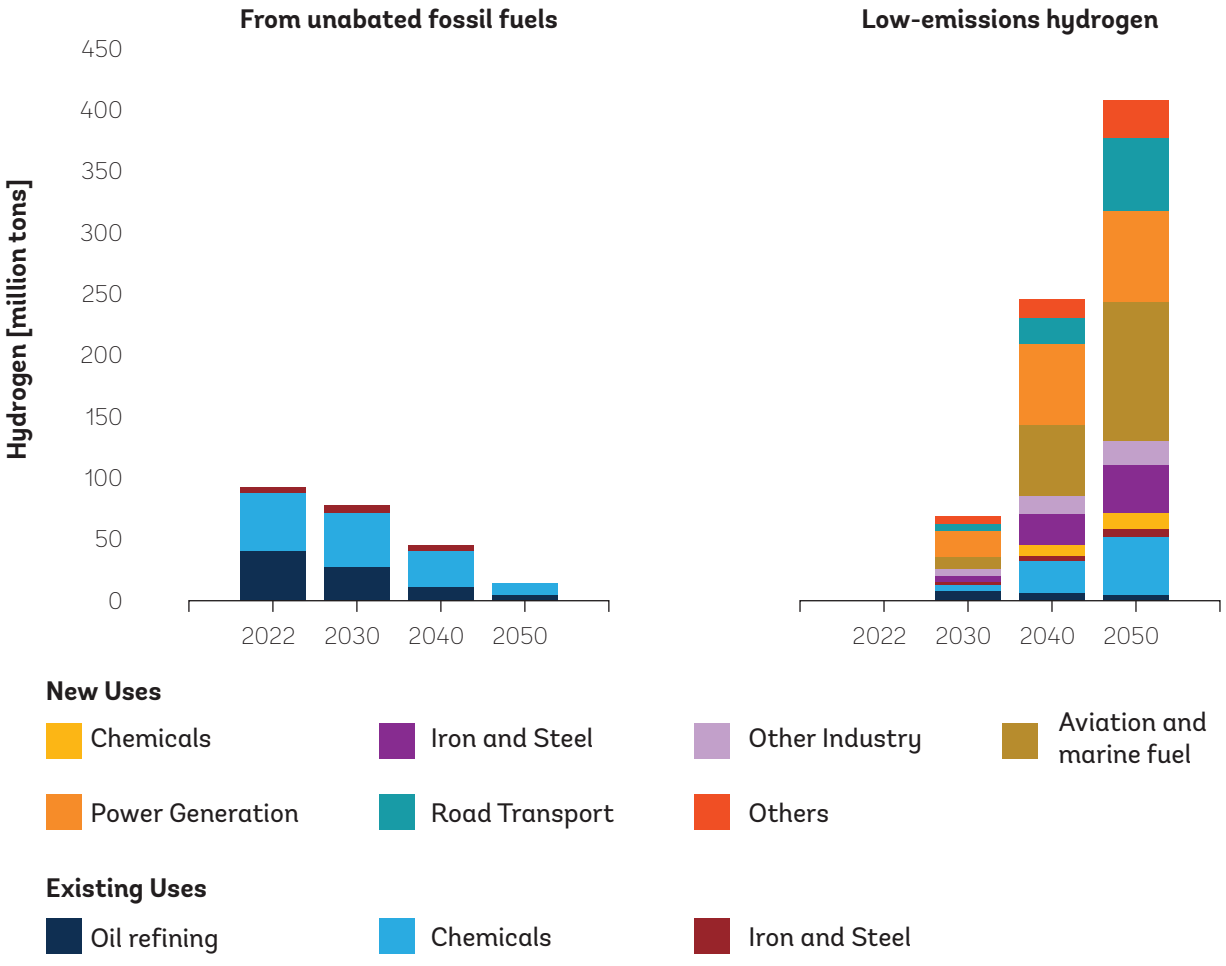
Upgrading public infrastructure in coordination with private sector investments in renewables and green hydrogen production can create new skilled jobs during construction and operation. The government needs to align its actions with the private sector to keep up with the rapid pace of development. Additionally, government facilitation is necessary to ensure the private sector maintains its "social license to operate" and sustains overall progress.

Despite the significant opportunities, only a few projects successfully secure funding. One of the main challenges is the cost-competitiveness of green hydrogen compared to fossil-based alternatives. Green hydrogen projects require access to affordable capital, which is particularly difficult in emerging economies where the lowest production costs can be achieved through favorable solar and wind resources. In contrast, GH2 production relies on a significant increase in renewable electricity, which requires efficient integration into the power grid. For grid-constrained countries, this poses a challenge. Electrolysis, the process of producing GH2, also requires access to reliable water sources, preferably through seawater desalination. This presents both opportunities and challenges alike.

The bankability of projects is further complicated by the reliance on long-term contracts in export markets such as Europe and East Asia, and the absence of a global spot market. Notwithstanding the considerable potential for climate mitigation, climate investing in green hydrogen does not seem to outweigh credit risk considerations. To mitigate investment risk, large-scale projects need to identify multiple and diverse demand sources for hydrogen. For example, as marine fuel in the transport sector or green steel in heavy industry applications.

The financial challenges of green hydrogen projects are further discussed in Chapter 7.

Figure 2.1. Global hydrogen demand in the NZE Scenario, 2022-2050



Source: IEA (2023).

2.3. Risk spotlight: Electricity crisis

Developing green hydrogen investment projects at scale in South Africa stands in the context severe electricity availability constraints. In 2022, load shedding – a forced intervention to cut electricity supply when generation capacity cannot meet demand – reached record levels. The severity of load shedding only increased over 2023. The country faced the equivalent of 157 days of blackout spanning over more than 200 days during the year (CSIR 2023). Estimates show that such power outages result in a negative economic impact of up to -3.2 percentage of the country's GDP (South African Reserve Bank 2023). While ranking amongst the largest power utilities in the world (by generation capacity), the aging coal-fired power plant fleet of South Africa's state-owned utility ESKOM, accounting for over 70 percent of nominal generation capacity (40 GW), is struggling to provide reliable power supply (CSIR 2023).[5] Adding alternative generation capacity is therefore paramount to reduce damage to the country's economy and society.

Since 2011, private sector participation in power generation was pursued by the government with an overall successful Renewable Energy Independent Power Producer Program (REIPPP). In recent years however, the pace of adding renewables, developed by the private sector, slowed down. From 2013 to 2022, around 6,230 MW of renewable generation assets became operational. Still renewables only make up around 11 percent of total nominal generation capacity (CSIR 2023).

Estimates show that loadshedding result in a negative economic impact of up to -3.2 percentage of South Africa's GDP.

In South Africa, generation capacity is only one challenge. Similarly, the transmission grid is under pressure, too. The country has an extensive power grid, but an asymmetric distribution of transmission capacity and poor grid maintenance over the last two decades has resulted in capacity deficits. While transmission grid capacity is comparably strong in provinces like Mpumalanga and Gauteng, where many power-hungry industrial consumers are situated, the most favorable renewable resources are mainly found in the three Cape Provinces. A robust expansion of transmission capacity in these areas is critical to harness cost-efficient renewable power generated in Northern, Western and Eastern Cape Provinces and must keep pace with a significant pipeline of green hydrogen projects. An effective expansion of generation and transmission cannot be shouldered by South Africa's financially troubled state-owned power utility ESKOM alone.

The question arises as to how to ensure that renewable electricity needs for green hydrogen production does not compete with resolving the country's electricity supply gap?

Green hydrogen is power hungry but can offer a net-benefit

Green hydrogen production requires significant amounts of renewable energy. As a rule of thumb, per one million tons of annual hydrogen production, around 20 GW of renewable electivity is required. In the context of South Africa's GHCS, aiming for a production target of around four million tons by 2050 (50 percent for export and 50 percent for domestic consumption), this would equate to around 80 GW of renewables. It is widely recognized that green hydrogen development must come with additional renewable power, both to maintain its social operating license, particular in countries where generation capacity is heavily constrained, but also due to certification requirements, ultimately producing a truly green molecule. Today, South Africa's

[5] Followed by wind power (6%), diesel and gas (6%), pumped storage (5%) and solar PV (4%). Smaller than 1 GW contributions come from hydro and concentrated solar power (CSP).

electricity generation mix consists of 80 percent of coal, mined domestically and feeding some of the world's largest coal-fired power stations (CSIR 2023). If renewables for alleviating the electricity crisis and green hydrogen are build out in parallel one objective doesn't have to be deprioritized over the other (DTIC 2023). To optimize electrolyzer utilization (the capacity factor), renewable power for green hydrogen production must be overbuild, due to the variation of sun and wind during 24 hours of the day. Such excess electricity production from green hydrogen projects ultimately adds net generation capacity to the grid. Plus, hydrogen as an energy storage medium can address intermittency issues inherent to renewable power.

To date, South Africa's long-term planning for electricity generation is largely build on prescribed targets. The Integrated Resource Plan (IRP) of 2019 (DMRE 2019), for example, set out renewable energy targets totaling around 30 GW by 2030 (from wind, solar, hydro, concentrated solar). The draft 2023 IRP revised those mid-term expectations further downwards, contrasting the 80 GW renewable requirement for green hydrogen production significantly.

The South African power market liberates

To attract private companies who are willing to invest into both generation and transmission, state-owned ESKOM would need to separate its generation, transmission and distribution businesses, which until recently were operating under a single entity. This follows proposed and international best practices – to establish a wholesale power market with transparent, non-discriminatory access for sellers and buyers (World Bank 2022).

In March 2024, the national assembly passed the Electricity Regulation Amendment (ERA) bill, which supports the restructuring of ESKOM into three separate entities. Further, the bill aims to aims to increase competition, and private sector participation in South Africa's power sector.

The state-owned National Transmission Company of South Africa (NTCSA), already unbundled from ESKOM in 2023, will act as the transmission system operator (TSO) under the oversight of the National Energy Regulator (NERSA). Under the amended Electricity Act, private companies can apply to NERSA for transmission, trading and import/export licenses.

The TSO has two important technical- and two market-related functions: First, it will act as the transmission company (transmitter), which will maintain and expand the electricity grid, ensuring fair access and keep the integrated power system working safely (system operator). Second, under the oversight of the regulator (NERSA), it shall establish a non-discriminatory trading platform and set the appropriate criteria for power market participants. To transition to a fully competitive power market, the TSO will also perform the functions of a central purchasing agency (CPA), which will buy and trade power from existing and new independent power producers (IPP) and will finalize power purchase agreements (PPAs) with ESKOM's generating assets, amongst others.

Aside from enabling full private sector access to the electricity grid, the legislative revision will also pave the way for private investments into transmission infrastructure by way of public-private partnerships (PPPs). In its current form however, the state will retain ownership of the transmission infrastructure.

While the generation cap for IPPs had been lifted already back in 2022 (from initially 1MW, then 100MW), the sale of electricity was only possible via ESKOM (i.e., through the REIPPP process) or by way of bilateral PPAs in the kilovolt range (i.e., industrial consumers). In practice, the liberation now means that IPPs could connect their generation assets to the grid independently and sell electricity to a host of customers directly. Also, households and smaller consumers would be enabled to buy from IPPs. For an open electricity market to function, power trading between generating entities spread across the country and customers would

be enabled by the introduction of a virtual wheeling platform (VWP), which is currently under development by ESKOM. The roll out of bidirectional smart meters would in addition enable households to feed excess electricity into the grid, from e.g., rooftop solar, offsetting parts of their electricity bill.

Reforms are showing promise for the green hydrogen economy

Until recently, electricity regulation did not offer an ideal enabling environment for private hydrogen developers, putting the competitiveness of the South African hydrogen price under pressure. Also the renewable generation targets prescribed within South Africa's Integrated Resource Plan (IRP) appear far off a pathway to 80 GW renewable power generation capacity for green hydrogen in 2050. However, under the revised legislative framework, green hydrogen development stands in a different light. Open access and competitive electricity wheeling can increase the load hours of electrolyzers for distributed renewable generation assets, translating into lower levelized cost of hydrogen (LCOH). Also, the sale of excess electricity would be facilitated under the new framework - another critical component to offer a competitive hydrogen price – and add critically needed generation capacity to the grid. Further, opening the transmission grid to private sector investments can remove transmission bottlenecks and connect renewables generation clusters with consumption centers.

Hydrogen pathway for international shipping

In the pursuit of the shipping's decarbonization, hydrogen emerges as a prime fuel pathway. It isn't a radical shift; many of the required technology advancements exist already. What needs to be done: implementation. South Africa has a developed maritime industry. While the sector is facing multiple challenges, the country is charting a course to overcome them with a bold vision to become a maritime center of excellence. And green hydrogen can become part of that.

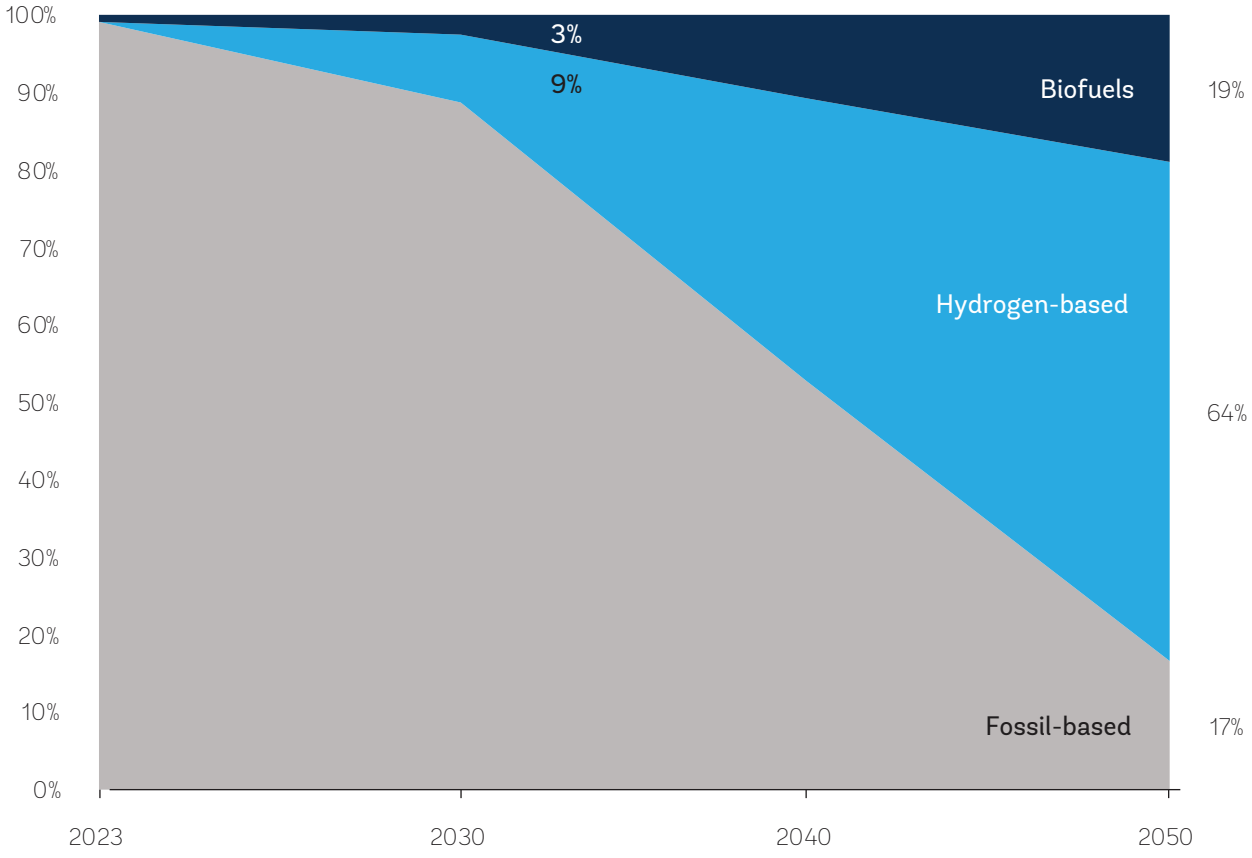
03

3.1. How fuel and policy pathways converge

The maritime sector can catalyze the hydrogen economy both as a consumer and an enabler. Deep-sea shipping, being globally regulated, can provide stable demand for GH2. To reduce the shipping fleet's carbon footprint, around 64 percent of its fuel mix is projected to be hydrogen-based by 2050 (Raucci, McKinlay and Karan 2023). World Bank analysis concluded that green ammonia or methanol are the most promising hydrogen-derived fuels for ocean-going ships. Biofuels and liquefied natural gas (LNG) however are expected to only play a limited role in decarbonizing maritime transport (Englert, Losos, et al. 2021).

> Green ammonia or methanol are the most promising hydrogen-derived fuels for ocean-going ships.

Figure 3.1. Fuel Pathways for international shipping



Source: Raucci, McKinlay, and Karan (2022).

Using green shipping fuels onboard ships does not constitute a technology shift, but rather an evolution of existing, established technology. Almost all ocean-going ships use internal combustion engines for propulsion and auxiliary power. Today, methanol engines are already on the water, while hydrogen or ammonia engines for ships are gradually becoming commercially available. Several leading marine engine manufacturers have ongoing development programs to manufacture ammonia- and hydrogen-fueled combustion engines (MAN Energy Solutions 2023a).

The first ammonia burning two-stroke engines are expected to be installed onboard a vessel during 2024 and become commercially available in 2025 (MAN Energy Solutions 2023b; WinGD 2023). The first four-stroke ammonia engine has been available since late 2023 (Wärtsilä 2023). From a technology maturity point of view, confidence in the uptake of hydrogen-based shipping fuels is reflected in most recent vessel orders from shipyards. As of early 2024, 29 methanol fueled ships are in operation, with an orderbook standing at 229 vessels.

For the sector to meet its 2030 target, around five million tons of green hydrogen would be needed for shipping alone.

Meanwhile, the orderbook for ammonia-fueled vessels currently tracks 13 assets (DNV 2024). So-called ammonia-ready ships, i.e., vessels which have a design prepared for an easy-switch over to run on ammonia, have recently outpaced methanol-ready ships at 322 units against 272 (Atchison 2024). In terms of vessel types, orderbooks signal that long-haul or liner shipping dominates alternatively propelled ships. While methanol orders dominate the container ship segment, ammonia fueled ships are either bulk carriers or large gas carriers. The latter, aside from running on ammonia themselves, will carry ammonia to hydrogen import markets.

For the shipping sector to meet its 2030 climate target (Figure 3.2), around five million tons of green hydrogen would be needed for shipping alone (Englert, Rojon, et al. 2023). As 40 percent of fossil fuels are transported by seaports (UNCTAD 2022), green fuels will inevitably be moved through ocean transport. Ports will serve as critical gateways for exporting and importing hydrogen derivatives and will function as clean energy hubs (Clean Energy Ministerial 2024). Port clusters will also aggregate demand from hydrogen consumers such as ships, aircrafts, heavy-duty vehicles, and heavy industries. For shipping's energy transition to succeed, green shipping fuel projects must be developed at scale and along all major trade lanes.

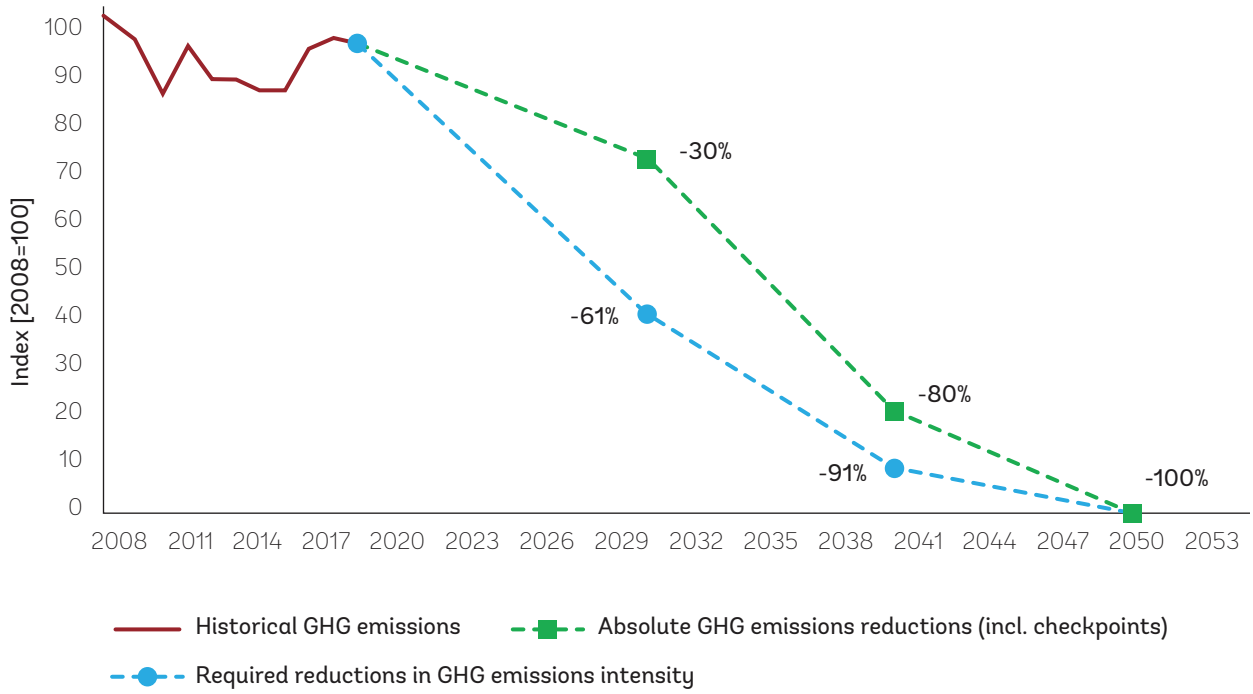
The secretariat of the International Maritime Organization (IMO) in London, United Kingdom, where governments develop climate policy for global shipping.

Shipping's energy transition, based on experience with previous maritime regulation, is expected to be policy driven. In July 2023, member states of the International Maritime Organization (IMO) unanimously adopted the 2023 IMO Greenhouse Gas Strategy. This landmark agreement replaced the 2018 Initial Strategy and significantly strengthened shipping's greenhouse gas (GHG) reduction targets. IMO member states agreed to:

- Reach net-zero GHG emissions from international shipping by around 2050, with interim checkpoints of 20-30 percent by 2030 and 70-80 percent by 2040; and
- Make zero- or near-zero GHG energy, fuels, and technologies 5-10 percent of shipping's energy mix by 2030

The IMO further agreed to develop a basket of policy measures consisting of both technical and economic elements. The technical element will be a marine fuel standard mandating the phased reduction of the GHG intensity of the energy used on board ships. This is expected to be key to driving the effective uptake of zero- and near-zero GHG fuels in a timely manner. The specific nature of the economic element still needs to be defined. Yet, it will be based on GHG emissions pricing. Simply put, it is a carbon price on maritime emissions. Many IMO member states highlight that carbon pricing will not only reduce GHG emissions but can also make shipping's energy transition more equitable. This is the case when revenues are strategically channeled back to those countries which may struggle the most with shipping's decarbonization and/or climate change — most often, this includes developing countries, specifically small island states and least developed countries. These measures are to be adopted in 2025, with an envisaged entry into force in 2027.

Figure 3.2. Greenhouse gas reduction targets for international shipping



Source: World Bank[6].

[6] Historical emissions data from Climate Action Tracker; linear reduction paths assumed for simplicity; GHG emissions intensity in tCO2e per ton-miles based on a medium trade growth scenario (OECD RCP2.6 L) of roughly 2 percent growth per year, as used in the 4th IMO GHG Study.

As a specialized agency of the United Nations, the World Bank is an observer to the IMO. World Bank analysis describes possible options to use revenue from an emissions pricing instrument (Dominioni, et al. 2023). Most notably, it could be used to speed up decarbonization in the shipping industry. For countries looking to advance their hydrogen economy, these discussions present an opportunity as they could potentially benefit from such a measure if money is used to support investments into alternative marine fuels within their economies.

3.2. South Africa's maritime sector

South Africa's maritime transport sector plays a pivotal role in the nation's economic development. The maritime industry facilitates international trade and connects the country to global markets. With a 2,798 km coastline along the Atlantic and Indian Oceans, South Africa is the largest container, bulk, and break-bulk hub in sub-Saharan Africa. The country boasts a network of well-established maritime ports that serve as vital gateways for both domestic and international trade. Eight main commercial seaports enable the efficient movement of goods for the import and export of raw materials like minerals, manufactured goods, and other bulk commodities.

With a 2,798 km coastline along the Atlantic and Indian Oceans, South Africa is the largest container, bulk, and break-bulk hub in sub-Saharan Africa.

The strategic location of the Cape of Good Hope is a crucial waypoint for maritime trade routes. Over time, the development of ports and shipping routes has always been closely tied to the economic growth and geopolitical dynamics of Southern Africa. The sector's impact extends beyond the ports, influencing various industries such as logistics, shipbuilding, and marine services.

South Africa has developed a comprehensive policy framework to govern the strategic role of its maritime sector. In 2017, the cabinet adopted the Comprehensive Maritime Transport Policy (CMTP) which aims to invigorate the maritime transport sector. An implementation plan aims to set up the basic principles for carrying out the CMTP, envisioning South Africa as an "International Maritime Centre (IMC)" by 2030.

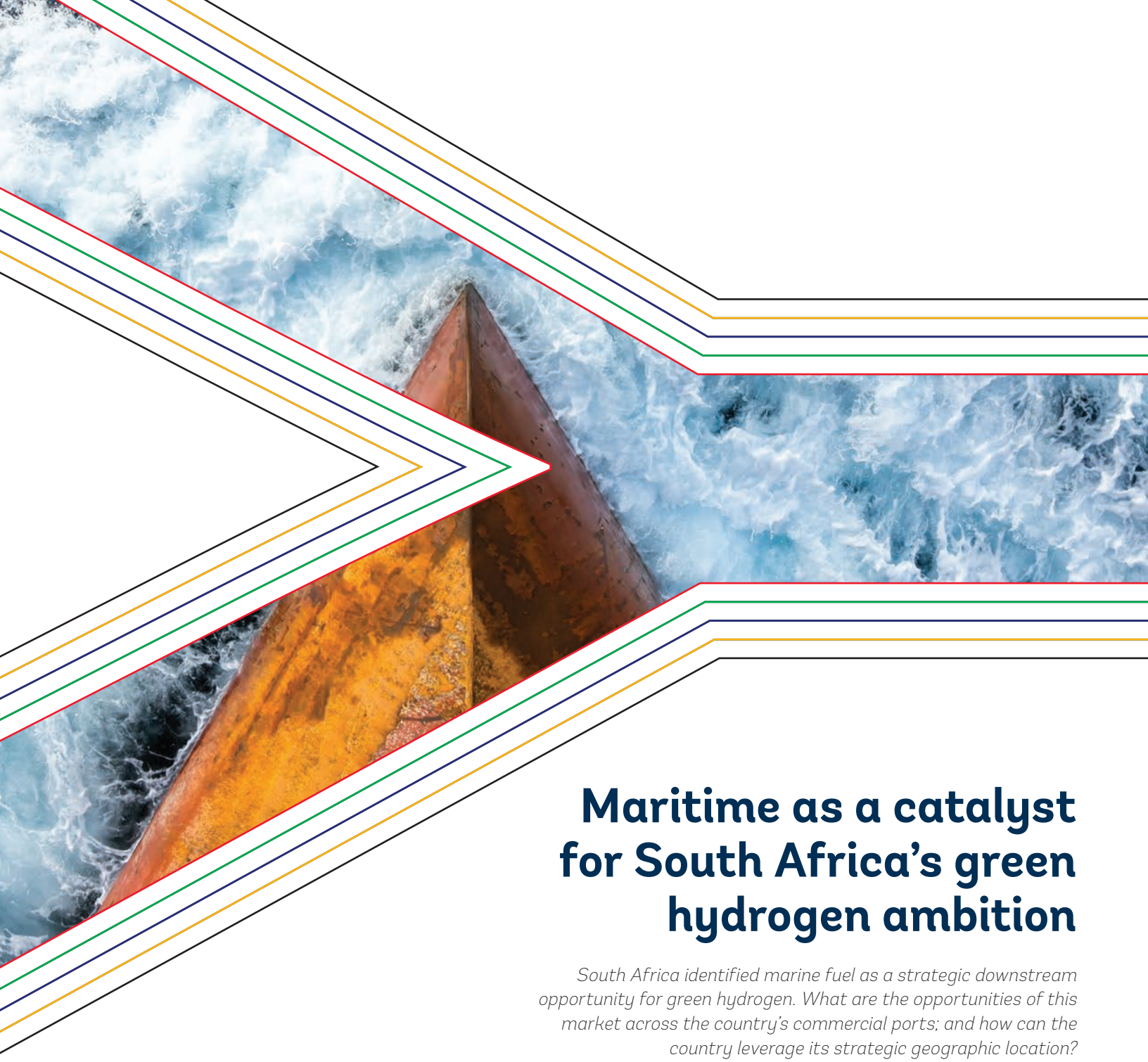
The CMTP also sets out key objectives which aim to harness socio-economic benefits for the South African economy. One target, for example, is to increase the number of ships refueling in South African territorial waters. These objectives aim to develop and grow South Africa to be an IMC in Africa, serving its maritime transport customers, and world trade in general, while contributing to the government's efforts of ensuring the country's competitiveness in international trade. The Department of Transport (DOT) recognized the role of the private sector for a successful maritime industry and in 2022, established the private sector led Maritime Industry Development Task Force Network (MIDTFN).

3.3. Risk spotlight: Maritime sector challenges

The South African port system has experienced significant challenges and is currently struggling to efficiently facilitate maritime trade. Operational problems, notably insufficient maintenance and theft have eroded the performance of ports and rail. This especially affected commodity exports, where in 2023, the volume of coal exports was about 30 percent lower than in 2018, despite the global coal price being three times as high

(World Bank 2023d). The country's main container ports have suffered from operational inefficiencies, too. In the World Bank's Container Port Performance Index (CPPI), the ports of Gqeberha, Durban, and Cape Town only ranked between 291 and 344 of 348 global ports (World Bank 2023c). Recognizing the importance of efficient ports and hinterland transport, the South African government adopted a freight logistics roadmap in 2023. This policy document outlines a strategic direction for the reform of the rail and port sectors, including aspects of private sector participation, competition, and the role of state-owned enterprises (DOT 2023).

South Africa's largest offshore bunkering location, Algoa Bay, has faced several challenges, too. The bunker hub off the coast of Gqeberha (and Coega), usually supplies around 50,000 to 100,000 tons of marine fuel in ship-to-ship operations every month. In September 2023, the South African Revenue Service (SARS) has imposed a moratorium on bunker operations due a tax and customs-related investigation and detained several vessels as a means of enforcement (SA News 2023). Previously, a moratorium on new licenses, required to commercially offer bunker services, was put in place by the South African Maritime Safety Authority (SAMSA), due to pending environmental risk assessments. While the cap on licenses caused concerns for industry representatives, civil society groups have long been criticizing the practice of bunkering in the environmentally sensitive marine area (The Maritime Executive 2022).



Maritime as a catalyst for South Africa's green hydrogen ambition

South Africa identified marine fuel as a strategic downstream opportunity for green hydrogen. What are the opportunities of this market across the country's commercial ports; and how can the country leverage its strategic geographic location?

04

4.1. The marine fuel market and green hydrogen

The Green Hydrogen Commercialization Strategy (GHCS) outlines two main markets for South Africa's green hydrogen economy. The first is a domestic market that focuses on the use of hydrogen in hard-to-decarbonize value chains such as liquid fuels and chemicals, as well as iron and steel production. The second is an export market for hydrogen to feed into the future global green energy trading markets. The maritime sector's hydrogen demand is considered an export sub-market. According to the government, South Africa's access to both the Indian and Atlantic Oceans could enable the country to secure an 8-10 percent market share of the global clean fuels market for shipping, equivalent to 0.8 to 1.0 million tons per year of hydrogen by 2050 (DTIC 2023). Marine fuel supply is a key downstream opportunity, which, unlike hydrogen exports, is not hampered by geographic distance to import markets. The South African government anticipates the commercialization of green marine fuel, a key direct use source for ammonia, from 2028.

This chapter provides a detailed analysis of the demand for hydrogen-based marine fuels produced from renewable electricity in South Africa's eight commercial seaports. The objective is to provide a complementary analysis of the global hydrogen demand from international shipping and to delve into the potential of marine fuel as a demand source for green hydrogen produced in South Africa.

The analysis considers the hydrogen demand from international shipping, considering the departing voyages from the eight commercial seaports in South Africa, namely Cape Town, Richards Bay, East London, Mossel Bay, Ngqura, Gqeberha (formerly Port Elizabeth), Durban, and Saldanha Bay. It also accounts for the by-pass traffic. The estimated uptake of green hydrogen-based marine fuels is projected for each port and aggregated by domestic region. The potential demand from maritime traffic passing by South Africa's coastline is estimated at a national level. The results are a crucial puzzle piece to understand the role of international shipping demand in the national hydrogen strategy and can help to further advance South Africa's green hydrogen ambition.

Box 1: Bunkering

FYI: What do ships "bunker"?

Bunkering in shipping refers to the process of supplying fuel to a ship. This crucial operation ensures ships have the necessary energy and fuel type to operate efficiently and safely during their voyages. The bunker supply chain involves multiple entities, including fuel suppliers or traders, bunker barge operators, and the ship's crew.

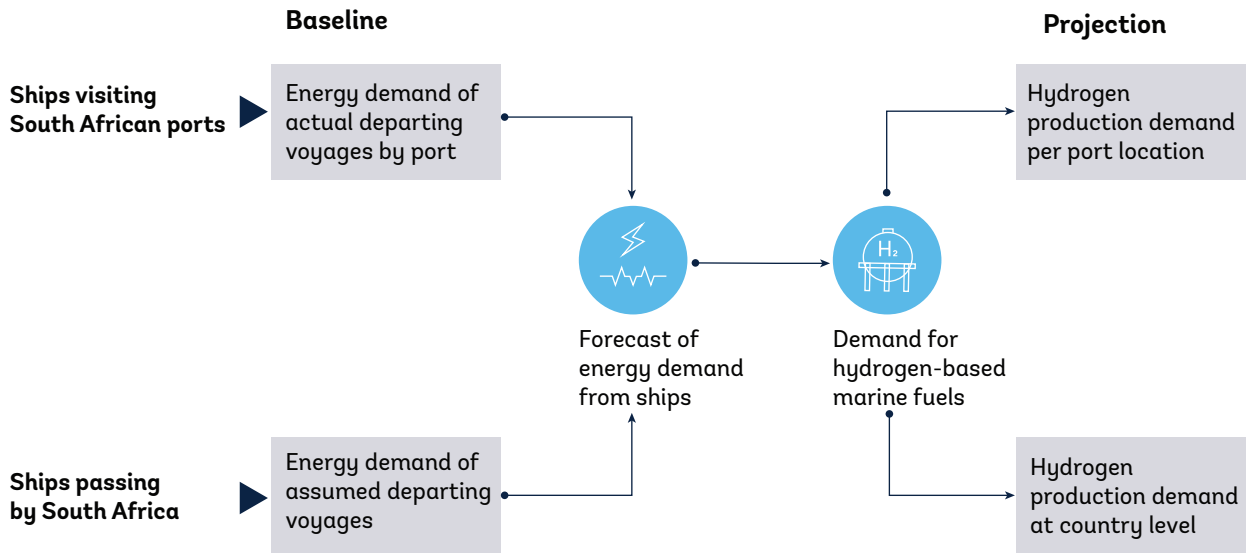
Ocean-going ships use different types of fuels, depending on vessel type and propulsion system. The most common ship fuels today are fossil fuels, namely Heavy Fuel Oil (HFO), Marine Gas Oil (MGO), and to some extent, Liquefied Natural Gas (LNG).

Ports around the world serve as key locations for bunkering operations and provide facilities and infrastructure to facilitate the efficient transfer of fuel from bunker barges to ships. Fuel is normally loaded onto bunker barges at refineries or storage facilities and then transported to ships docked at berth or anchored within port limits or in sheltered waters offshore. Usually, fuel is transferred through ship-to-ship operations.

4.2. Approach

The overall approach to estimating the maritime hydrogen demand is based on four steps. This methodology approach was used for the fleet calling at South African ports and for the fleet passing by South Africa (Figure 4.1.).

Figure 4.1. Quantifying hydrogen demand from ships



Source: World Bank.

Ships visiting South African ports

The energy demand for international voyages departing from South Africa is used as an indicator to estimate the potential sales market for marine fuel. This was estimated using Automatic Identification System (AIS) data for the years 2020, 2021 and 2022. A baseline fleet has been established for each port. This was based on an initial screening of 15,628 vessels that spent time in the region during the analysis years. After that, the base fleet was determined for each port by collecting all voyages that departed the port and went onto an international destination. The total fuel consumption for these departing voyages was accumulated to indicate the total demand per port.

Ships passing by South Africa

All vessels with AIS recordings within a defined region in proximity of the South African coastline were collected for each year 2020, 2021, and 2022. The passing by fleet was identified for each year by deducting the vessels from the fleet calling in the eight ports. The total annual fuel consumption of the fleet passing by was estimated using AIS data. To estimate the potential energy demand, the ratio between the total fuel consumption and the fuel consumption of the departing voyages for the fleet calling at South African ports was used.

Forecast of energy demand from ships

The forecast is based on the three-year baseline energy demand of international departing voyages estimated in the previous step. It considers the transport demand growth, and associated increases in shipping activities, and therefore, energy demand. It factors in a constant annual growth of 1 percent for all ship types due to an increase in shipping transport demand.

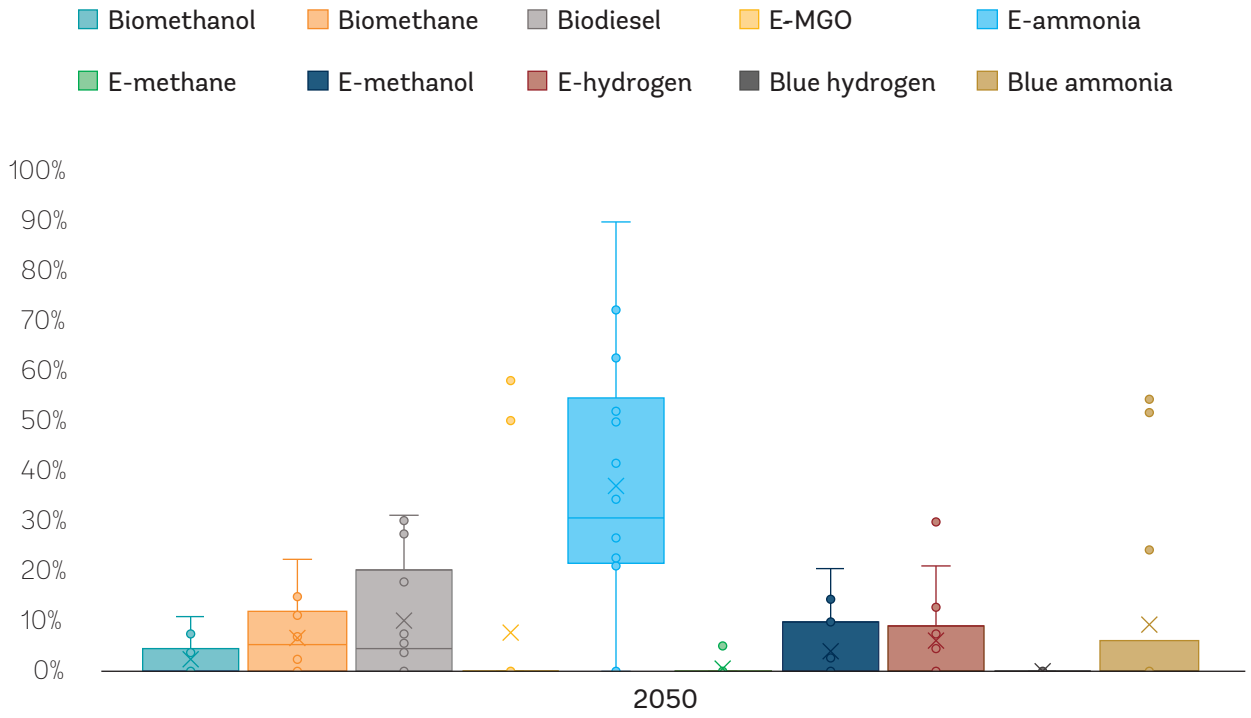
Demand for hydrogen-based marine fuels

To estimate the uptake of e-fuels for each of the eight ports, it is assumed that all ship vessels, except for liquified gas tankers, oil tankers, and cruise ships, will switch to green hydrogen-based marine fuels.[7] Since the overall energy demand from the excluded ships is relatively low (10 percent of the total), this has a minor impact on the overall hydrogen demand.

The share of hydrogen-based marine fuels is based on three scenarios showing lower, average, and upper bounds. These scenarios are based on 14 different 'hydrogen-based fuels scenarios' collected in analysis by Raucci et al. (2023). The lower bound is set as the lower quartile, while the upper bound is the upper quartile (Figure 4.2.). The advantage of using the average and upper and lower quartile of 14 different scenarios is that the assumed shares are representative of the consensus across different organizations and methodologies.

[7] The rationale of excluding this vessel group is mainly due to their distinct operating profiles, and operational requirements (e.g., LNG carriers use boil-off gas for propulsion).

Figure 4.2. Distribution of fuel types for multiple hydrogen scenarios



Source: Raucci, McKinlay and Karan (2023).

Table 4.1. Shares of green hydrogen-based fuels based on 14 different scenarios

	2030	2050
Lower quartile	1%	33%
Average	6%	55%
Upper quartile	7%	82%

Source: Raucci, McKinlay and Karan (2023).

For each port, the uptake of the fuels is projected, and demand for hyrogen-based marine fuel is expressed in TWh/year. The same approach is used for the fleet passing by South Africa.

Hydrogen production demand

To estimate the hydrogen required to meet the demand for hydrogen-based marine fuels from international shipping, it was assumed that green ammonia will be used onboard vessels. The energy demand for hydrogen-based fuels was converted in tons of green ammonia for which the necessary green hydrogen feedstock was calculated.

These calculations were applied to both the fleet calling at each of the South African ports, as well as the fleet passing by the country.

The study did not consider the breakdown of hydrogen-based fuel demand into each type of fuel (e.g., green methanol) as it is beyond the scope of this research. Therefore, to simplify the analysis, green ammonia was chosen to calculate the hydrogen production demand. Since hydrogen is a crucial raw material for all green hydrogen-based fuels, the estimate using green ammonia can be used as a proxy for other hydrogen-based marine fuels. A sensitivity analysis was performed to evaluate the differences.

4.3. Energy demand

Energy demand from international ship traffic

Overall, the energy demand for international departing voyages for vessels calling in all eight commercial ports is approximately 26.8 TWh. This is based on a three-year average from 2020 to 2022.

Durban, Richards Bay, Cape Town, and Saldanha Bay are the ports with the highest demands respectively with 24 percent, 21 percent, 17 percent, and 15 percent of the total. Gqeberha and Nqgqura combined also have one of the highest demands, reaching 21 percent of the total. This leaves East London and Mossel Bay with relatively low energy demand, accounting for approximately two percent of the total.

Bulk carriers, containerships, and chemical tankers require the most energy for their international departures from these ports. The highest energy demand is from bulk carriers with a size range of 60,000 to 100,000 dwt in Richards Bay port, which accounts for almost 40 percent of the total demand there.

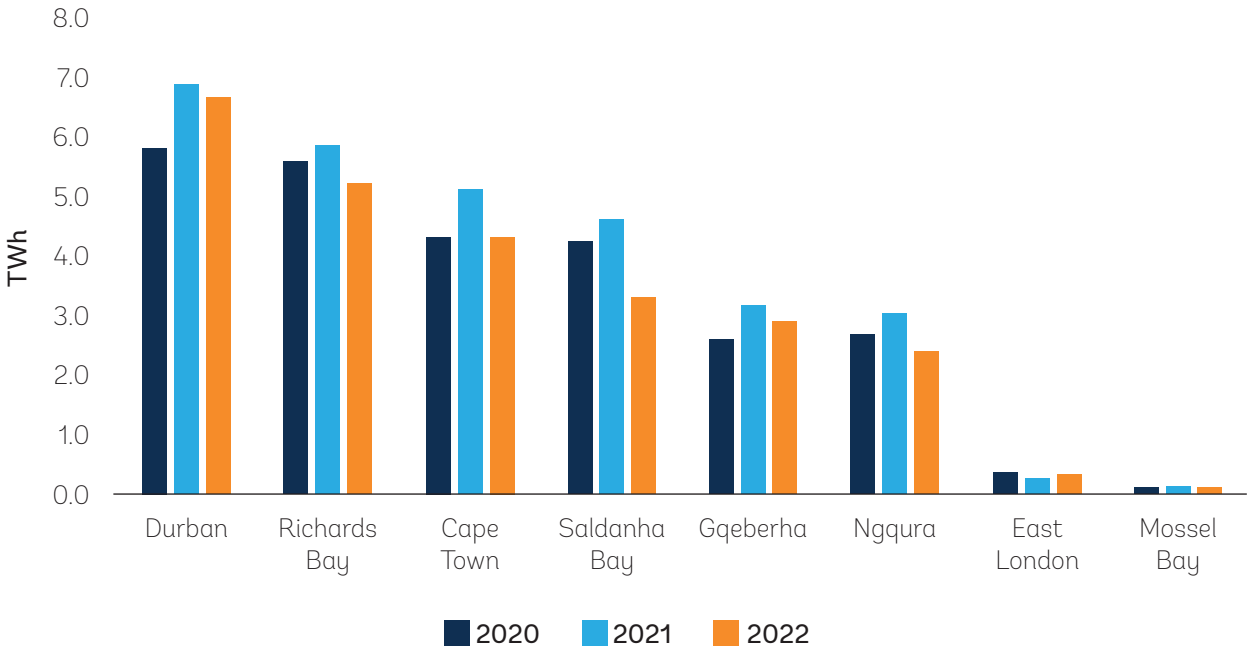
The maritime traffic that is passing by South Africa's coastline is offering the country a unique opportunity to strategically place itself as a supplier of green shipping fuels. For this fleet, the potential demand for international departing voyages is estimated to be 52.9 TWh, which is approximately two times the energy demand for vessels calling in all eight ports.

Energy demand by port

The study uses the energy demand of international departing voyages to estimate the potential market for marine fuel sales in South Africa's eight seaports. This estimation is used as a baseline to understand the potential demand for hydrogen-based marine fuels.

The energy demand for international departing voyages from eight commercial seaports in South Africa is presented in Figure 4.3.

Figure 4.3. Energy demand for international departing voyages from South Africa's commercial ports



Source: World Bank.

In general, the energy demand for international departing voyages increased in 2021. The demand decreased in 2022 for most ports, except for East London, which had an opposite trend, and Mossel Bay, which remained relatively stable. Table 2 provides the three-year average energy demand for international departing voyages, which serves as a baseline for assessing the potential demand hydrogen-based fuels. On average, the annual changes in energy demand were around +9 percent and -10 percent.

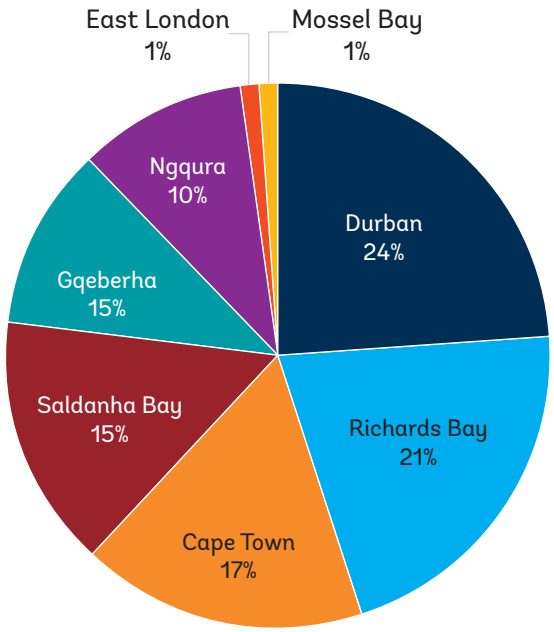
Table 4.2. 3-year average energy demand for int. voyages for main ports and annual changes

	3-year Average 2020-2022	Change 2020-2021	Change 2021-2022
	TWh		
Durban	6.5	18%	-2%
Richards Bay	5.6	5%	-11%
Cape Town	4.6	19%	-15%
Saldanha Bay	4.1	9%	-28%
Gqeberha	2.9	22%	-8%
Ngqura	2.7	14%	-22%
East London	0.3	-31%	22%
Mossel Bay	0.1	18%	-18%
Total eight ports	26.8	9%	-10%

Source: World Bank.

Figure 4.4. presents the breakdown of the eight ports in terms of their shares of the total energy demand. Durban, Richards Bay, Cape Town, and Saldanha Bay are the ports with the highest demands. Gqeberha can be grouped with Nqgqura, also known as Coega, due to their proximity, which means that combined, they also have one of the highest demands. This leaves East London and Mossel Bay as the two ports among others with relatively low energy demand for international departing voyages.

Figure 4.4. Breakdown by ports of energy demand for int. voyage as share of total ports



Source: World Bank.

Considering only the demand for departing voyages may have some limitations. For instance, some ships calling in South Africa may not need to refuel for each voyage, which could lead to an overestimation. Conversely, some vessels refueling in South Africa may take more fuel than necessary for the next voyage, which could lead to an underestimation. Nonetheless, this study assumes that the demand for international departing voyages is still representative of the potential bunker sales market. This is particularly true when estimating the future uptake of hydrogen-based fuels, for which vessels may need to refuel more often due to the lower energy density of the fuel.

Moreover, it should be noted that of the 15,628 vessels tested, AIS data was unavailable for 855 of these vessels. For each of the missing vessels, a projected departing energy demand was calculated based on results for other vessels in the same size category (totaling 5.7 TWh for all 855 vessels). In the original screening, 23.4 percent (3,658 vessels) of the vessels tested made it into a baseline fleet of one of the ports. Assuming the same proportion of the missing fleet would join, - this would increase the total fuel demand for South Africa by 1.34 TWh in 2022, from 25.4 TWh to 25.7 TWh. This means that there could be a potential underestimation of 5.3% which is lower than most of the annual variations across the years 2020 and 2022.

Energy demand by vessel type

Table 4.3. shows the energy demand for international departing voyages according to vessel type and size grouping. This data does not include domestic voyages and vessels smaller than GT 300.

As noted earlier in this chapter, it is not surprising that bulk carriers, containerships, and chemical tankers have the highest energy demand for international departing voyages in these ports. They are followed by oil tankers and vehicle carriers. The highest contribution comes from bulk carriers of 60,000 to 100,000 dwt in Richards Bay port, representing almost 40 percent of the entire demand in this port.

Table 4.3. Energy demand for international departing voyages by vessel type/size

		Durban	Richards Bay	Cape Town	Saldanha Bay	Gqeberha	Ngqura	East London	Mossel Bay
Ship Type	Size range	GWh	GWh	GWh	GWh	GWh	GWh	GWh	GWh
Bulk carrier	0-9,999 dwt	4	3		1	1			
Bulk carrier	10,000-34,999 dwt	140	82	25	39	74	31	3	1
Bulk carrier	35,000-59,999 dwt	603	877	58	547	467	281	9	5
Bulk carrier	60,000-99,999 dwt	651	2,205	102	187	855	657	22	12
Bulk carrier	100,000-199,999 dwt	138	1,702	95	1,881	46	25	18	13
Bulk carrier	200,000-+ dwt	16	50	49	959	18	2	22	7
Chemical tanker	5,000-9,999 dwt	7	6	9	0	2	1	0	
Chemical tanker	10,000-39,999 dwt	43	43	4	0	30	12		
Chemical tanker	40,000-+ dwt	815	351	176	3	237	149	15	38
Fully Cellular Container	0-999	3		4					
Fully Cellular Container	1,000-1,999	35	7	181		4	2		
Fully Cellular Container	2,000-2,999	229		164		65	22		
Fully Cellular Container	3,000-4,999	897	2	702	7	130	117		4
Fully Cellular Container	5,000-7,999	286		760		45	166		
Fully Cellular Container	8,000-11,999	387		969	3	4	611	4	
Fully Cellular Container	12,000-14,499	11		9	4		63		
Fully Cellular Container	14,500-19,999						6		
Fully Cellular Container	20,000-+				3				
General cargo	0-4,999	5	0	10		3	1		0
General cargo	5,000-9,999	22	8	20	1	5	2	0	
General cargo	10,000-19,999	68	10	36	2	13	10		
General cargo	20,000-+	182	88	89	31	76	51	5	1
Liquefied gas tanker	0-49,999	21	47	5	20	16	15		2
Liquefied gas tanker	50,000-99,999	44	60	21		183	66		
Liquefied gas tanker	100,000-199,999	49		45	4	59	71		
Liquefied gas tanker	200,000-+					12	11		
Oil tanker	0-4,999	0	0						

		Durban	Richards Bay	Cape Town	Saldanha Bay	Gqeberha	Ngqura	East London	Mossel Bay
Ship Type	**Size range**	GWh	GWh	GWh	GWh	GWh	GWh	GWh	GWh
Oil tanker	5,000-9,999	7	15	3	0	1	1		
Oil tanker	10,000-19,999	1		1					
Oil tanker	20,000-59,999	186	14	38		47	8	4	11
Oil tanker	60,000-79,999	6	1	144		38	14		
Oil tanker	80,000-119,999	6		45	6	56	61		4
Oil tanker	120,000-199,999	346	1	72	115	117	165	4	
Oil tanker	200,000-+	109		132	223	23	44	24	13
Cruise	0-1,999								
Cruise	2,000-9,999	1							
Cruise	10,000-59,999	3	6	18		9			
Cruise	60,000-99,999	36		8					
Cruise	100,000-149,999								
Cruise	150,000-+								
Refrigerated bulk	0-1,999			1					
Refrigerated bulk	2,000-5,999	5	1	30					1
Refrigerated bulk	6,000-9,999	27	0	107		25			
Refrigerated bulk	10,000-+	73		282		18	3		
Ro-Ro	0-4,999								
Ro-Ro	5,000-9,999	0							
Ro-Ro	10,000-14,999			2					
Ro-Ro	15,000-+	66				11		0	
Vehicle	0-29,999	859	0	7		192	6	141	
Vehicle	30,000-49,999	19						1	
Vehicle	50,000-+								
Miscellaneous - fishing				10		0	1		
Offshore		34		91	2	3	23	0	1
Miscellaneous - other		39	14	91	25	15	19	22	2
TOTAL		6,480	5,593	4,614	4,062	2,901	2,717	296	115

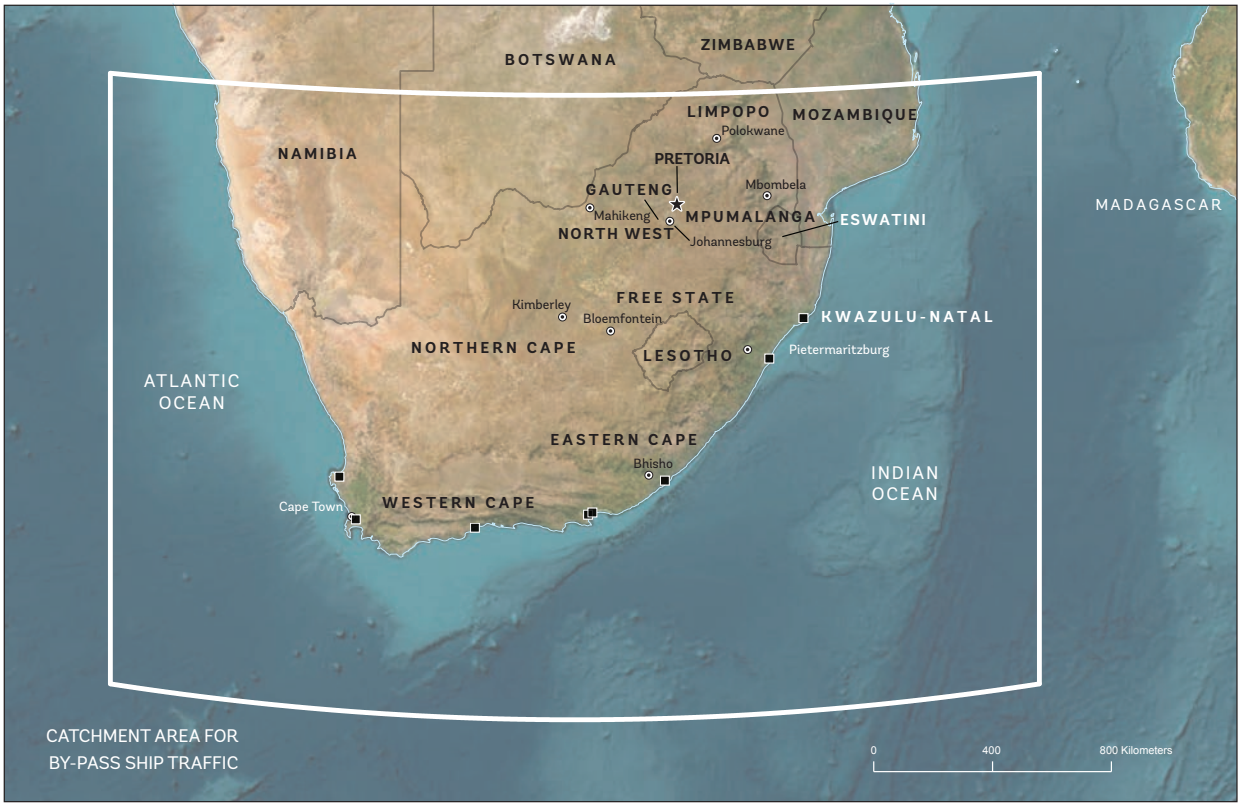
Source: World Bank.

Energy demand of passing fleet

The development of a bunker supply market for marine fuel from green hydrogen could attract additional bunkering stops from ships sailing in the vicinity of the South Africa coastline. Therefore, this section aims to broadly outline the size of this additional potential market.

For this analysis, all vessels with AIS recordings within the defined region (shown in Figure 4.5.) were collected for the years 2020, 2021, and 2022. The crossing fleet could then be adapted for each year by deducting the vessels from any of the eight ports' baseline fleets.

Figure 4.5. Catchment area for the passing fleet



Source: World Bank, (latitude min -39.2, max -22.1; longitude min 11.1, max 38.0). For accessibility purposes, the locations of ports are approximate only.

Results are shown in Table 4.4. On average 5,171 vessels pass through this region each year without stopping at any of the eight selected ports. These vessels consume 41 million tons of marine diesel oil (MDO) equivalent per year (490 TWh). Results for the baseline fleet calculated a fuel demand of 10.8 percent for South African ports compared to the annual fuel consumption, using the energy demand of departing voyages. By assuming this same ratio, then the total size of this additional market is 52.9 TWh, equivalent to 1.83 million tons of hydrogen.

Table 4.4. Potential demand from the passing fleet

Year	Baseline Fleet				Passing Fleet		
	No of vessels	Total annual fuel consumption	Fuel consumption from departing voyages	% captured	No of vessels	Total annual fuel consumption	Fuel consumption from departing voyages*
	Count	TWh	TWh	%	Count	TWh	TWh
2020	3,154	240	25.8	10.7	4,887	482	51.6
2021	3,489	271	29.2	10.8	5,424	526	56.8
2022	3,201	233	25.4	10.9	5,127	462	50.3
Average	3,281	248	26.8	10.8	5,146	490	52.9

*Assuming the same ratio of global annual fuel consumption to port demand

Source: World Bank.

4.4. Forecast of demand for hydrogen-based marine fuels

A demand forecast for hydrogen-based fuels is key to quantify the demand and possible local hydrogen production. This analysis estimates the demand in terawatt hours (TWh) for each of the ports, aggregated by region. In addition, the demand of the passing fleet was also estimated for the whole country.

Overall, the demand for hydrogen-based fuels from international departing voyages is estimated to be approximately 4.9 TWh in 2035, reaching 16.5 TWh in 2050. The passing fleet is estimated to demand 11.2 TWh in 2035, reaching 38.5 TWh in 2050.

> The eastern ports of Durban and Richards Bay are projected to be the largest consuming region of these new fuel types.

The eastern port region is expected to be the largest consumption hub, followed by the western port region and the central region. For instance, the eastern ports of Durban and Richards Bay are projected to be the largest consuming region of these new fuel types. The annual projection for both combined ports is of 2.3 TWh in 2035, reaching 7.8 TWh by 2050. The western region, including Cape Town and Saldanha Bay, is estimated to demand 1.6 TWh of hydrogen-based fuels in 2035, increasing to 5.3 TWh in 2050.

The central port region, driven mainly by the demand in Gqeberha and Ngqura, is estimated to demand 1 TWh of green hydrogen-based fuels in 2035, increasing to 3.4 TWh in 2050.

Table 4.5. Projected annual energy demand for hydrogen-based fuels (in TWh)

	Lower quartile			Average			Upper quartile		
	2030	2040	2050	2030	2040	2050	2030	2040	2050
Durban	0.1	1.1	2.3	0.4	2.0	3.9	0.4	2.9	5.9
Richards Bay	0.1	1.1	2.3	0.4	2.0	3.8	0.4	2.9	5.7
Eastern Region	0.1	2.2	4.6	0.8	4.0	7.8	0.8	5.8	11.6
Cape Town	0.1	0.8	1.6	0.3	1.4	2.7	0.3	2.0	4.1
Saldanha Bay	0.0	0.8	1.6	0.3	1.3	2.6	0.3	1.9	3.9
Western Region	0.1	1.5	3.2	0.5	2.7	5.3	0.6	4.0	7.9
Gqeberha	0.0	0.5	1.0	0.2	0.8	1.6	0.2	1.2	2.4
Ngqura	0.0	0.4	0.9	0.2	0.8	1.6	0.2	1.2	2.3
East London	0.0	0.0	0.1	0.0	0.1	0.2	0.0	0.1	0.3
Mossel Bay	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1
Central Region	0.1	1.0	2.0	0.3	1.8	3.4	0.4	2.6	5.1
Total eight ports	0.3	4.7	9.8	1.6	8.5	16.5	1.7	12.3	24.7
Passing fleet	0.7	10.7	22.8	3.7	19.5	38.5	3.9	28.2	57.4

Source: World Bank.

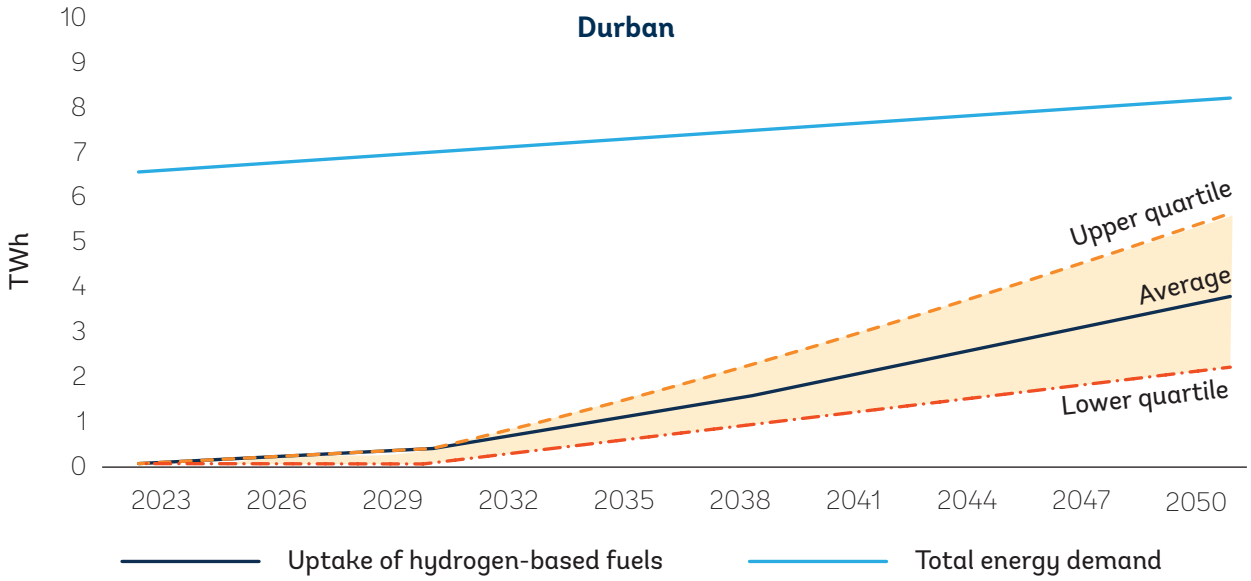
4.4.1. Expected market demand by port

The following sections show the range of demand for hydrogen-based fuels at each of the eight South African ports, projected from 2023 to 2050, broken down by shipping segments. The first chart includes the total port's energy demand. The difference between the total port energy demand (blue line) and the projected demand ranges (in orange) is accounted for by the projected demand for other fuels (not green hydrogen based).

Durban

Durban is projected as the biggest consumer of hydrogen-based marine fuels among the South African ports. The average demand is estimated to start at 0.4 TWh in 2030, which then multiples five-fold by 2040. From 2040-2050, the port's energy demand for marine fuels from hydrogen doubles to 3.9 TWh per annum.

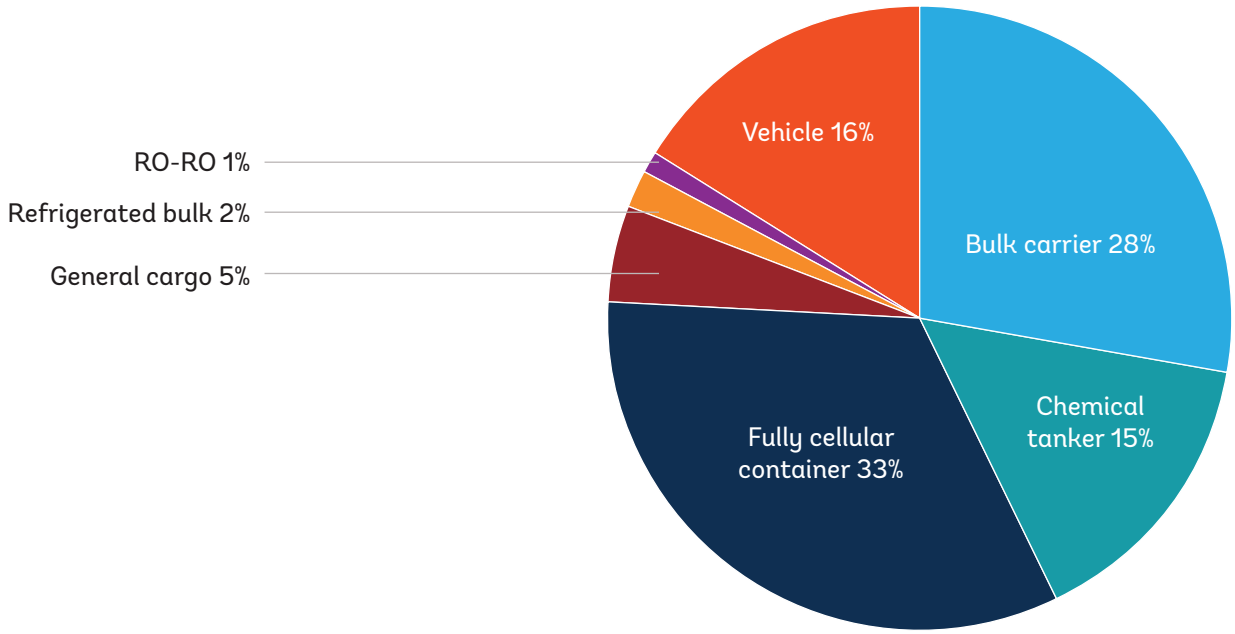
Figure 4.6. Demand projections in TWh versus the total energy demand at the port of Durban



Source: World Bank.

At Durban, the vessel activity is split across containerships, bulk carriers, vehicle carriers and chemical tankers. This diversifies the end consumer of fuels across fleet segments, safeguarding from excessive exposure to a specific market.

Figure 4.7. Breakdown of shipping segment for the port of Durban

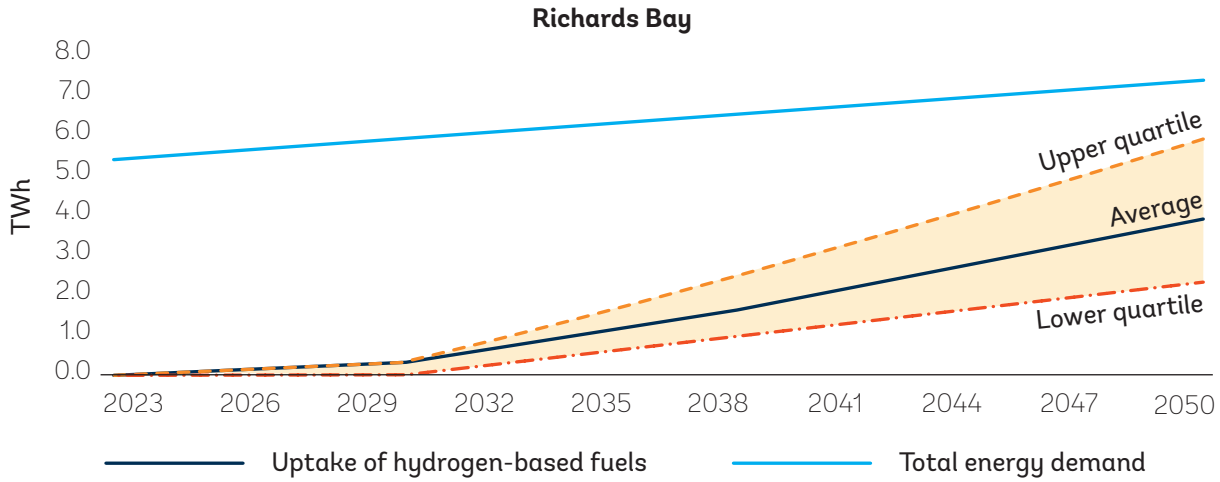


Source: World Bank.

Richards Bay

Demand at Richards Bay is almost equivalent to Durban (Figure 4.8.) but is solely centered around the activity of bulk carriers, as illustrated in Figure 4.9.

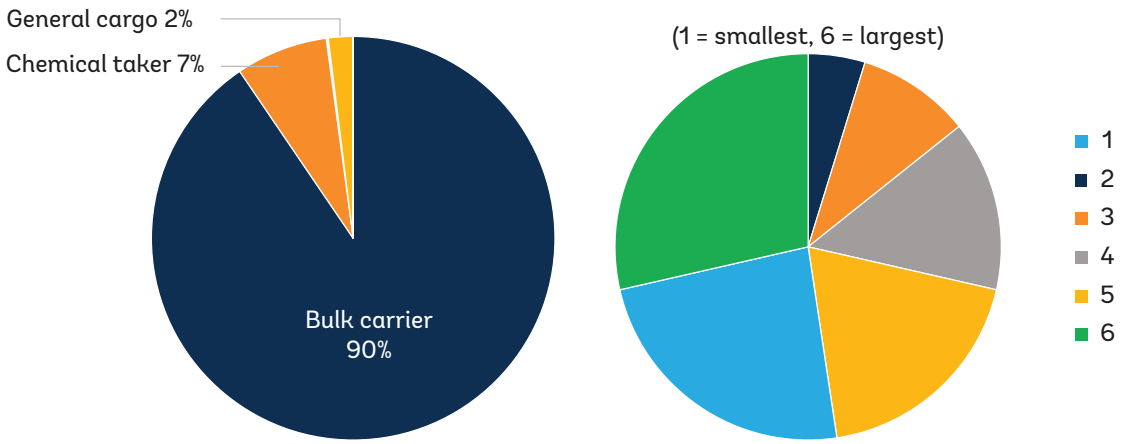
Figure 4.8. Demand projections in TWh versus the total energy demand at the port of Richards Bay



Source: World Bank.

The port of Richards Bay is the country's largest coal export location. Bulk carrier activity might change as the world shifts away from fossil fuels. But the port also sees break-bulk activity involving pig iron and base metals. This could get a boost from future needs for energy transition infrastructure and critical minerals. Therefore, differentiating between the bulk carrier activity at the port is important. One way of differentiating the market is by the vessel size of the bulk carriers, as broken down in Figure 4.9. Nevertheless, historic commercial data, such as from ship operators and terminals, would be helpful to clearly differentiate cargoes.

Figure 4.9. (left) Breakdown of shipping segment for the port of Richards Bay (right) Breakdown of energy demand by bulk carrier segment size



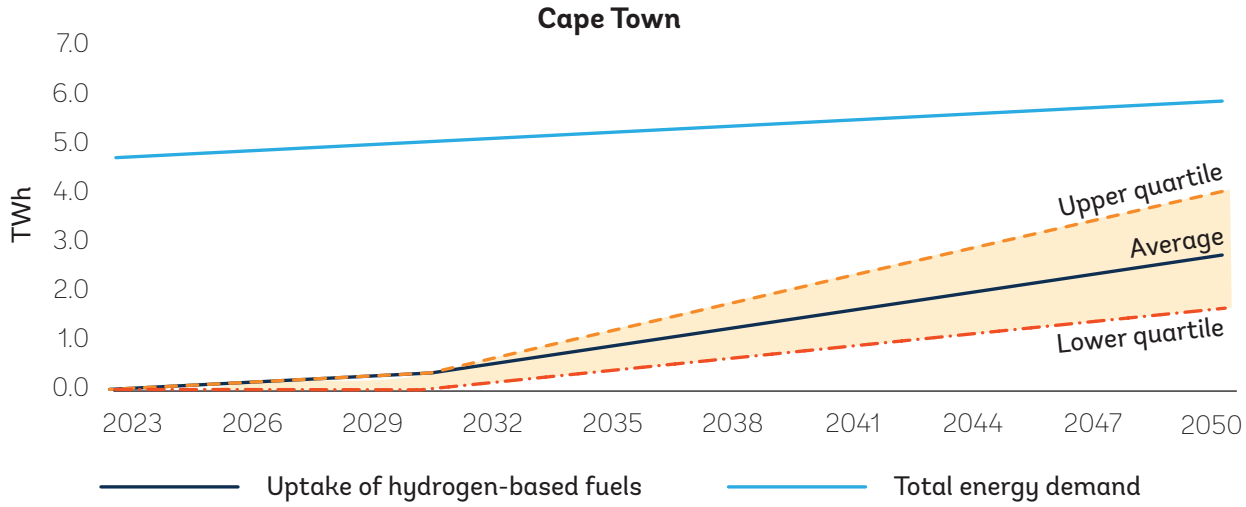
Source: World Bank.

Cape Town

The consumption of hydrogen-based fuels in the western port region is projected to be split evenly between Cape Town and Saldanha Bay, with annual demand estimates for Cape Town ranging from 1.6-4.1 TWh by 2050.

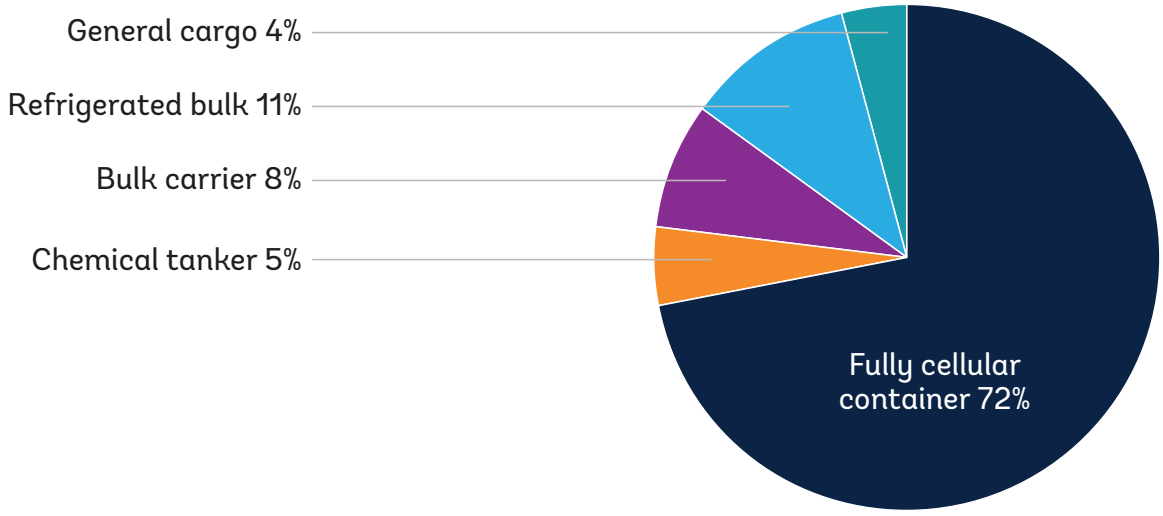
The port of Cape Town operates as a major container terminal with more than 72 percent of vessel activity from containerships, and the remainder split between general cargo, liquid and dry bulk.

Figure 4.10. Demand projections in TWh versus the total energy demand at the port of Cape Town



Source: World Bank.

Figure 4.11. Breakdown of shipping segment for the port of Cape Town



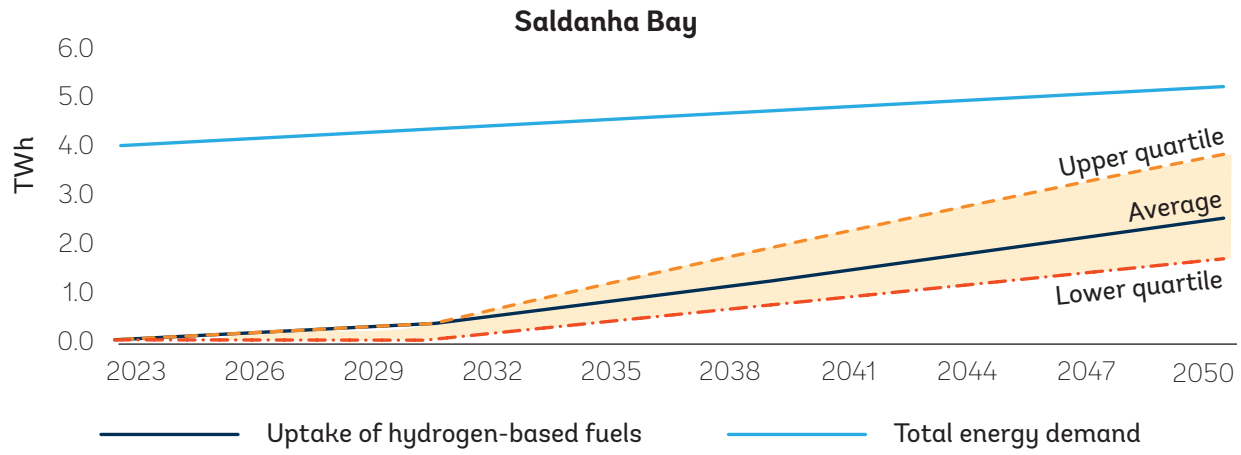
Source: World Bank.

Saldanha Bay

Saldanha Bay almost exclusively focuses on bulk carrier activity, given its substantial iron ore exports. The port's annual demand for hydrogen-based marine fuels is estimated to range between 1.6-3.9 TWh by 2050.

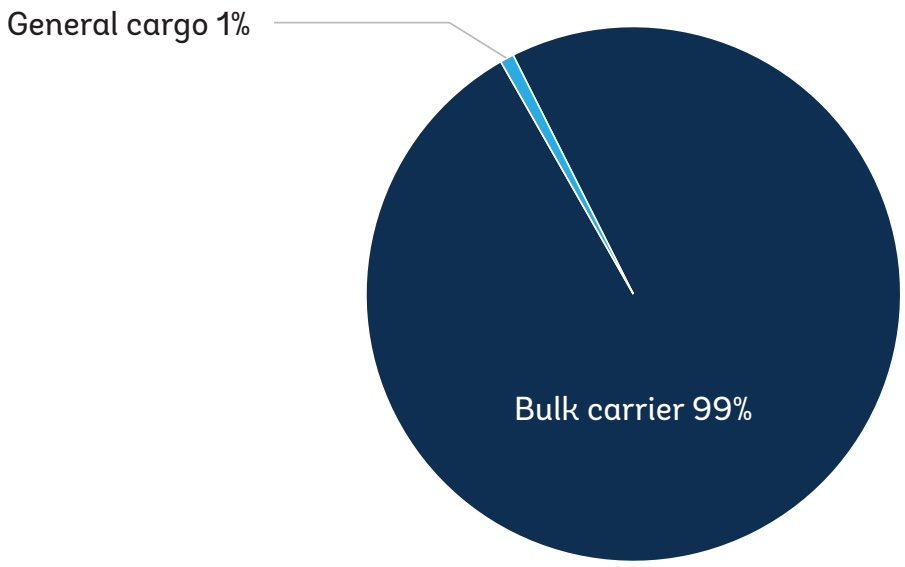
The diametrically opposing shipping activity profiles of Saldanha Bay and neighboring Cape Town offers an opportunity to diversify offtake risks across the port demand centers.

Figure 4.12. Demand projections in TWh versus the total energy demand at the port of Saldanha Bay



Source: World Bank.

Figure 4.13. Breakdown of shipping segment for the port of Saldanha Bay

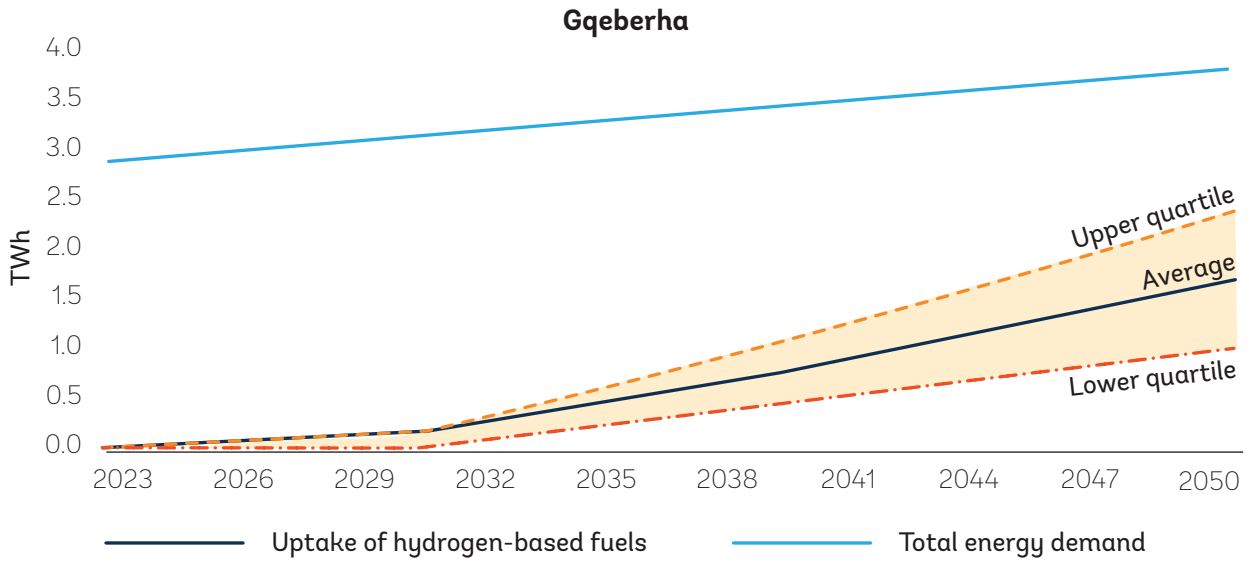


Source: World Bank.

Gqeberha

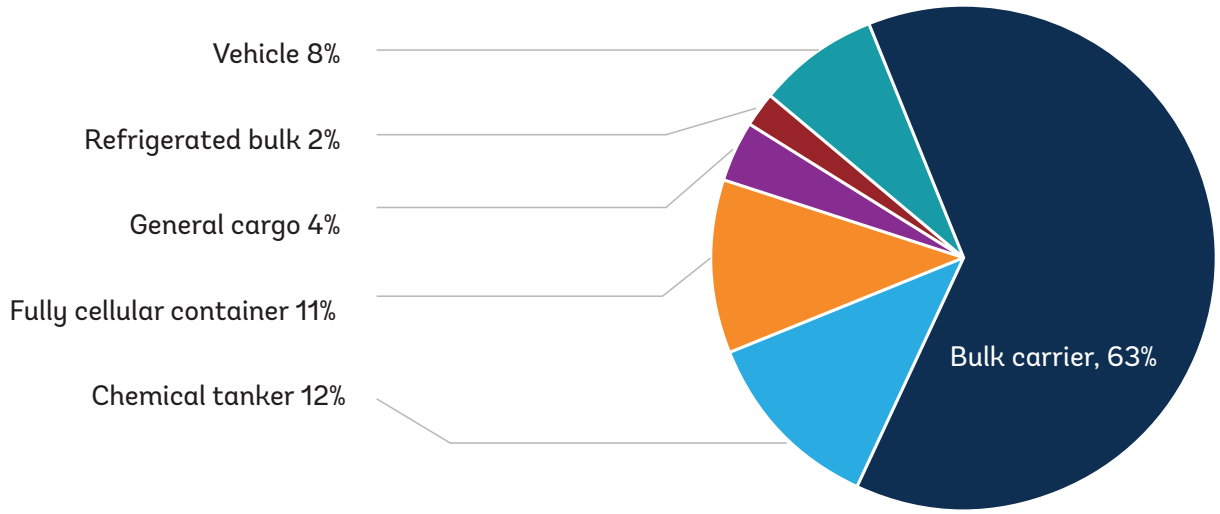
As one of the ports with the largest demand for hydrogen-based fuels in the central port region, demand at Gqeberha is estimated to average 0.2 TWh by 2030, rising to 0.8TWh and 1.6 TWh in 2040 and 2050 respectively. Bulk carriers make up 63 percent of the vessel activity at the port, followed by chemical tankers and containerships at 12 percent and 11 percent respectively.

Figure 4.14. Demand projections in TWh versus the total energy demand at Gqeberha



Source: World Bank.

Figure 4.15. Breakdown of shipping segment for Gqeberha

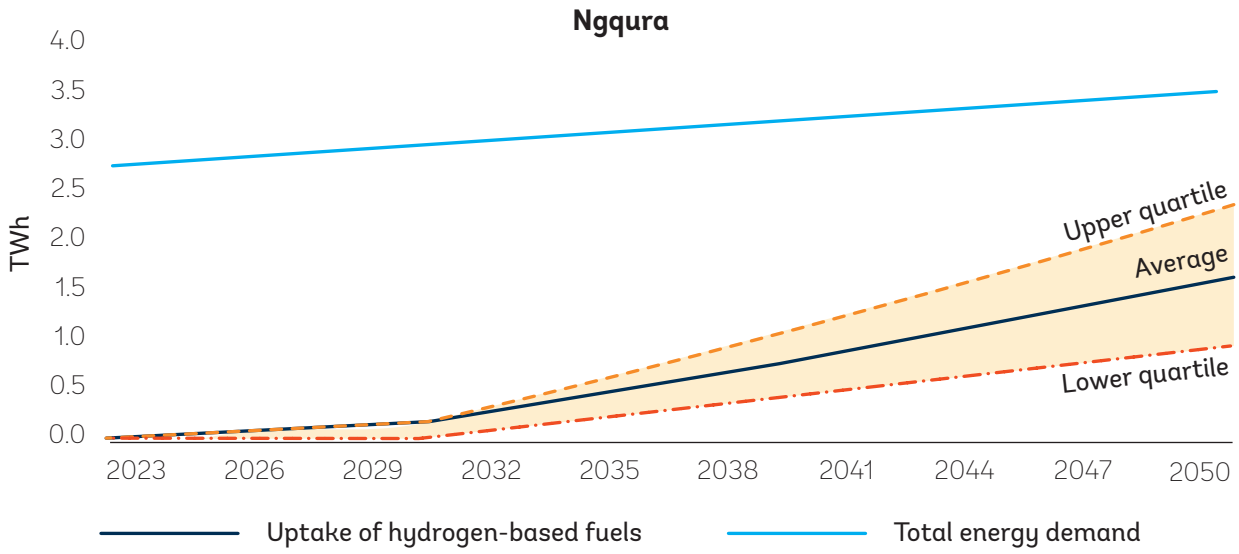


Source: World Bank.

Ngqura

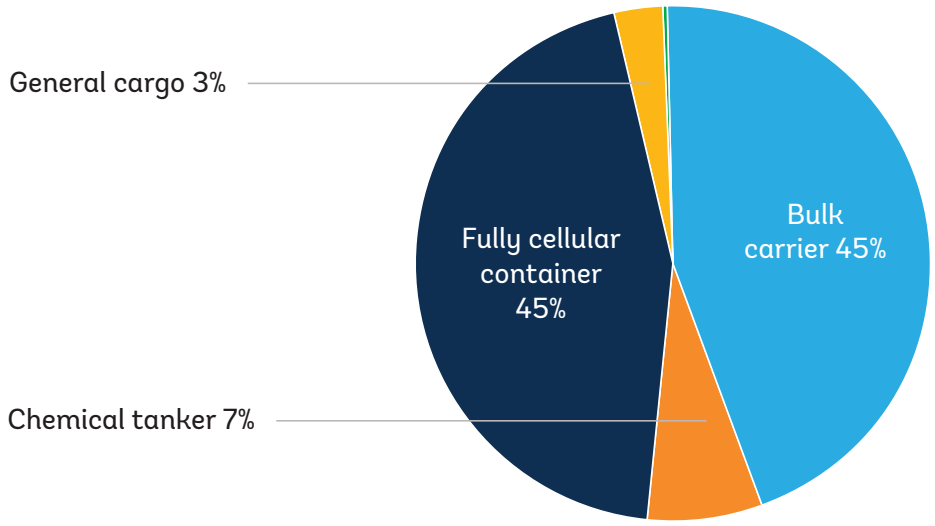
The demand projections for the port of Ngqura mirrors estimates for neighboring Gqeberha. Their cumulative demand is expected to make up 93 percent of the central port region's requirements in 2050. Vessel activity at the port of Ngqura is evenly split at 45 percent apiece between containers and bulk carriers, with the remaining 10 percent attributed to general cargo ships and chemical tankers.

Figure 4.16. Demand projections in TWh versus the total energy demand at the port of Ngqura



Source: World Bank.

Figure 4.17. Breakdown of shipping segment for the port of Ngqura



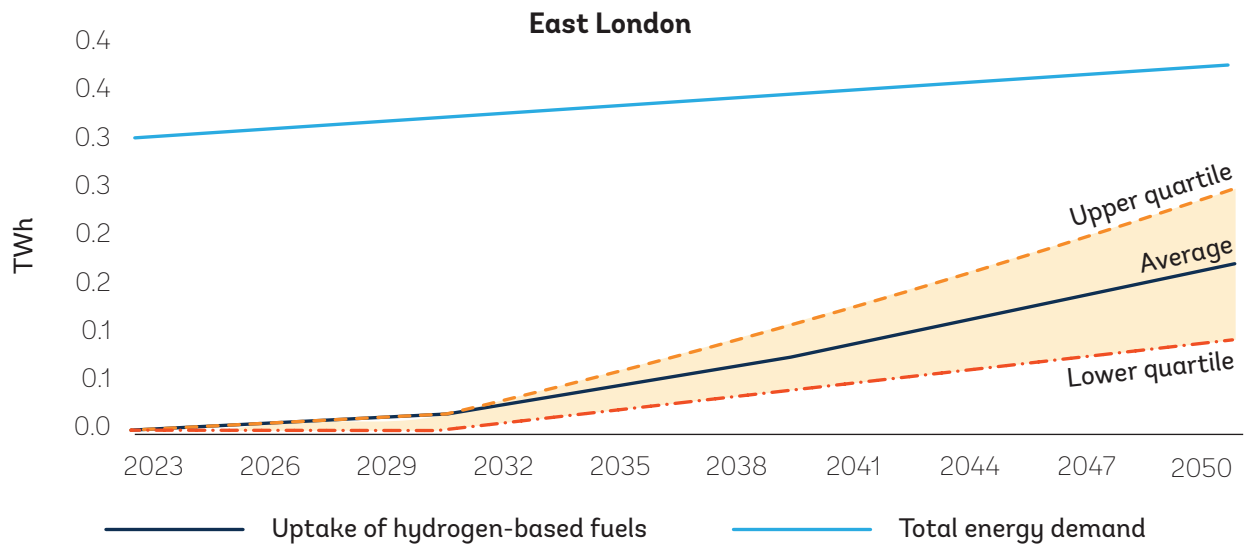
Source: World Bank.

East London

Demand for hydrogen-based fuels at East London is among the lowest of the country's ports, with annual demand only averaging 0.2 TWh by 2050.

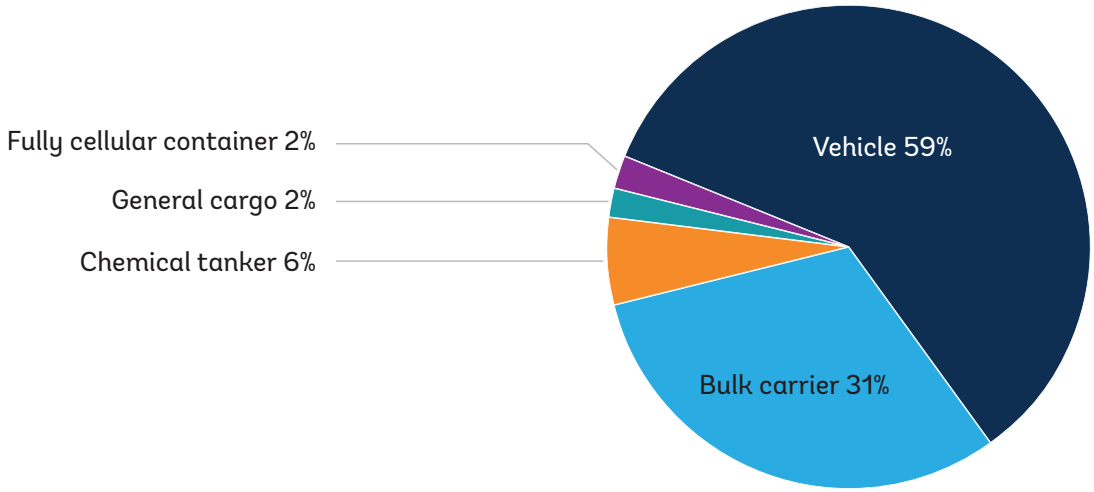
Despite this, there is a high proportion of activity at the port from vehicle carriers (59 percent). Vehicle carriers typically follow set routes, providing a consistent market demand for new fuel types. This aids in planning regional infrastructure since it identifies projected demand streams that can be relied upon with greater certainty. Combining demand from various ports can create scale and give central planners a clearer signal of demand.

Figure 4.18. Demand projections in TWh versus the total energy demand at the port of East London



Source: World Bank.

Figure 4.19. Breakdown of shipping segment for the port of East London

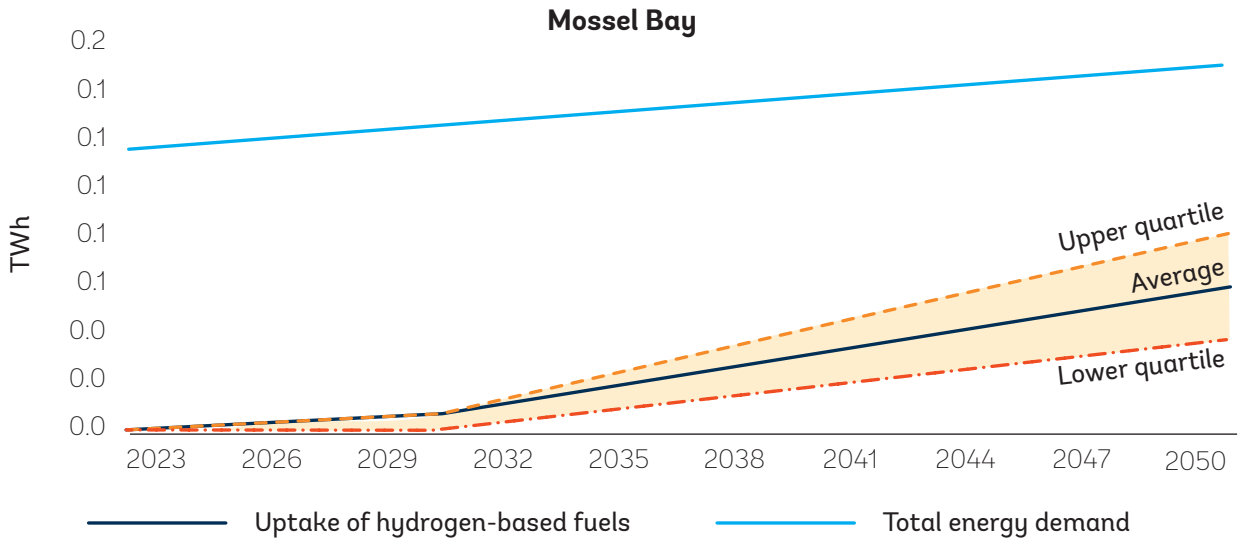


Source: World Bank.

Mossel Bay

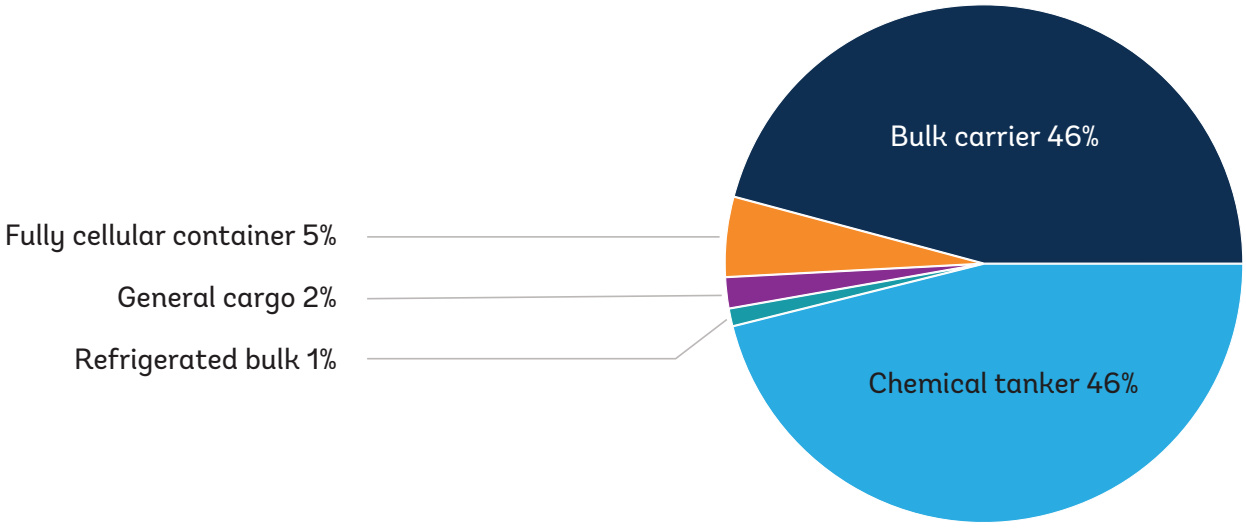
Demand for hydrogen-based marine fuels is estimated to the be smallest here, not exceeding 0.1 TWh in even the upper bound scenarios. Activity is predominately focused on chemical tankers (46 percent) and bulk carriers (46 percent).

Figure 4.20. Demand projections in TWh versus the total energy demand at the port of Mossel Bay



Source: World Bank.

Figure 4.21. Breakdown of shipping segment for the port of Mossel Bay

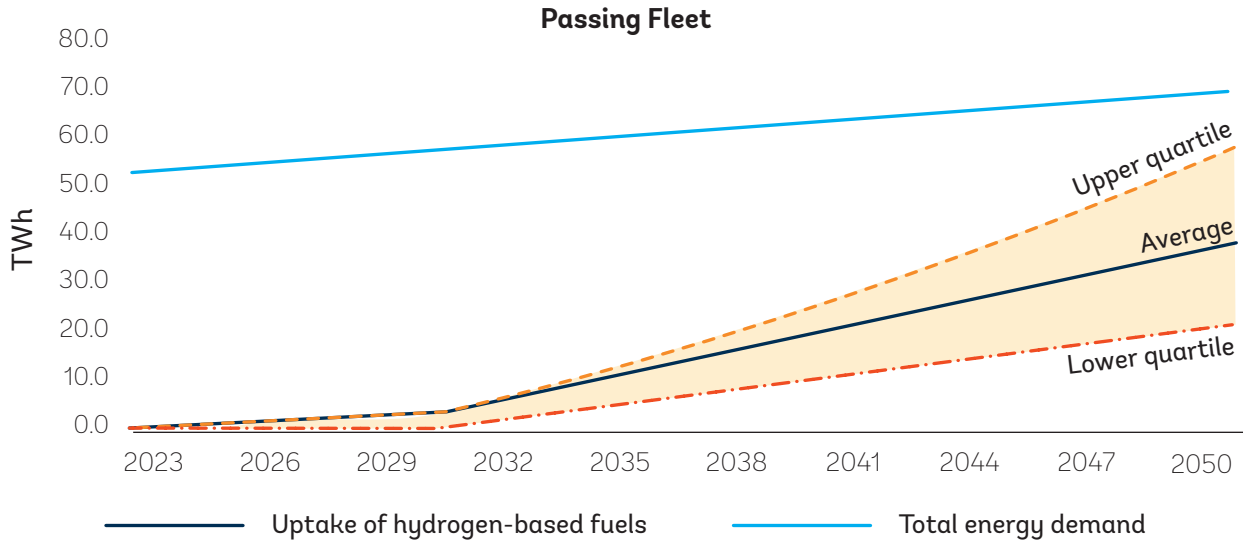


Source: World Bank.

Passing fleet

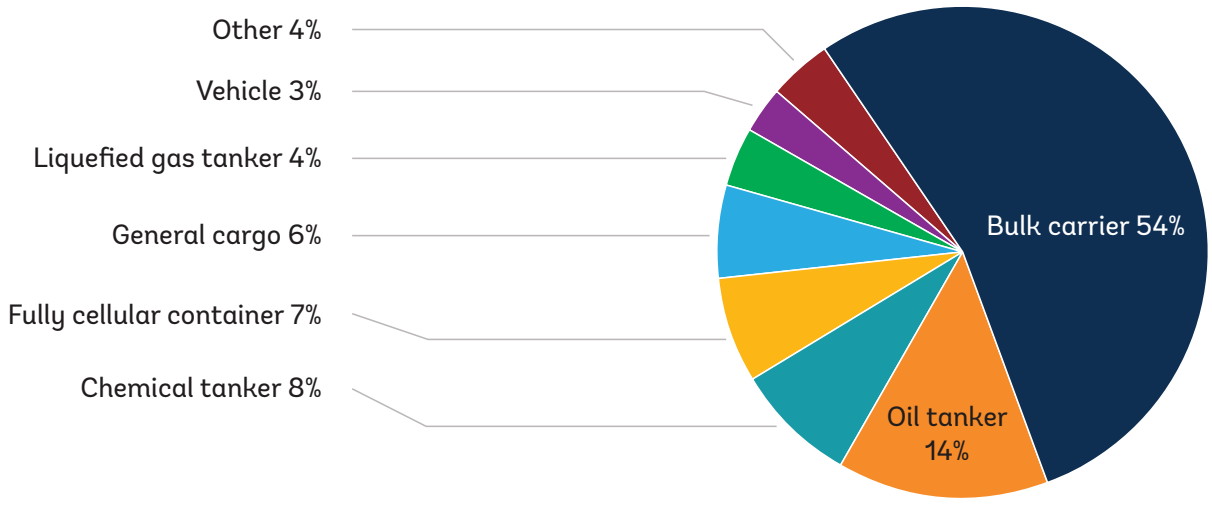
The total energy demand of departing voyages for the passing fleet is estimated to be 52.9 TWh (Figure 4.22.). The proportion of this demand that calls for hydrogen-based marine fuels is expected to grow over time. Using the same methodology applied to each port, projections for this new market from the passing fleet are shown Figure 4.22.. Also included is a breakdown by ship type in Figure 4.23., which shows that more than half of this traffic consists of bulk carriers.

Figure 4.22. Demand projections in TWh versus the total energy demand for the Passing by Fleet



Source: World Bank.

Figure 4.23. Breakdown of shipping segment for the Passing-by-Fleet



Source: World Bank.

4.5. Hydrogen production demand

To assess the demand and potential for green hydrogen production, the demand for hydrogen-based fuels is translated into green ammonia demand (which is measured in tons of hydrogen per annum (tpa)). Targeting the year 2035, there could be the opportunity for green hydrogen production in each port region.

In 2035, for instance, to balance the demand from international shipping in the eastern port region, 80,000 tpa hydrogen could be produced for shipping. In addition, around 55,000 tpa hydrogen production for shipping in the western region, and 35,000 tpa hydrogen in the central region.

For the passing fleet, additional hydrogen production could be considered across the country. The combined production capacity target could be approximately 386,000 tpa of hydrogen by 2035.

The sensitivity analysis shows that if green methanol would be preferred as fuel instead of green ammonia, the hydrogen production would need to increase capacity by around 12 percent. For instance, 90,000 tpa hydrogen production in the eastern region; 62,000 tpa hydrogen production in the western region; and 39,000 tpa hydrogen production in the central region.

4.5.1. Results by port and region

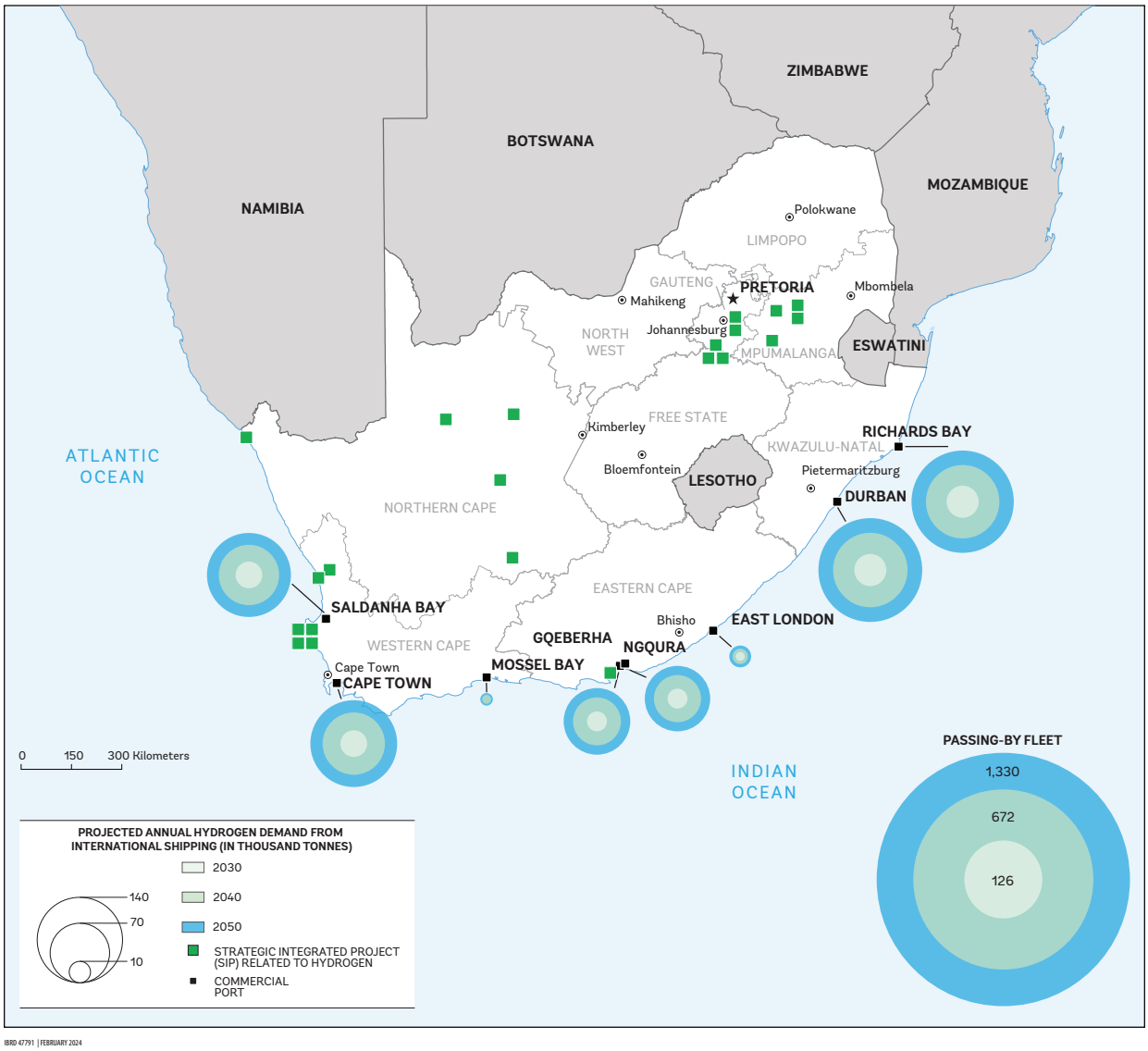
Table 4.6 shows the projected hydrogen demand for all eight commercial ports. These estimates assume that green ammonia is the preferred fuel. The energy density of ammonia is assumed to be 5.14 MWh/ton and the ratio ton of hydrogen per ton of ammonia is assumed to be 0.177.

Table 4.6. Projected hydrogen demand from international shipping (,000 tons)

	Lower quartile			Average			Upper quartile		
	2030	2040	2050	2030	2040	2050	2030	2040	2050
Durban	3	38	80	13	70	136	14	101	203
Richards Bay	2	37	78	13	68	132	14	98	197
Eastern port region	5	76	158	26	138	268	28	199	399
Cape Town	2	27	56	9	49	95	10	70	141
Saldanha Bay	2	27	56	9	46	89	9	66	133
Western port region	3	53	111	18	94	184	19	137	274
Gqeberha	1	16	33	5	29	56	6	42	84
Ngqura	1	15	31	5	28	54	6	40	80
East London	0	2	3	1	3	6	1	4	9
Mossel Bay	0	1	1	0	1	2	0	1	3
Central port region	2	33	69	12	61	118	12	88	176
	0	0	0	0	0	0	0	0	0
Total eight ports	11	162	339	56	293	570	60	424	849
Passing fleet	25	370	787	126	672	1,330	136	973	1,984

Source: World Bank.

Figure 4.24. Hydrogen demand from international shipping in South Africa's commercial ports



Source: World Bank. For accessibility purposes, the locations of ports and projects are approximate only.

4.5.2. Sensitivity analysis

The amount of hydrogen needed to produce green ship fuels depends on various factors. The main factor is the assumption of uptake. In this study, the lower and upper quartiles derived from 14 studies are used to determine the consensus and exclude any outliers. This approach reduces the range of uncertainty.

To illustrate, let's consider the western port region, where the total hydrogen demand in 2040 is approximately 94 ktons. The lower and upper bounds of 53 kton hydrogen and 137 kton hydrogen respectively indicate a possible range of -44 percent and +46 percent.

However, other factors also affect the hydrogen demand, such as the type of fuel (green ammonia vs green methanol), the annual growth rate, and the efficiency of the production plant (i.e., the amount of hydrogen required to produce one ton of fuel). Table 4.7. summarizes the sensitivity of hydrogen demand to different assumptions. The analysis shows that by changing these assumptions, the hydrogen demand could range from -44 percent to +87 percent.

This sensitivity analysis shows that these other factors could impact the total hydrogen demand. Considering the average values only, the hydrogen production demand could increase by up to +29 percent compared to the value estimated in this analysis.

Table 4.7. Sensitivity analysis of fuel uptake

			Example: Western port region by 2040					
Energy demand growth, annual	Hydrogen - based fuel used onboard	Ratio ton hydrogen /ton fuel	%			'000 tons hydrogen		
			Lower	Average	Upper	Lower	Average	Upper
1%	Ammonia	0.177	-44%	0%	45%	53	94	137
2%	Ammonia	0.177	-35%	15%	67%	61	109	158
1%	Ammonia	0.195	-38%	10%	59%	59	104	150
1%	Methanol	0.213	-37%	12%	62%	60	106	153
2%	Methanol	0.213	-27%	29%	87%	69	122	176
1%	Methanol	0.234	-31%	23%	78%	66	116	168

Source: World Bank.

What if annual growth of energy demand is two percent instead of one percent?

If the annual energy demand growth is higher, that means that transport demand is higher. In the western port region, the demand for hydrogen production would, as a result, increase by approximately 16 percent, from 94 ktons to 109 ktons.

What if the efficiency of the production plant is lower?

If the efficiency of the production plant is lower, it indicates that the ratio of hydrogen to ammonia is higher than the theoretical value of 0.177 tons of hydrogen per ton of ammonia (inverse theoretical conversion factor of approx. 5.6). Assuming a 10 percent increase in this ratio, i.e., 0.195 ton H2/ton NH3, the demand for hydrogen production would increase by the same percentage, i.e., 10 percent.

For instance, the total hydrogen demand in the western port region is 94 ktons in 2040. If the ratio of hydrogen to ammonia increases by 10 percent, it means that the hydrogen demand would also increase by 10 percent, reaching up to 104 ktons.

What if green methanol, instead of green ammonia, was the predominant fuel choice of ships?

This indicates that fuel energy density and the efficiency of the production plant would be different. Assuming an energy density of 5.53 MWh/ton and a ratio of hydrogen to methanol of 0.213 (inverse theoretical conversion factor of approx. 4.7), the demand for hydrogen production would increase by around 23 percent. Compared to ammonia, it means that the higher energy density of methanol offsets only in part the higher ratio of hydrogen to methanol. In other words, if methanol, produced from green hydrogen, was the predominant fuel of choice for the shipping industry, more green hydrogen would be needed.

For instance, consider the total hydrogen demand in western port region of 94 ktons in 2040. If green methanol would be used instead of green ammonia, the hydrogen demand would increase by 12 percent, reaching up to 106 ktons. If the annual growth would also increase to two percent, then hydrogen demand would reach 122 ktons.

Producing green shipping fuels: The Saldanha case study

The port of Saldanha was identified as a prime candidate, where a comprehensive green shipping fuel project could be implemented. Five key drivers of the case study lead to a deeper understanding of the technical, financial, and economic aspects of such an undertaking. Delivering a project with multiple large-scale infrastructure components can be achieved by detailed planning and risk mitigation.

05

The objective of this work was to carry out a pre-feasibility case study at the port of Saldanha, to understand the technical requirements as well as the cost drivers to produce and supply hydrogen-based fuels at a South African port.

The case study takes a step-by-step approach to describe how to develop a large-scale investment project to produce green shipping fuels. It is unpacked by explaining the design basis, high-level technical development, and a financial and economic analysis. Considering that this is a novel, still maturing market, a risk management lens was applied throughout the analysis.

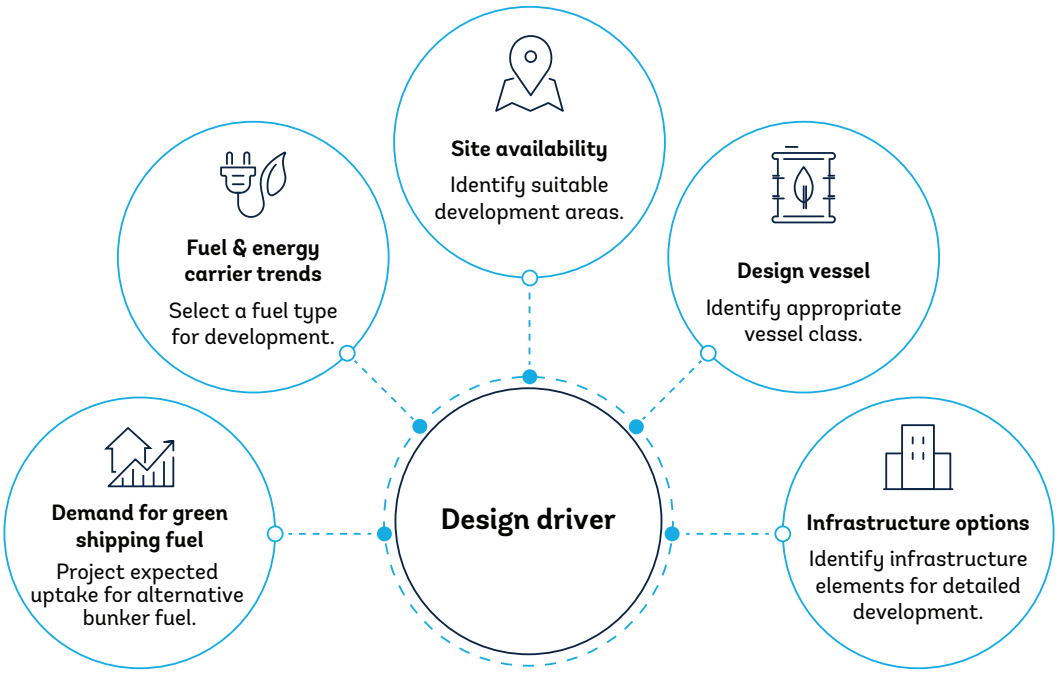
If a developer advances a project, the layout, scale, and siting of key infrastructure components like renewable power plants and facilities for hydrogen production and transmission may be different. However, this technical knowledge can support policymakers, industry, and financiers in understanding and overcoming challenges in developing green hydrogen projects. At the same time, it aims to equip the South African maritime sector with a detailed understanding of how green shipping fuels can be produced and supplied in the country.

The port of Saldanha is the largest natural deep-water port in the Southern hemisphere, approximately 100 kilometers northwest of Cape Town. The port is primarily an export port for iron ore (around 60 million tons per annum), which is sourced from mines in the Northern Cape province and transported to the export terminal via a dedicated, 860 km-long railway line (the Sishen-Saldanha railway line, also known as the Ore Export Line (OREX)). The port also imports crude oil and handles various cargoes including project cargo.

5.1. Approach and design basis

Five design drivers were considered to identify the functional requirements of the study and to provide a basis for the development of a sizable fuel production plant, integrated in an end-to-end value chain.

Figure 5.1. Five design drivers



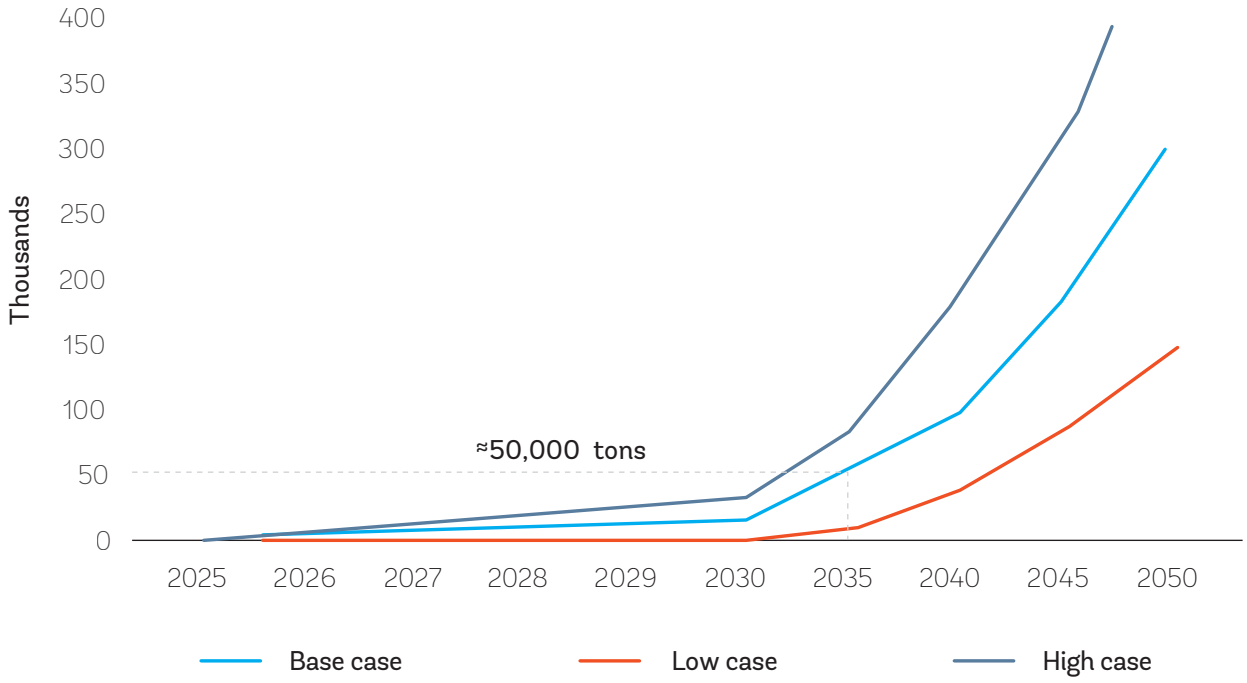
Source: World Bank.

5.1.1. Driver 1: Market demand

A demand estimation for green ship fuels in the ports of Saldanha and Cape Town forms the basis for the scaling of the study case. Offtake from vessels on international voyages and departing from the ports of Saldanha Bay (for vessels carrying iron ore) and Cape Town (for container vessels) is the key determinant for the demand projection. Based on historic numbers, the projection assumes an average growth in vessel calls, green fuel uptake in the market, and the ships' capability of bunkering green ammonia.

As one of the largest iron ore export ports globally, demand in Saldanha is dominated by Capesize vessels (180,000+ DWT) on mainly long-haul voyages, while demand for green fuels in Cape Town is driven by container ships on regional and currently served trades, such as to Europe, the Middle East, and the Americas. The demand for hydrogen-based fuels is expected to ramp up significantly after 2030, reaching around 50,000 tons of green hydrogen feedstock by 2035. This is the volume determined as the reference demand for the size of the fuel production facility.

Figure 5.2. Hydrogen demand from marine fuel in the Western Cape Province



Source: World Bank.

The demand projections for the case study established the baseline energy demand based on historic vessel call data from port authorities as well as a single fuel uptake scenario. A comparison of this demand forecast with the projections in Chapter 4, verifies the robustness of the reference demand assumed for the development of the fuel production (compare Table 5.1.).

Table 5.1. Robustness check of results

							Example: Western port region by 2040		
Energy demand growth, annual	Hydrogen-based fuel used onboard	Ratio ton hydrogen/ ton fuel		%			'000 tons hydrogen		
				Lower	Average	Upper	Lower	Average	Upper
Robustness check with projections in Chapter 4									
2%	Ammonia	0.177	Chapter 4	-44%	0%	45%	61	109	158
2%	Ammonia	-	Saldanha case study	-63%	0%	90%	38	104	198

Source: World Bank.

As demand for hydrogen for marine fuel use only increases gradually, other complementary sources of demand are needed until there is enough demand from the shipping industry to justify a sizable plant. Other demand sources, such as local industry or exports, can bridge the gap until bunker demand matches production capacity.

Market demand

What you should know

The size of the example green hydrogen plant is based on a predicted bunker demand of approximately 50,000 tons of green hydrogen in 2035. Given that bunker demand is subject to the gradual expansion of the market, other demand sources were considered to bridge the gap till that time. For Saldanha, the demand could come from green steel production at a local steel plant. This allows the project to develop into a hydrogen hub with multiple hydrogen consumers (more on this in Chapters 6 and 7).

5.1.2. Driver 2: Fuel & energy carrier trends

A successful energy transition for international shipping will require green fuels for ships. Hydrogen-based fuels, such as ammonia and methanol, are critical to reduce greenhouse gases from the sector. In turn, biofuels and LNG will only play a limited role (Englert, Losos, et al. 2021).

Besides the development of the green bunker fuel market, there is a growing momentum for large-scale trade in hydrogen and hydrogen derivatives to meet the global demand for renewable energy sources. Due to the low volumetric energy density of compressed hydrogen, molecule carriers and derivatives of hydrogen are needed to overcome storage, transport, and distribution challenges. There are several hydrogen carrier options, including ammonia and methanol. For example, most large-scale, export-orientated green hydrogen projects have chosen ammonia as the preferred carrier (Hydrogen Europe 2023). The selection of one bunker fuel, in other words, hydrogen carrier type, is an essential design driver which forms the basis of the infrastructure and process design for this study.

All candidate fuels are referred to as "green" as they are derived from hydrogen that is produced using renewable energy through water electrolysis. Table 5.2. provides detailed considerations for individual fuels.

As an energy source for ships, the characteristics of the most promising future bunker fuels are different considering factors such as technology maturity, emissions, fuel characteristics, and on-board storage.

Table 5.2. Characteristics of hydrogen-based marine fuel types

		Green Methanol	Green Ammonia	Green Hydrogen
Key properties	Volumetric Energy Density (MJ/l)	15	12.5	8.5
	Flammability	Highly flammable	Narrow flammability range (18-25 % in air)	Highly flammable
	Toxicity	Toxic	Highly toxic	Non-toxic
Emissions	Carbon Dioxide (CO2)	≈ net-zero emissions during combustion	≈ zero emissions during combustion	zero emissions during combustion
	Nitrous Oxides (NxO)	Controllable by engine optimization	Controllable by engine optimization, but require increased scrutiny to maintain carbon dioxide reduction benefits	none
	Sulfur Oxides (SOx)	None	None	None
	Nitrogen Oxides (NOx)	Controllable by Selective Catalytic Reduction (SCR) systems	Controllable by SCR systems	Controllable by SCR systems

		Green Methanol	Green Ammonia	Green Hydrogen
Production	Resource requirements for production	sustainable carbon source, renewable power, sustainable water source	renewable power, sustainable water source	renewable power, sustainable water source
Operations	Storage	under ambient conditions	Stored at - 33°C (or compressed to 8 bar)	Stored at -252.8 °C
	Ship engines	available	Commercialization from 2025+	Commercialization from 2025+
	Port infrastructure	established systems for commodity handling	established systems for commodity handling	not established
	Bunkering experience	yes	pilot	Pilot for small-scale
	Fuel guidelines	Interim IMO fuel guidelines developed	IMO guidelines under development	IMO guidelines under development

Source: IRENA (2022), IEA (2019), IRENA and Methanol Institute (2021).

The comparison indicates that pure green hydrogen is the least favorable due to technological immaturity, complexities in handling, and low energy density. Nevertheless, pure hydrogen remains the critical feedstock for green methanol and green ammonia production. Green ammonia and methanol on the other hand display clear potential to develop. While methanol comes with the challenge of sustainable carbon sourcing, operational guidelines and regulations are more advanced and the immediate availability of engines are an advantage.

Despite its toxicity and gas dispersion issues – two key safety concerns – ammonia is a commonly traded commodity. Ship engines are expected to be introduced to the market, including the relevant regulations, at a potential production start of the case project. The report elaborates in more detail on safety and environmental concerns of ammonia handling in Chapter 5.

The major difference between the three candidate fuel types is the required carbon source for methanol production. A closed carbon cycle must be available for the fuel to be ultimately carbon neutral. Three possible carbon sources can be considered for methanol synthesis (International PtX Hub 2022):

- Biogenic sources: Biogenic residue from agriculture or forestry
- Industrial point sources: Production of mineral products (e.g., cement), metals (e.g., iron and steel), and chemicals and petrochemical products
- Direct Air Capture (DAC): The extraction of CO2 directly from the atmosphere

Table 5.3. Potential carbon sources for green methanol production

Criteria	Biogenic Source	Industrial Source	Direct Air Capture (DAC)
Closed-Carbon Cycle	Yes	No	Yes
Costs	Low (subject to availability)	Low	High
Technology Maturity	High	Medium-high	Low
Scalability	Depends on utilization potential	Likely reducing over time	High
Carbon source available at Saldanha?	×	✓	✓

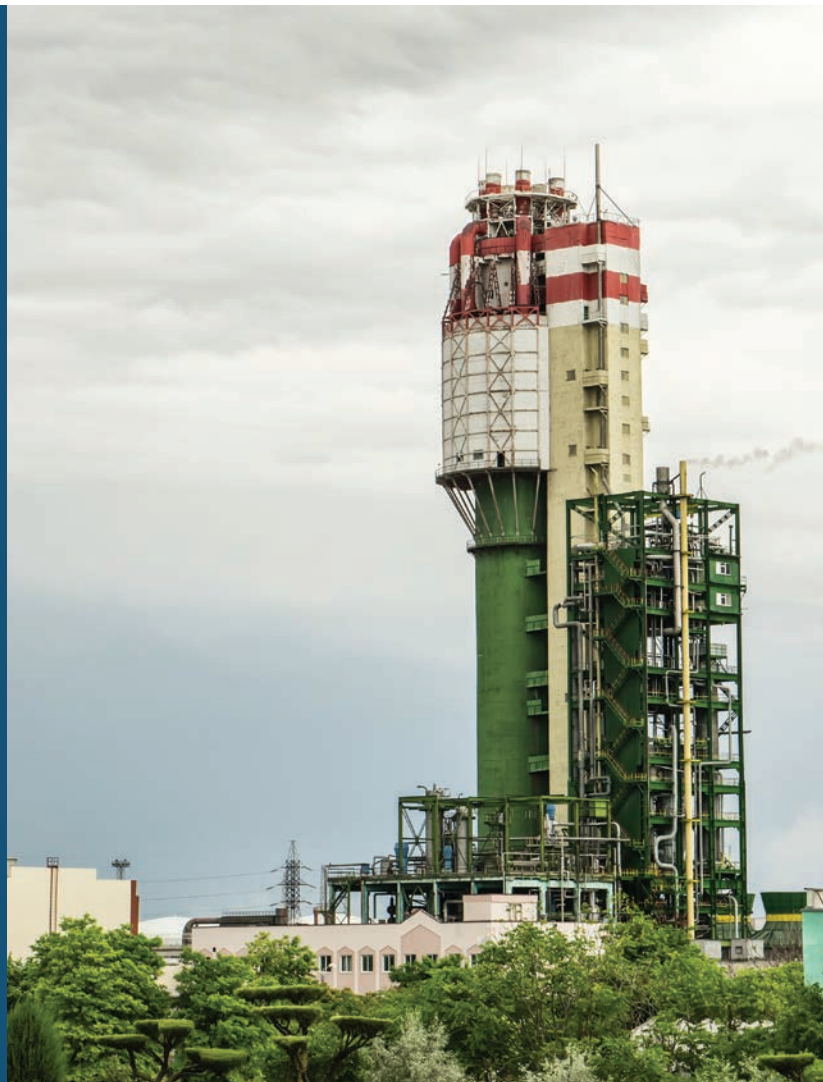
Source: International PtX Hub (2022).

According to Table 5.3., the study project has no preferred carbon source. DAC is an unattractive option due to the lack of technological maturity and high costs. Carbon from biogenic sources is not available. While industrial carbon sources exist within the Saldanha Bay region, mainly from cement production, the lack of a closed-carbon cycle limits the sustainability of methanol produced from these industrial sources.

Fuel & energy carrier trends

What you should know

Green ammonia was selected as the preferred green shipping fuel for this study due in part to ammonia's dual use as an energy carrier for hydrogen exports. The pending finalization of safety rules for onboard safety, bunkering and emissions management are core issues which must be addressed. However, these are not viewed as unresolvable barriers to the large-scale implementation of green ammonia as a marine fuel (Mærsk Mc-Kinney Møller Center 2023). This selection does not rule out green methanol as a suitable bunker fuel. Both fuels share similarities in their individual production pathway, allowing ammonia in large parts, to serve as a proxy for developing methanol production.



5.1.3. Driver 3: Site availability

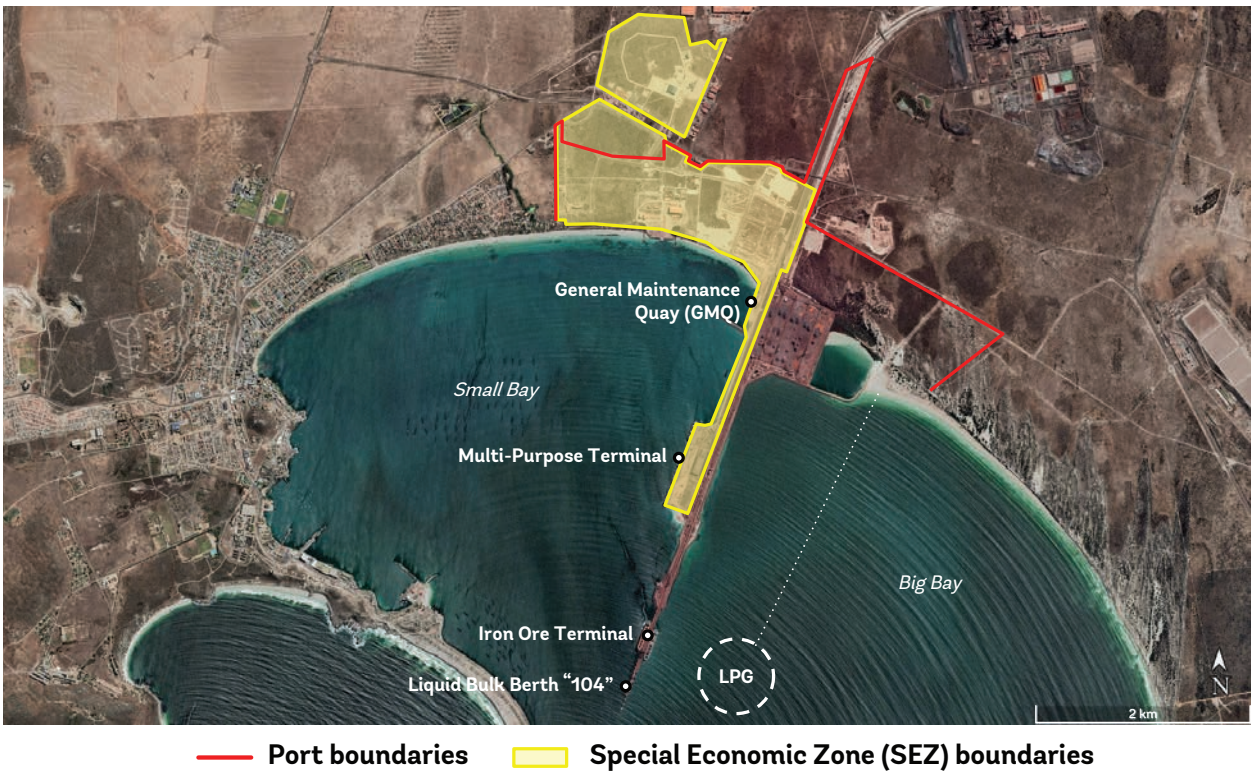
Land availability, current land use, and the future location of infrastructure elements represent key design drivers, too. Therefore, both the port area and the larger region must be assessed regarding suitable land parcels for ammonia production, storage, bunkering, the production of renewable power and hydrogen to identify and select project developments sites.

Note: The status of the land areas corresponds to the status at the time the report was prepared. This does not represent actual availability, nor does it claim accuracy over actual lease or ownership rights. The selection of potential development sites serves as example only.

Port area

The landside port facilities are owned by Transnet National Port Authority (TNPA) and are operated in conjunction with the Freeport Saldanha Industrial Development Zone (IDZ). Previously known as the Saldanha Bay IDZ, it was the first Special Economic Zone (SEZ) located within a port. Furthermore, the IDZ is designated as a customs controlled Area (CCA). The 365-ha footprint of the IDZ has regulated boundaries, which determine whether a project developer can take advantage of tax benefits, amongst others. While the land is classified industrial, it neighbors and sometimes overlaps with residential areas, heavy industry, and environmentally sensitive areas. These spatial constrains require cautious development.

Figure 5.3. Spatial zoning of the greater Saldanha port area



Source: Freeport Saldanha (2022).

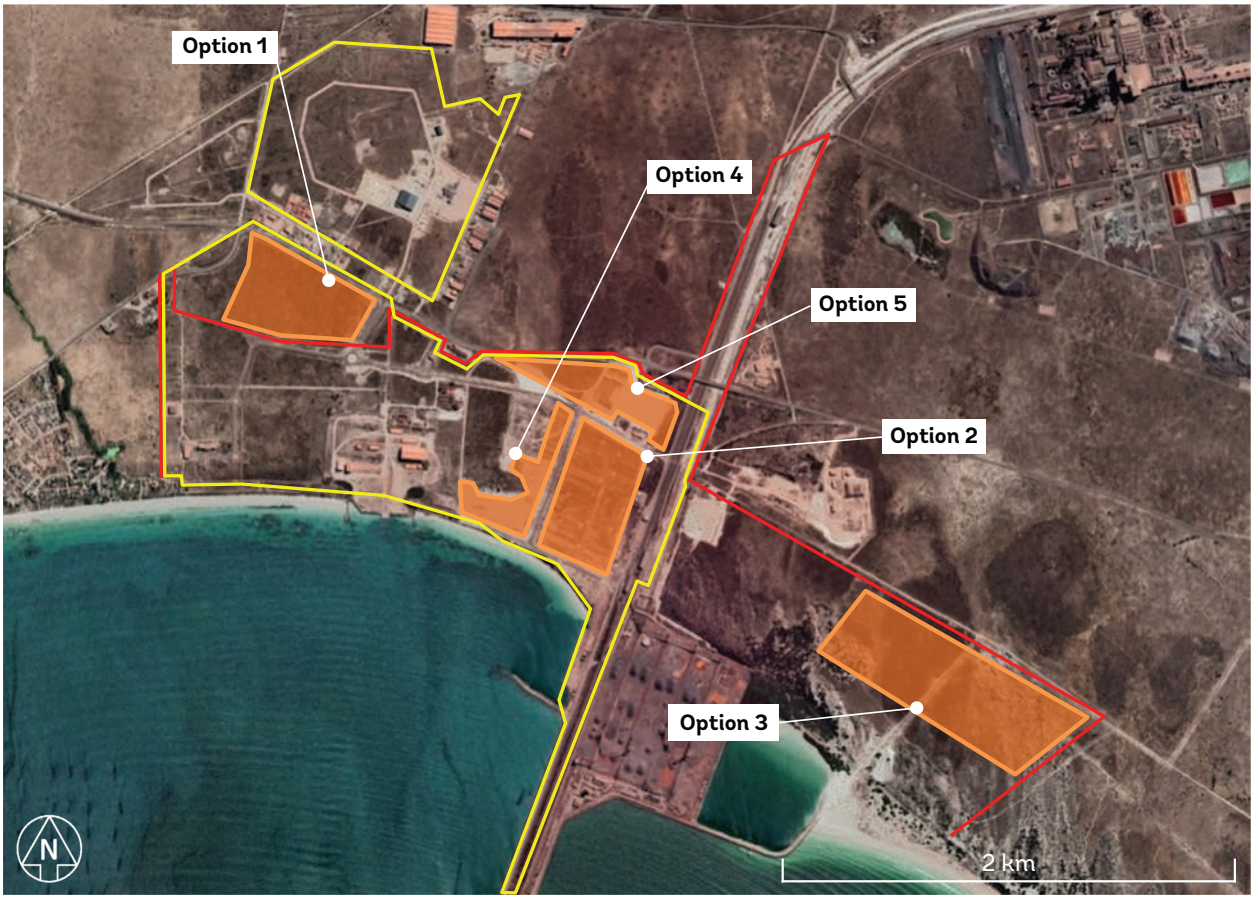
Large amounts of undeveloped land within the Saldanha Bay region have been identified as Critical Biodiversity Areas (CBA). These areas, identified through a conservation planning process, help preserve ecosystems and species, and maintain long-term ecological functioning of the landscape. While development of these areas should be avoided where possible, the presence of a CBA does not remove the potential for land parcel development. Instead, it introduces developmental requirements to assist in ecological preservation. One such requirement is a biodiversity offset. This entails protecting a parcel of CBA land in another location, where the size needed is a certain proportion of the developed land, which could be up to 1:10. Development should be approached with forethought to prevent project delays and requires further, more detailed study.

Figure 5.4. Greater Saldanha port area and Critical Biodiversity Areas (CBAs)

Source: Google Earth, edits by World Bank based on input from TRIPLO4 (2022).

To capitalize on available incentives, the "ease of doing business" model offered by the IDZ, and the port proximity, the site identification process aimed to narrow down a 10-15 ha plot within or adjacent to the zone's boundaries. Five options with differing ownership, lease, and suitability conditions were assessed further. In this case, any land within the port boundaries is owned by the public port landlord and would be leased to the IDZ for onward use by a potential developer.

Figure 5.5. Candidate development sites



Source: Google Earth, edits by World Bank.

Table 5.4. Comparison of candidate development sites

Site	Site area (ha)	Landowner	Availability and suitability
Option 1	20	IDZ	This land has CBA status and would require offset land to be obtained before development.
Option 2	20	Public port landlord	The operational rights are owned by a private concessionaire (lease).
Option 3	20	Public port landlord	Land is undeveloped, with undulating terrain requiring extensive earthworks and infrastructure development. There is also a high likelihood that critical biodiversity pockets could constrain the site.
Option 4	8.6	Public port landlord	Has no critical environmental constraints but does not meet the estimated spatial requirements for the development and would need to be used in conjunction with another site.
Option 5	12.3	Public port landlord	The land has no critical environmental constraints but is not within the signed-off lease footprint of the IDZ. Further discussions between the public port landlord and IDZ would be required.

Source: World Bank.

For this study, Option 5 has been selected as the main location for developing the ammonia synthesis plant and storage facility. Furthermore, Option 4 has been identified as a secondary site to locate any ancillary elements that cannot be housed in the main facility due to spatial constraints, providing a buffer for the development. Option 1 can also be considered as a development location. However, its status as a CBA may hinder the process.

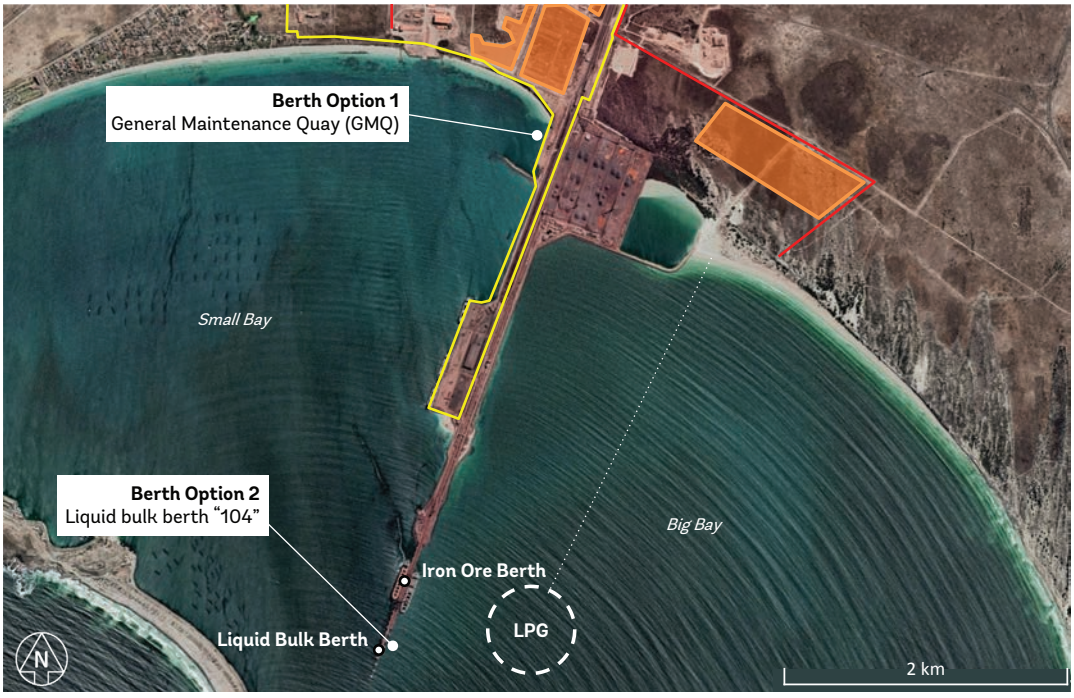
For bunkering vessel loading operations, two indicative berth options were evaluated in consultation with port stakeholders.

Table 5.5. Comparison of two berth options to accommodate a bunker vessel

Berth option	Name	Operational status	Suitability
Berth option 1	General Maintenance Quay (GMQ)	Private operator will operate the facility. The berth pocket was recently dredged to 8.5m.	Private operator could lease for use by bunker barge. A barge with a small draft has been selected to mitigate dredged depth limitations.
Berth option 2	Berth "104"	Existing plans to develop this berth further.	The project size is currently insufficient to warrant the use of Berth "104". However, the berth is planned as a common user facility, allowing for the potential for larger exports if needed in the future.

Source: World Bank.

Figure 5.6. Berth options at the port of Saldanha



Source: Google Earth, edits by World Bank.

Site for renewables

As a first step, to determine a feasible site for renewable energy generation, a list of candidate sites is identified using renewable energy resource and social-environmental constraint mapping.

Potential solar and wind resources in the Northern and Western Cape provinces, electric grid infrastructure and Renewable Energy Development Zones (REDZ) are mapped as a first step. REDZs are geographical areas identified by the South African Department of Forestry, Fisheries, and the Environment (DFFE), where wind and solar developments can occur in concentrated zones. These areas have the potential to act as energy generation hubs, providing anchor points for grid expansion where regulatory processes can be streamlined. For the Saldanha case study, a notable REDZ zone is the Komsberg REDZ (Zone 2). This zone encompasses 8,850 sq km of land on the border between the Western and Northern Cape and has the potential for large-scale wind and solar PV energy facilities. The analysis shows that wind resources are better around Saldanha, with solar resources improving significantly further inland.

Figure 5.7. Solar and wind energy resource in the Northern Cape and Western Cape Provinces

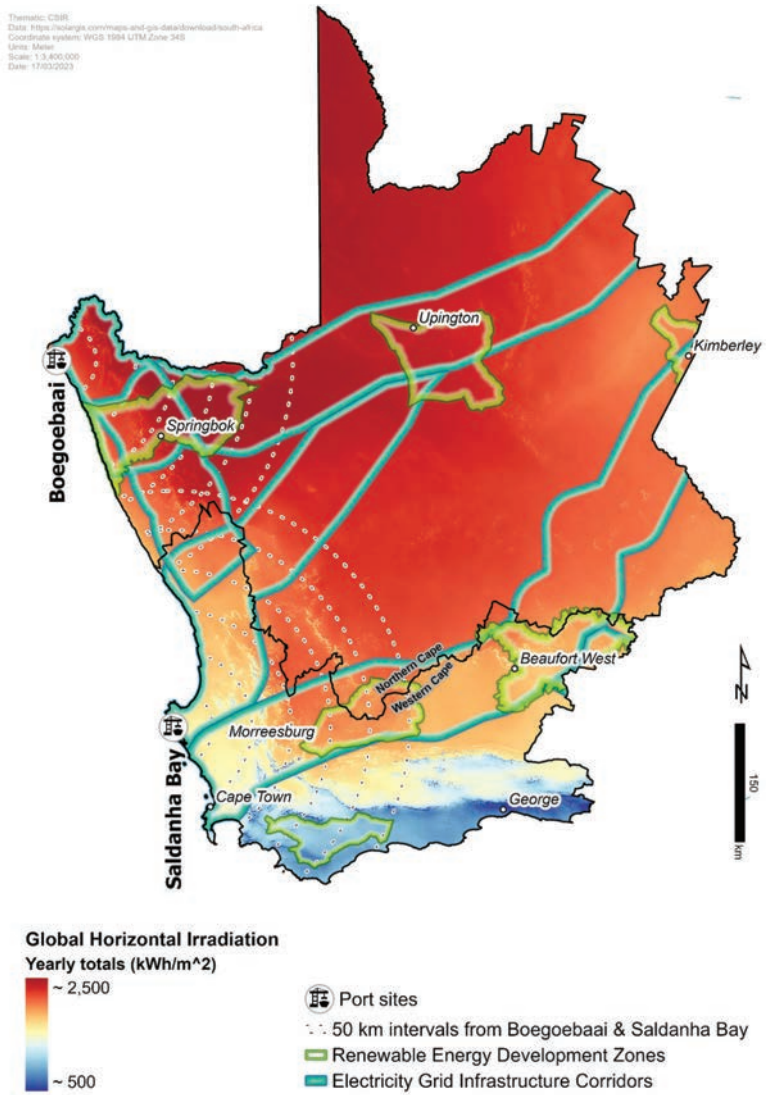
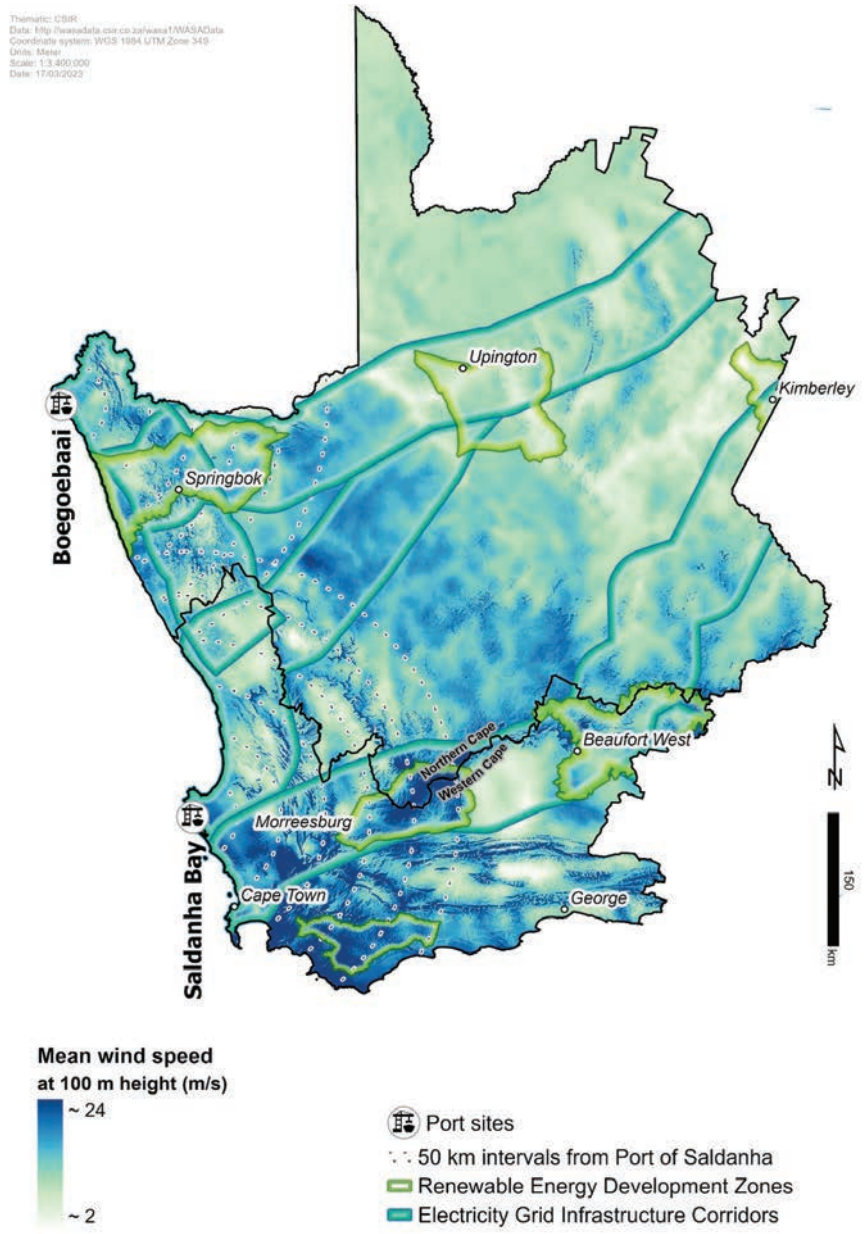


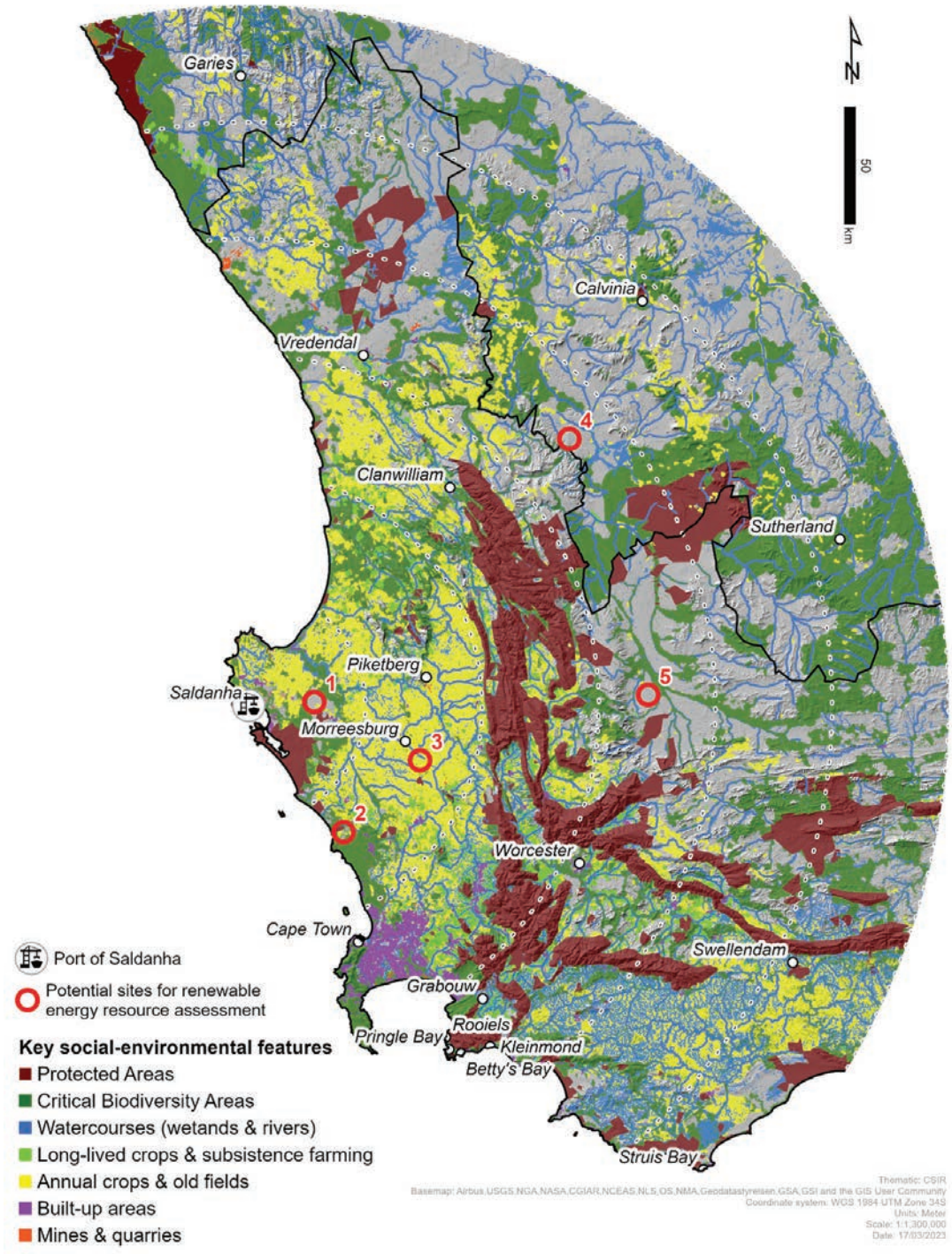
Figure 5.7. Solar and wind energy resource (continued)



Source: CSIR (2023).

As a second step, a high-level social-environmental constraints mapping analysis is conducted to support the selection process of concrete candidate sites. The mapping is based on environmental and land use feature that may prevent or constrain the development of large-scale renewable energy and includes data relating to conservation planning, such as law-protected areas like CBAs; aquatic ecosystems, such as watercourses, wetlands and rivers; and land use, such as urban-, built-up areas, agriculture, mines, and quarries. The analysis within a 300 km radius shows that agricultural activities constrain the Saldanha Bay area. The extensive transformation of the natural landscape by agricultural activities has resulted in many areas being protected or identified as CBAs, especially where typical South African vegetation, like fynbos and renosterveld is present.

Figure 5.8. Key social-environmental features in a 300 km radius from the port



Source: CSIR (2023).

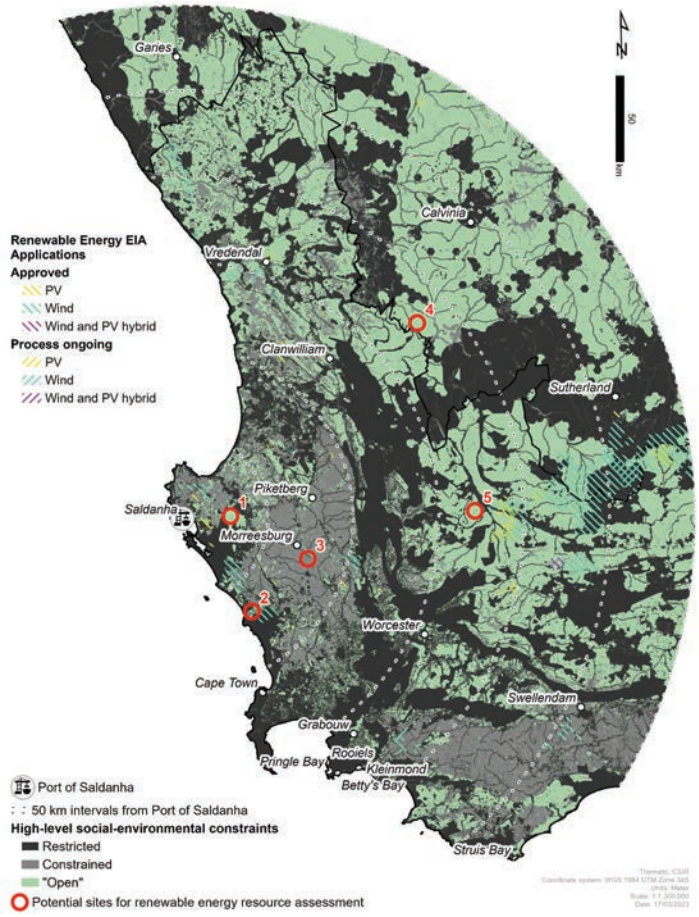
The analysis identifies potential areas that should be avoided, or which are constrained for development. Conversely, initial sites for subsequent resource assessment could also be identified. According to their sensitivity to renewables development, the data was divided into restricted, constrained, and open areas.

Table 5.6. Constraint rating and their social-environmental features

Category		Land Use
	Restricted / Avoid	**Protected areas:** national parks, nature reserves, private nature reserves, mountain catchment areas, protected environments, wilderness areas
		Critical biodiversity areas (CBA category 1 and 2)
		Aquatic features: watercourses, rivers, and wetlands
		Agriculture: horticulture, viticulture, pivot irrigation (incl. Rooibos), shade nets, subsistence farming, food gardens
		Land use/cover: built-up areas, urban, residential, villages, smallholdings, commercial
	Constrained	**Land use/cover:** mines and quarries
		Agriculture: rainfed annual crops/planted pastures; strip field cultivation, smallholdings, old fields
		Aquatic features: watercourses, rivers, and wetlands (32 m buffer)
	Open	**Remaining areas**

Source: CSIR (2023).

Figure 5.9. Constraint rating in a 300 km radius from the port



Source: CSIR (2023).

The constraint mapping shows that areas available for renewables development open significantly as the distance from Saldanha Bay increases (> ~150 km). The solar resource also increases further inland. This is also evident from the actual number of renewables development projects proposed south of the town of Sutherland. However, delivering electricity to Saldanha Bay from the areas further inland would require more extensive energy transmission. Based on larger tracts of preferably open or constrained land, the screening identified five candidate sites which are potentially "open" for renewables and green hydrogen developments (see Figure 5.9. and Table 5.7. for further feasibility considerations). While the list of candidate sites is not exhaustive, and must be subjected to further technical, environmental, safety and social perspective, the sites serve as a departure point for further detailed assessment.

Site availability

What you should know

For the case study, a site, around 70 km inland from Saldanha was selected due to its favorable renewable energy potential, relative short distance to the port and previously approved environmental impact assessment. While within a matrix of agricultural fields, farming land can offer an opportunity for land use change or co-existence. The inland location will house the solar and wind farm as well as the electrolysis plant. The ammonia synthesis and storage facilities can be situated within the limits of the special economic zone, adjacent the port of Saldanha. Two berth options can serve the loading of a bunker vessel or potentially accommodate larger vessels for fuel exports.

Table 5.7. Candidate sites for further assessment

Site		Latitude	Longitude	Approx. Distance to Saldanha	Potential Unconstrained Area Available (ha)	Comments on Potential Feasibility
1	East of Saldanha Bay	-32.9932	18.24893	20	8 500	Any further wind farm developments in the direct vicinity of Saldanha Bay are largely constrained due to nearby radar antenna.
2	North of Cape Town, near Atlantis	-33.4990	18.36825	70	5 800	Proximity to Cape Town and its surrounding coastal settlements, private nature reserves and tourism may present challenges.
3	Near Moorreesburg	-33.2284	18.72773	70	25 000 +	Likely the most feasible site due to its proximity to Saldanha Bay and potential for land use change or co-existence.
4	East of Clanwilliam	-31.9965	19.43567	170	100 000 +	Less constrained but distance from Saldanha Bay is a challenge.
5	Between Worcester and Sutherland	-32.9901	19.78011	160	60 000	Less constrained but distance from Saldanha Bay is a challenge.

Source: CSIR (2023), World Bank.

5.1.4. Driver 4: Design vessel

Market demand (Driver 1) and the selection of a fuel type (Driver 2) determine suitable design vessels for the project case, including navigational and infrastructure requirements and possible limitations. Indicatively, reference vessels for two different operational requirements were selected: Firstly, a bunker vessel (or barge) to supply fuel at Saldanha and Cape Town. And secondly, an ammonia carrier for long-haul export trades, to account for a fallback market for green ammonia, if or until not supplied to the shipping industry as a marine fuel.[8]

Bunker vessel

Bunker barges can transfer fuel to vessels alongside at berth or at anchorage near ports. Therefore, functionality requirements for barge operations must consider the following.

[8] For both vessels, it is assumed that ammonia will be transported in a liquified state, pressurized to 1 atm, and refrigerated at 33°C.

- **Sea-going ability:** Typically, bunker barges are limited in size and act in protected port environments or inland waters. Bunker vessels for open waters are typically larger and are therefore preferred for this study, increasing its operating radius (including the port of Cape Town).
- **Draft restrictions:** To dock at shallower berths within the port, a draft of less than 8 m is required.
- **Ammonia handling capabilities:** Most bunker barges currently serve vessels running on fossil-based fuels. While less common, a barge with capabilities to handle ammonia is required for the case study.

Since ammonia bunker barges are scarce, mid-size liquefied petroleum gas (LPG) tankers have been considered for barge sizing. Although these semi-refrigerated vessels are designed to carry LPG, many can also transport ammonia and petrochemical gas. These vessels often operate on intra-regional trade routes, particularly in Europe (see example in Table 5.8.). Given potential water depth restrictions at berth and the project scale (Driver 1), a small LPG carrier was selected.[9]

Table 5.8. Reference bunker vessels

Parameter	Small Size	Medium Size
Vessel Type	LPG Carrier	LPG Carrier
Example Vessel	IMO 9870513	IMO 9734836
Capacity (DWT)	8,025	18,446
Capacity (cbm)	9,145	20,600
LOA (m)	120	160
Beam (m)	20	26
Draft (m)	7.21	9.50

Source: World Bank.

Ammonia carrier

Ammonia is typically shipped using compatible LPG vessels. An ammonia-carrying vessel must be accommodated within the port to facilitate the potential export of ammonia to overseas demand centers. Today the ammonia trade is commonly performed by three different vessel types:

- **Fully refrigerated**, with a carrying capacity of up to 30,000 cbm
- **Fully pressurized**, with a carrying capacity commonly up to 10,000 cbm
- **Semi-pressurized and refrigerated**, with a carrying capacity of up to 20,000 cbm

Most of the current ammonia trade within Europe occurs using small vessels with a carrying capacity between 5,000 and 10,000 cbm. Currently, in the ammonia trade, relatively small vessel types are used due to the characteristics of feedstock logistics. However, in the future, for the large green ammonia trades over longer haul distances, large LPG vessels suitable for ammonia will likely be used. Therefore, the current LPG fleet gives a good indication of the potential design vessel for the export of green ammonia and aligns with recent announcements for large ammonia carrier orders (Martin 2023, The Maritime Executive 2023) (compare Table 5.9.).

[9] The vessel is preferably fueled by ammonia to ensure a truly zero-carbon operation under carbon accounting methodologies.

Table 5.9. Common LPG tanker size range

Dimension	Range
Load capacity (cbm)	40,000 – 91,000
Capacity (DWT)	30,000 – 60,000
Length (m)	190 – 230
Beam (m)	30.0 – 36.6
Draft (m)	10.5 – 12.2

Source: Thoresen (2014).

Today, the largest LPG carriers have a load capacity of approximately 90,000 cbm. A future increase in ammonia trade may result in larger ammonia carrying vessels in the future and could be as big as the so-called Long Range 2 (LR2) tanker vessels. Hence, a vessel size equivalent to an LR2 tanker is selected as the reference vessel to export ammonia (see Table 5.10.).

Table 5.10. Dimensions of an LR2 tanker, the reference vessel for exporting ammonia

Dimension	Value
Capacity (DWT)	120,000
Length (m)	265
Beam (m)	45.8
Draft (m)	15.8

Source: PIANC (2016).

Design vessel

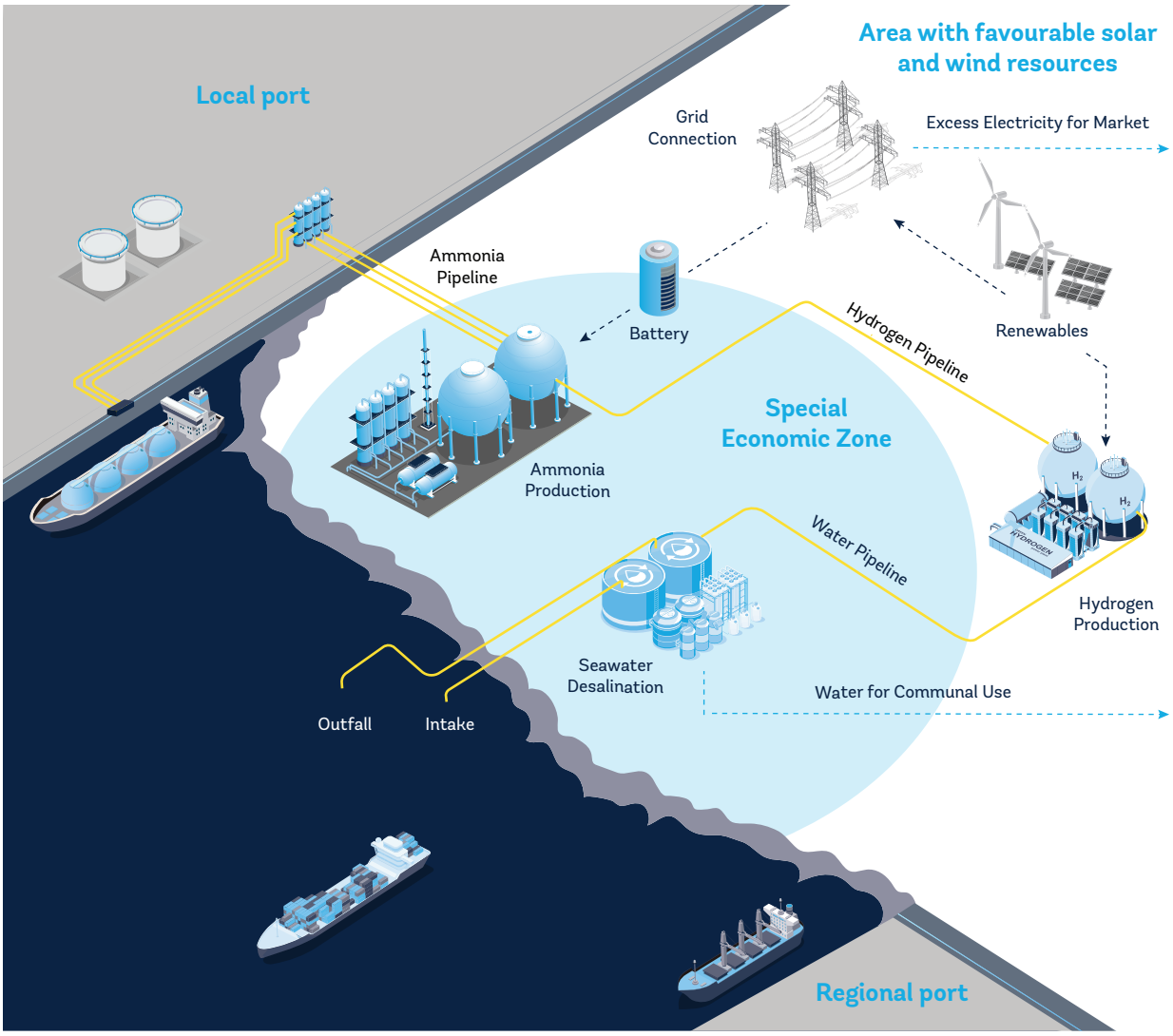
What you should know

To supply green ammonia to ships regionally, a smaller sized bunker vessel with a carrying capacity of 8,000 tons was selected. For exports, the reference vessel equals a Long Range 2 tanker, with a deadweight of 120,000 tons.

5.1.5. Driver 5: Infrastructure options

The high-level infrastructure outline depicts the geographic separation of process components and its link with the port interface.

Figure 5.10. High-level infrastructure outline

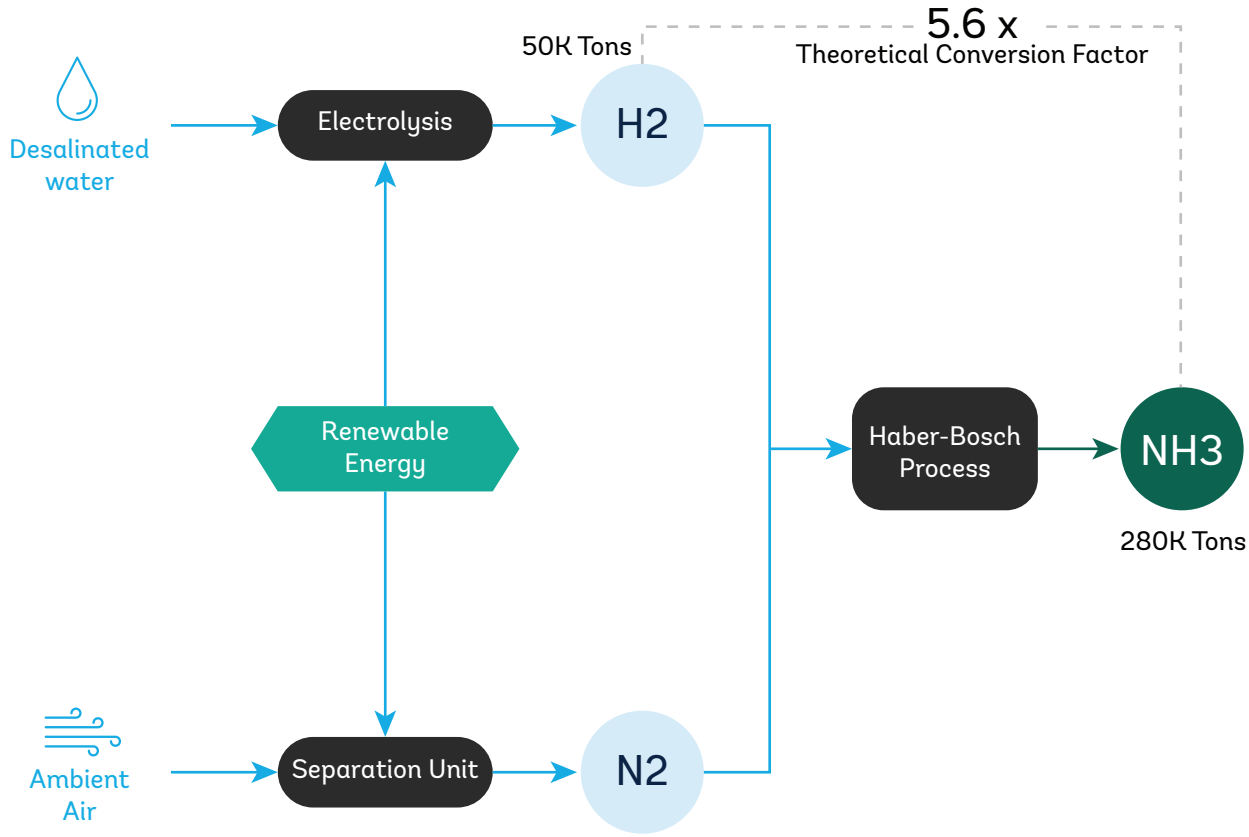


Source: World Bank.

5.2. Technical development

Based on the five design drivers, the example project can be technically developed in more detail. The detailed technical parameters will inform the project costing and the financial and economic viability of the study case. This section unpacks the individual technical components and rationalizes the operational approach for their selection.

Figure 5.11. Green ammonia production



Source: World Bank.

5.2.1. Production plant optimization

An iterative optimization process was applied to determine the most efficient and effective technical layout. A core component for business case optimization and public acceptance of large-scale green hydrogen or green ammonia projects is the decision whether to connect the additional renewable generation capacity to the grid or keep the renewables disconnect, i.e., islanded.

For each possible setup the optimization analysis aimed to achieve the lowest possible levelized cost of ammonia (LCOA), while balancing operational requirements.

Variable inputs to determine adequate component sizing are:

(i) potential oversizing of renewables and the split between solar and wind generation capacity,

(ii) the oversizing of electrolyzer, and

(iii) the use of battery storage.

The analysis concludes on a grid-connected set-up, which aims to produce a lower LCOA and adds net electricity to the power grid.

Table 5.11. Optimized sizing of process components

System element	Parameter	Value	Unit
Renewable energy generation plant	Solar Farm Capacity	957	MW
	Wind Farm Capacity	638	MW
	Combined Capacity Factor	0.267	No.
Electrolyzer	Capacity	492	MW
	Design Output	150.3	tH2/day
	Annual Capacity Factor	0.630	No.
	Annual Production	52,647	tH2/year
Air separation unit (ASU)	Capacity	699	tN2/day
	Design Power Demand	7.08	MW
	Annual Production	233,000	tN2/year
Ammonia synthesis plant	Capacity	310,000	tNH3/year
	Design Power Demand	15.57	MW
	Annual Production	280,270	tNH3/year
	Annual Capacity Factor	0.904	No.

Source: World Bank.

5.2.2. Renewable energy generation plant

The hybrid renewables plant was optimized to work at a split of 40 percent wind and 60 percent solar power. The selected technology directly affects the full load hours per year and, as a result, the capacity factor of the electrolyzers for the hydrogen production. A hybrid solar-wind plant optimizes the use of renewable resources. While wind alone is expected to yield a capacity factor of around 20-30 percent per night, combining it with solar results in capacity factors of up to 80 percent by day.[10]

[10] Based on wind turbines of 3 MW each at 150 m hub height and single axis tracking (SAT) photovoltaic (PV).

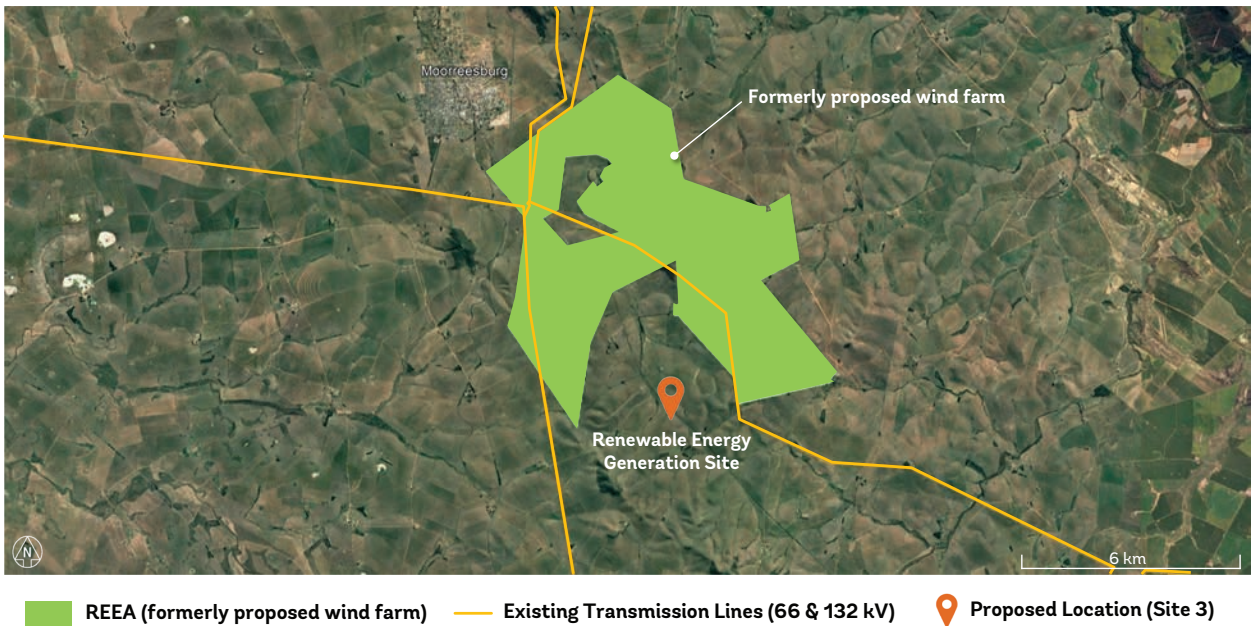
Table 5.12. Hybrid renewables plant

Plant Element	Capacity Split	Plant capacity (nominal) (MW)	Total capacity (MW)
Wind farm	40 percent	638	1,595
Solar farm	60 percent	957	

Source: World Bank.

The preferred candidate site identified in the spatial analysis (Design driver 3) lies within a matrix of farmland and in the vicinity of previously approved environmental impact assessments (EIA) for a wind energy project (DFFE 2022).[11]

Figure 5.12. Selected candidate site and boundaries of previously approved REEA (wind power)



Source: Google Earth, edits by World Bank.

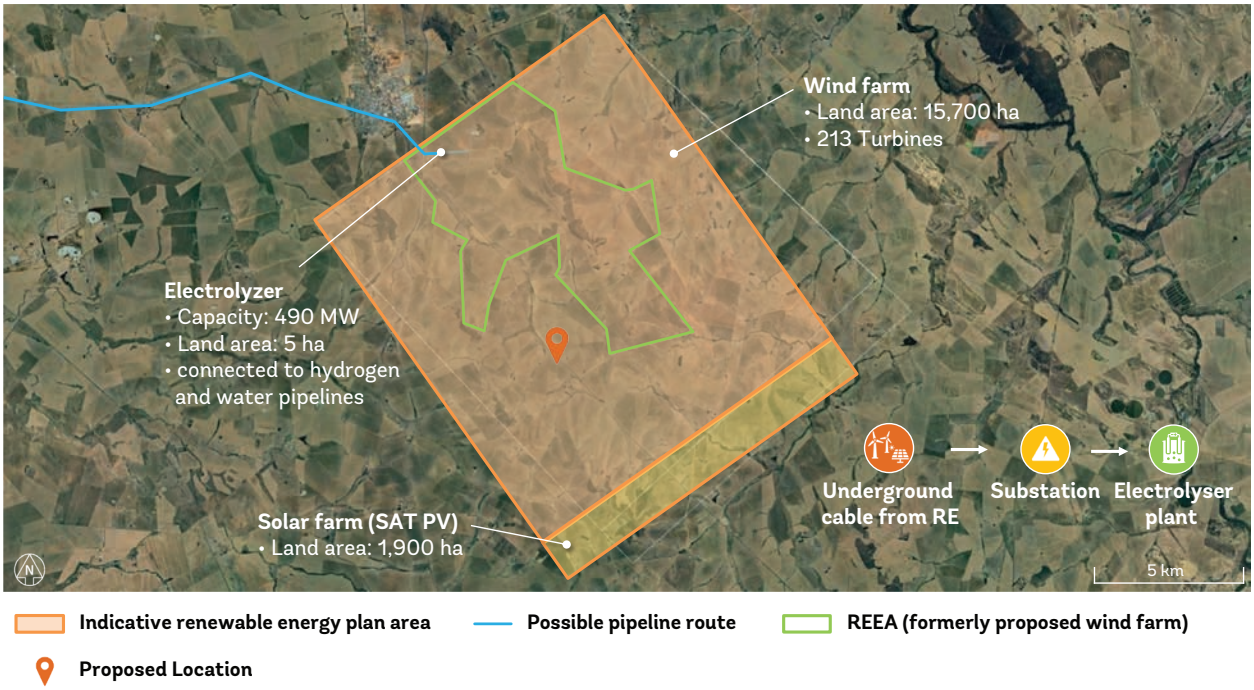
Table 5.13. Indicative spatial requirements for the hybrid renewable energy plant

Plant	Estimated Area Requirement (ha)	Comments
Wind Farm	15,700	A conservative area estimation to account for land use or potential geographical limitations. Estimated from the sizing of a previously proposed wind farm.
Solar Farm	1,900	The area was estimated using an approximation of 2 ha per MW. The estimated area relies on the assumption that an uninterrupted area is available and could be used.

Source: World Bank.

[11] Although the development of this wind energy facility does not appear to have progressed after approval of the EIA, it demonstrates the potential to develop a similar facility in the area. The EIA also provides useful site-specific information to assist with the infrastructure development of this example project.

Figure 5.13. Indicative spatial requirements for the hybrid renewable energy plant



Source: Google Earth, edits by World Bank.

As concluded in a previous Renewable Energy EIA Application (REEA), the location of the wind farm within an agricultural area could potentially create only minor interference with existing land users, as wind turbines only require small sections of land and can co-exist with agricultural activities easily once in operation (Savannah Environmental 2015). However, this is not the case for solar PV farms. These require the use of the entire indicated area to accommodate the PV panels. Therefore, developing a 957 MW solar farm would require the sterilization of approximately 1,900 ha of agricultural land in the case of the candidate site. Further studies should investigate the potential of locating the solar farm further inland to reduce the socio-environmental impact of the development, as well as to better exploit solar resources.

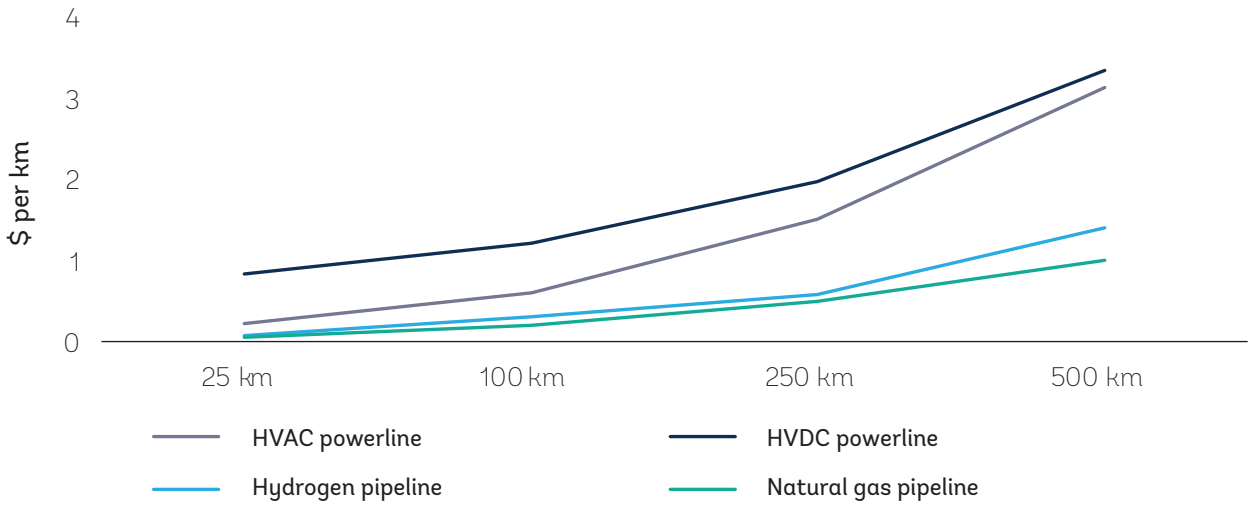
5.2.3. Energy transmission

Energy transmission can mean the transport of electrons, i.e., electric power, or the transport of molecules, e.g., hydrogen or ammonia. The inland location of the renewables introduces several challenges surrounding the transport of electricity, water, and hydrogen/ammonia to the portside facilities. The selection of energy transmission methods is vital to the technical development of the project and influences the siting of project components, significantly impacting expenditure.

Pipelines versus power lines

A report commissioned by the Australian Pipelines and Gas Association (APGA) found that energy transport and storage using natural gas or hydrogen pipelines is more cost-effective than electricity transport and storage for all other scenarios (APGA 2022). The study showed that energy transport through pipelines could be up to five times more cost-effective than via high-voltage powerlines. In addition to cost benefits (see Figure 5.14.) there are other benefits of using pipelines for the transport of energy (compare Table 5.14).

Figure 5.14. Comparison of energy transmission options



Source: APGA (2022)[12].

Table 5.14. Comparison of pipeline and powerline transmission options

Factor	Pipelines	Powerlines
Loss of supply (Reliability)	Length of high-pressure gas pipelines: 9,000 km No. of supply loss events per 1,000 km: 0.03.	Length of high voltage power lines: 43,000 km No. of supply loss events per 1,000 km: 0.42.
Environmental	Lower impacts on existing habitats, land use, and communities as they are often installed below ground.	Above-ground powerlines increase bushfire risk or interference with existing land use activities. Using below-ground powerlines reduces this risk but costs five to six times as much.
Visual appeal	Can run below ground, increasing safety and visual appeal.	May be viewed visually unappealing.

Source: World Bank.

Using a hydrogen pipeline can further reduce costs by eliminating the need for separate hydrogen storage. The IEA estimated the cost of hydrogen storage at $ 90,909/t in 2017 $ (IEA 2019). In comparison, a hydrogen pipeline costs $ 1,210,000/km (IEA 2019), which, although costly, can serve a double purpose of transport and storage.

[12] Original chart in Australian Dollars, converted into US Dollars at AU$/US$=0.65

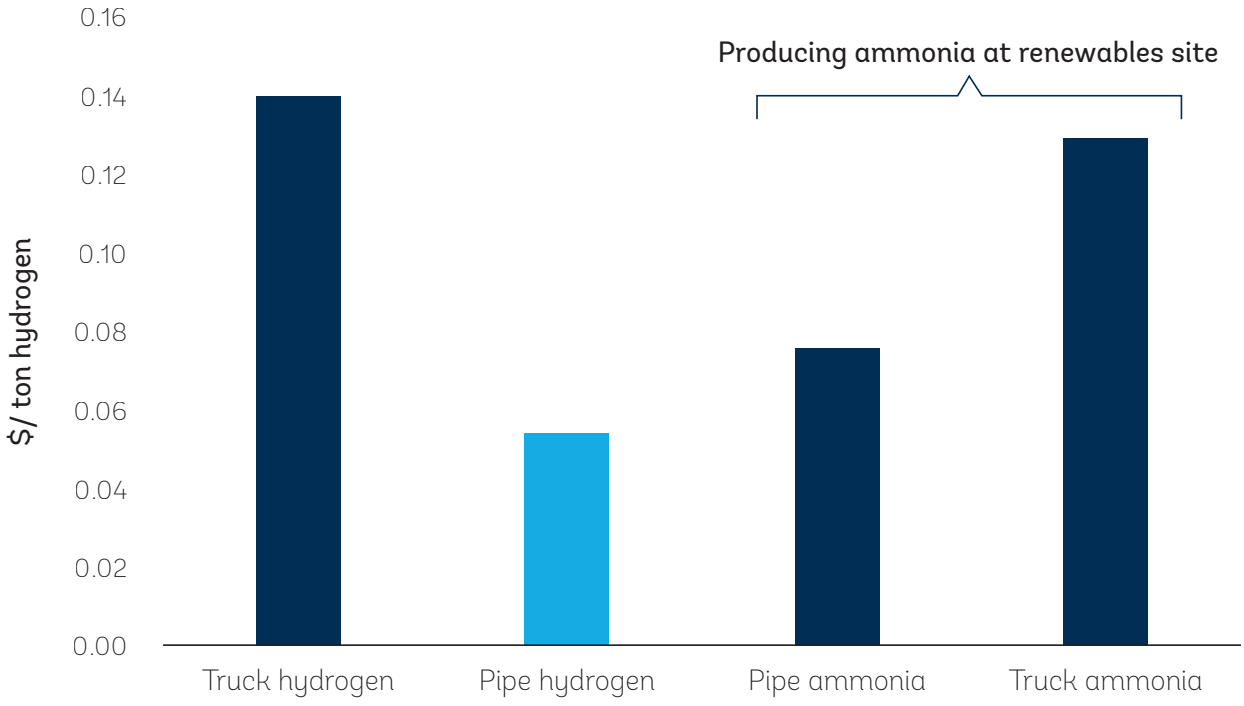
This comparison supports the decision to locate the electrolysis plant at the renewable energy generation site. Due to the high energy requirements of the electrolyzer to produce hydrogen, locating the plant close to the renewables will reduce the reliance on powerlines and allow pipelines to be integrated into the development. This will assist in increasing the overall efficiency and safety of the project while simultaneously lowering the cost.

Levelized Cost of Transport (LCOT)

The previous section concluded that molecules are the more cost-effective way to transport energy. This would mean to produce hydrogen at the renewable energy generation site inland. However, the question of how to ultimately move hydrogen molecules remains. One obvious option is to use a pipeline. Other options include transporting hydrogen by truck or converting hydrogen into ammonia and then moving the derivative by truck. Where pipelines fulfil the dual use of transport and storage, they outperform options where additional storage would be needed. This also applies to a case where ammonia would be produced inland, adding complexity to the project, as there is still a requirement to store hydrogen as a buffer.

However, the trade-off between a hydrogen pipeline and hydrogen storage deserves attention. Besides costs, there are physical risks related to hydrogen pipelines, including embrittlement and leaks, which opens areas of further analysis at more advanced stages in project development. For the case study, piping hydrogen is selected as the preferred approach. The ammonia production plant is located at the port and connected to the inland electrolysis plant through a hydrogen pipeline.[13]

Figure 5.15. Levelized cost of transport (LCOT) to move molecules in the example case



Source: World Bank.

[13] LCOT calculated based on site-specific input parameters.

Hydrogen pipeline

Hydrogen pipelines are already used at large scale globally. For example, The Netherlands has over 1,000 km of dedicated hydrogen pipelines alongside a 136,000 km network of natural gas pipelines (Invest in Holland 2022). In the whole of Europe there are around 15 dedicated hydrogen pipelines with a total length of 1,500 km, mainly serving petrochemical and chemical industries (Lipiäinen, et al. 2023). Although natural gas pipelines are more common and established, hydrogen pipeline networks require very similar components, creating an opportunity to repurpose existing pipelines, too (Guidehouse 2021).[14] However, the specific properties of hydrogen must be considered when designing or repurposing a pipeline system. A hydrogen pipeline has three high-cost elements: the capital expenditure of the pipeline and the compressors, and the compressor's electricity supply.

Table 5.15. Selected parameters for the hydrogen pipeline

Parameter	Requirement
Pipeline Length [km]	73
Hydrogen Storage Volume [t]	114
Operating Pressure [bar]	50
Pipeline Diameter [inch]	48

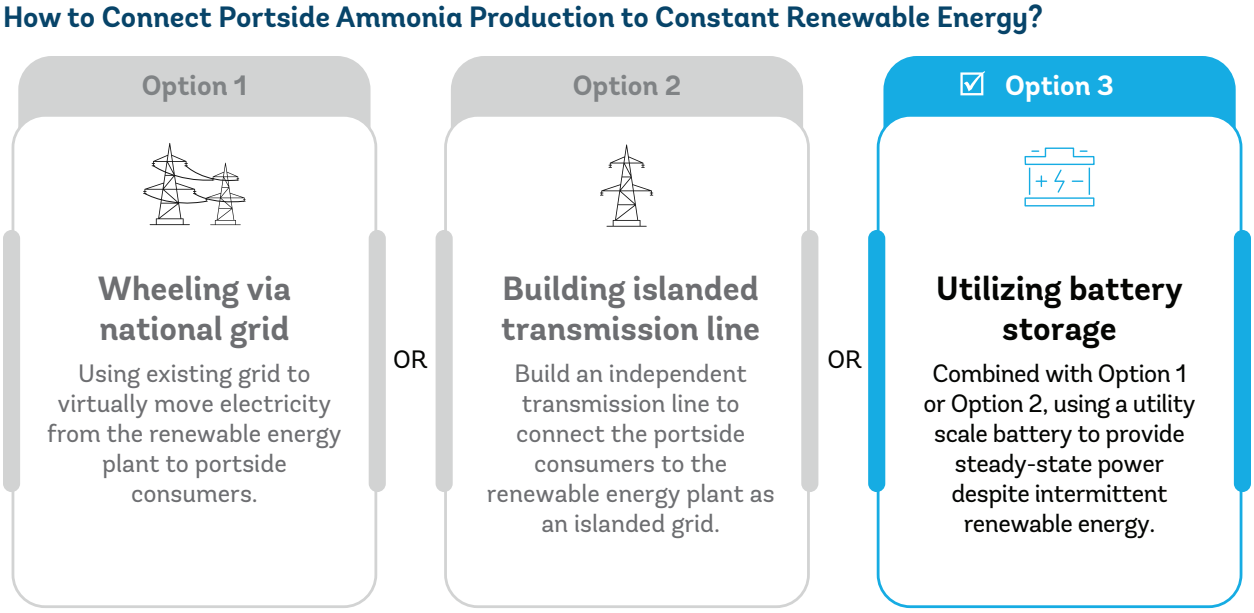
Source: World Bank.

Supplying renewable power to the ammonia production plant

The co-location of renewables and electrolyzers enable a direct connection between green electricity and hydrogen production without the need to wheel electricity via the national power grid. However, for the case study, the ammonia synthesis plant is located at the port. Spatially separating hydrogen production from ammonia production creates a challenge to guarantee uninterrupted renewable electricity supply to the entire production process. To connect the ammonia synthesis to uninterrupted green electricity, three options are conceivable:

[14] There are currently three common types of natural gas pipelines in Europe, 48-inch at 80 bars, 36-inch at 50 bars, and 20-inch at 50 bars (Guidehouse 2021).

Figure 5.16. Options to supply power to the ammonia synthesis plant



Source: World Bank.

- **Wheeling via national grid:** Utilizes existing public infrastructure. The South African grid is constrained, and transmission capacity issues would need to be looked at. Compliance with RED III guidelines of the European Union however is more of a concern.
- **Building islanded transmission line:** Extra cost for project development (estimated at around $130-260k per km) and additional regulatory implications for construction. Only offers intermittent power supply since not connected to the grid.
- **Utilizing battery storage:** Extra cost for project development (estimated at around $1,600/kW. Requires external power source for charging from Options 1 or 2. Improves the reliability of the system and addresses RED III compliance.

Ensuring that the final molecule can be certified green is paramount, for example considering the recently adopted stringent EU Renewable Energy Directive (RED III) (Directive (EU) 2023/2413 of the European Parliament and of the Council of 18 October 2023). The legislative package requires 42 percent of all ammonia production to be green by 2030. This applies to both production volumes in the EU as well as to imports.

In the technical setup, the principles of additionality, geography, and temporal correlation under RED III (and its associated delegated acts) are observed. Here, the two main energy consumers are electrolyzers, and the ammonia synthesis (i.e., Haber-Bosch process). While the electrolyzers are directly connected to the renewables plant and produce hydrogen when renewable power is available, the ammonia plant requires constant power supply to avoid thermal cycling.[15] A battery option would ensure that during the day, renewable energy is wheeled through the existing grid to charge batteries and provide eight hours of power

[15] Thermal cycling of the ammonia synthesis plant is the unwanted starting and stopping of the process due to intermittent power supply. This can cause damage on the technical equipment, mainly the catalyst in the Haber-Bosch reactor, which would alternate temperature between ambient and around 500 degrees Celsius and would lead to potential early replacement.

during the night to produce ammonia from hydrogen which is stored in the connected pipeline. The batteries could verifiably be charged with renewable electricity both for monthly and the more stringent hourly temporal correlation, effective from 2030 under RED III.

Utility-scale batteries are typically connected to transmission networks or other power sources and can support capacities up to hundreds of megawatt-hours (IRENA 2019). Lithium-Ion (Li-ion) batteries are the most mature and widely used and are considered the representative technology for utility-scale battery storage (NREL 2022).

Combining the use of the national grid and utility-scale battery storage is the most cost-effective scenario. It ensures constant power supply to the portside facilities and can comply with certification rules.

5.2.4. Hydrogen production

Green hydrogen is the feedstock for ammonia production and is produced through water electrolysis, where renewable electricity splits water into hydrogen and oxygen. Aside from renewable electricity, electrolyzers and a desalination plant are the main technical components. For the study, alkaline electrolyzers are favored for their lower costs, longer stack lifetimes and technology maturity, while not ruling out Proton-Exchange Membrane (PEM) or the more advanced Solid Oxide Electrolyzer Cell (SOEC) in future developments (compare 3).

Table 5.16. Comparison of electrolyzer technologies

Parameter	Alkaline	PEM	SOEC
System Efficiency [%]	51 – 60	46 – 60	76 – 81
H2 Production Rate [Nm^3/hr]	< 1400	< 400	< 10
1 GW Facility Footprint (ha)	10 – 17	8 – 13	Not given
CAPEX – 2030 [EUR_{2018}/kWe][1]	342	555	683
Operating Pressure [bar]	1 – 30	30 – 60	1
Load range, relative to nominal load [%]	10 – 110	0 – 160	20 – 100
Stack Lifetime [hrs]	60,000 – 90,000	30,000 – 90,000	10,000 – 30,000
Maturity	Mature and commercial	Commercial	Demonstration

[1] Optimistic values

Source: Adapted from IEA (2019), IRENA (2020), Roos (2021).

As opposed to using fresh water from the public water system, seawater desalination addresses prevalent water scarcity in the region and competition for freshwater resources. The theoretical minimum water requirement for electrolysis is nine kilograms of water per one kilo of hydrogen (9 kg H2O/kg H2O, plus a 25 percent add-on factor for a conservative assumption (Roos 2021). Consequently, the water requirement for the electrolyzers is determined at around 560 million liters per year.

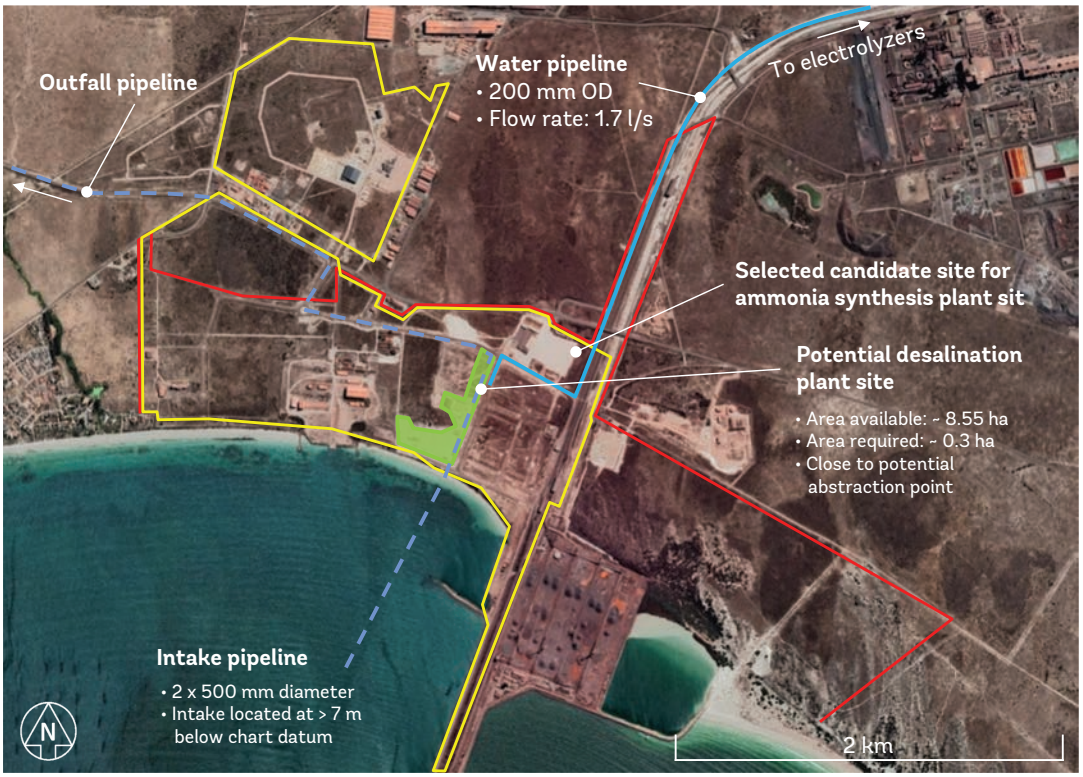
Table 5.17. Water requirements for the desalination process[16]

Parameter	Unit	Quantity
Electrolyzer annual water requirement	Ml/yr	560
Desalination plant daily output	Ml/day	1.5
Desalination plant daily intake	Ml/day	3.8

Source: World Bank.

Sheltered conditions within the harbor can provide a calm environment for seawater abstraction. The plant could be located within the IDZ boundaries, taking advantage of associated incentives and the proximity to the ammonia plant. For the case study, the water requirements are comparably low and so is the footprint of the desalination plant (around 0.3 ha), adding flexibility around the plant's location within the potential land parcel. There is also the potential to run the intake pipeline along the main jetty due to the small size of the pipes (2 x 500 mm diameter).[17] While the abstraction conditions are ideal, the brine outfall pipeline will need to be located elsewhere, preferably on an open coastline with better dilution potential, to reduce the environmental impacts of the saline discharge water.

Figure 5.17. Indicative layout of the desalination plant



Source: Google Earth, edits by World Bank.

[16] Based on the Strandfontein desalination plant commissioned in 2018.
[17] The intake velocity should be limited to 0.15 meters per second to protect marine life. Water quality within the port require further, detailed study. The intake should ideally be located where the seabed level is deeper than -7 meters to chart datum to limit sediment ingress.

5.2.5. Ammonia production

Ammonia synthesis combines green hydrogen (produced at the inland location) with nitrogen (separated from ambient air) to produce the final molecule, green ammonia. This would take place within the IDZ boundaries.

The core technical elements of an ammonia synthesis plant are:

- **Hydrogen,** which is moved via a pipeline from the inland electrolysis plant.
- **Air Separation Unit (ASU),** which separates ambient air into its main components, nitrogen (78 percent) and oxygen (21 percent) through a cryogenic process. Today's ASU systems have a capacity of 625-750 tons per hour. For instance, a unit in Secunda, (Mpumalanga province) commissioned in 2018 by French company Air Liquide, produces 5,000 tons of oxygen daily following a EUR200 million investment.

Table 5.18. Details for an Air Separation Unit (ASU)

Component	Unit	Quantity
ASU Capacity	tons N3/day	699
Power Demand	MW	7.1
Annual Nitrogen Production	tpa	233,000

Source: World Bank.

- **Ammonia synthesis unit,** which combines green hydrogen from the electrolyzers with nitrogen separated by the ASU using a Haber-Bosch reactor. To meet an output of 280,000 tons per annum of green ammonia, the unit requirements must account for a capacity factor smaller than one.

Table 5.19. Details for the ammonia synthesis plant

Component	Unit	Quantity
Ammonia Unit Capacity	tons NH3/year	310,000
Power Demand	MW	15.57
Annual Production	tons NH3/year	280,270
Annual Capacity Factor	No.	0.904

Source: World Bank.

Box 2: Haber-Bosch Process

FYI: How to manufacture ammonia?

The Haber-Bosch process allows for direct ammonia synthesis from hydrogen and nitrogen and was commercialized during the 1920s. This process is commonly applied in conjunction with steam methane reformation (SMR) of natural gas, producing grey ammonia. However, the approach is slightly different to produce green ammonia, as hydrogen is produced from renewable energy sources rather than fossil-fuel-based natural gas.

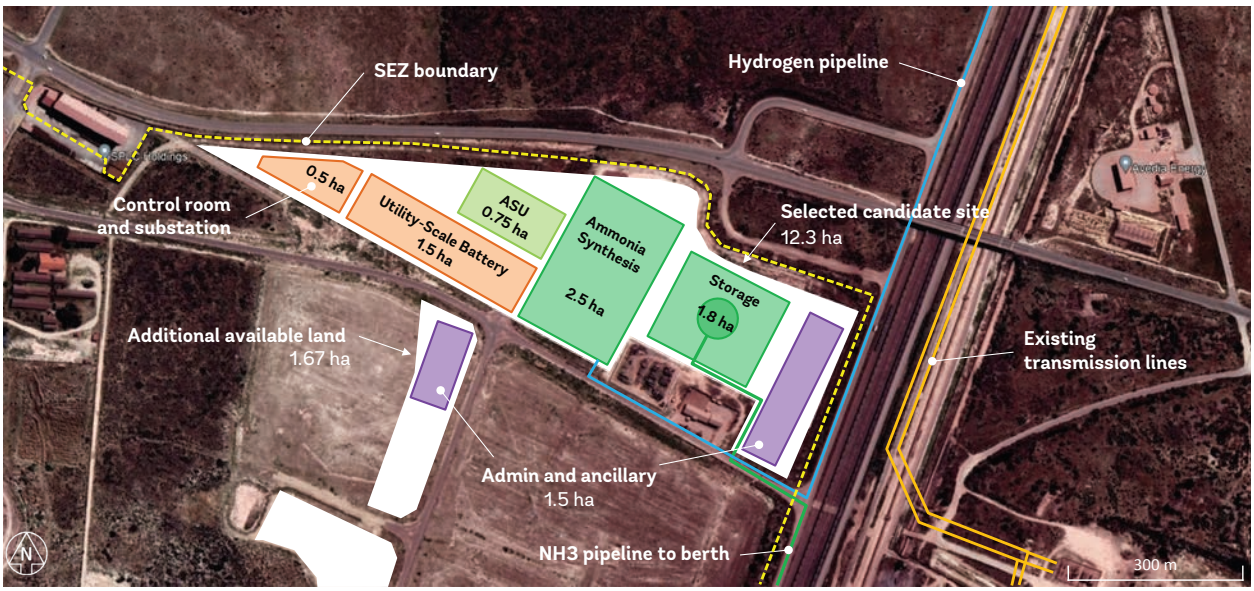
- **Ammonia storage** will provide the capacity to store ammonia after production until it is supplied to ships. The fuel can be directly pumped from the storage tanks to the loading facility at the berth, i.e., to fill a bunker barge. For a cost-effective tank option, onshore ammonia storage is somewhat influenced by the storage conditions used onboard the receiving vessel. For both large and mid-size ammonia carriers, refrigerated tanks are used onboard. As a result, cryogenic (-33°C) flat bottom tanks would be an appropriate landside storage solution.

  Refrigerated ammonia tanks differ the type of containment, insulation, and location of pump tanks. A single containment tank type is the most attractive for this project setup given the low storage requirement, low risk of seismic activity in the region, as well as the tank's comparably low investment cost. For a throughput of 280,000 tons per annum, one 30,000 cbm single containment tank would be needed to guarantee sufficient buffer storage for around 18 days. Ancillary facilities supporting the storage system can be in the peripheral servitude of the tank or within the ammonia synthesis unit space. If the case study was to primarily export ammonia through large LPG carriers, the required storage capacity would be larger, in line with the carrying capacity of larger vessels.

- **Utility-scale battery** primarily provides uninterrupted supply of renewable electricity to the air separation and ammonia synthesis unit, with approximate electrical loads of 7.08 MW and 15.57 MW respectively. Adding a 10 percent buffer to the electrical requirement results in a 25 MW utility-scale battery, which supports constant power supply (for around eight hours with a capacity of 200 MWh) and keeps ammonia synthesis cycling within acceptable levels.

The spatial requirements for ammonia production can be plotted onto 5.25 ha and fit within the candidate site identified within the port and IDZ boundaries.

Figure 5.18. Indicate spatial layout of the ammonia synthesis plant



Source: Google Earth, edits by World Bank.

5.2.6. Port infrastructure

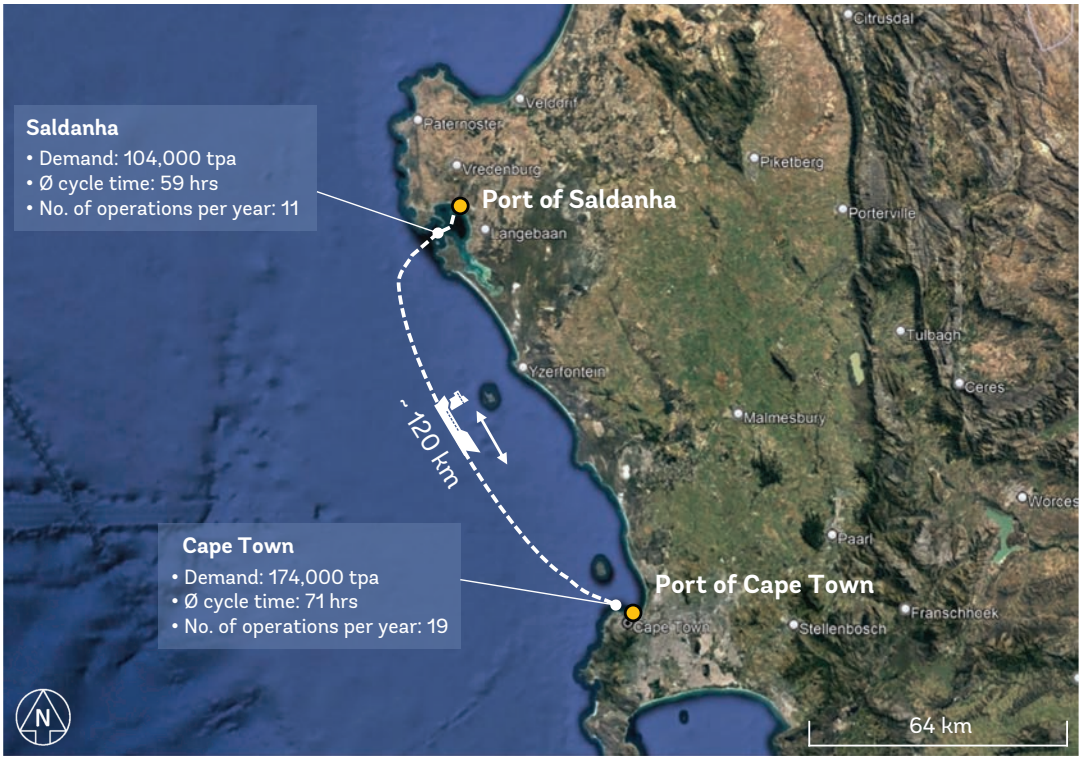
On the marine side, a bunker vessel must be able to load ammonia for further distribution. Based on the selection of a preferred berth, the case study could take advantage of existing pier infrastructure, namely the General Maintenance Quay (GMQ) which is operated as common-user facility by a private concessionaire. To transfer ammonia from the production and storage site to the quay, a 1.25 km cryogenic ammonia pipeline (with the allowance of multiple pipes) as well as berth handling equipment, such as loading arms for vessel loading would be needed as an upgrade to existing infrastructure.

Bunker supply

A bunker vessel would deliver the ammonia to ships requesting bunkering at anchor or at berth visa ship-to-ship (STS) transfer at the port of Saldanha and would serve the port of Cape Town, too. Ammonia will be transported to receiving vessels under refrigerated conditions (-33°C) using cylindrical horizontal semi-refrigerated storage tanks and transferred by way of composite flexible hose between barge and client vessel.

Based on the selected ocean-going bunker vessel (Table 5.8.), the carrying capacity of around 9,145 cbm, translates into a parcel size of 6,236 tons of ammonia. Based on the assumption that for both ports, each bunkered vessel receives a fuel parcel of around 4,573 cbm, two container vessels can be refueled in Cape Town during one trip at a ship-to-ship loading rate of 400 cbm/h. Based on the assumed ship fuel demand, the vessel would have an operating time of 2,060 hours per year, but can vary based on waiting time between bunker operations, decreasing the inevitable downtime.

Figure 5.19. Bunkering operations in Saldanha and Cape Town



Source: Google Earth, edits by World Bank.

Table 5.20. Summary of bunkering operations

Parameter	Port of Saldanha	Port of Cape Town
Forecast bunker throughput (tpa)	104,000	174,000
Average cycle time (hrs)	59	71
Number of cycles per year (No.)	11	19
Operational hours per annum (hrs)	700	1,360

Source: World Bank.

5.2.7. System Requirements Summary

Figure 5.20. Spatial overview of the system for the case study

Source: Google Earth, edits by World Bank.

Table 5.21. Spatial and system requirements for main system elements

System element	Location	Details	Spatial requirements
Renewable energy generation plant	Moorreesburg renewable energy site	Wind: 638 MW Solar: 957 MW Total: 1.6 GW	Wind: 15,700 ha Solar: 1,900 ha
Desalination	Port of Saldanha – Landside facility within the SEZ	Plant volume: 560 Ml/yr	Plant: 0.3 ha Pipeline to electrolyzer: 73 km
Electrolysis	Moorreesburg renewable energy site	Electrolyzer capacity: 492 MW H2 annual production volume: 52,600 tpa	5 ha
Hydrogen pipeline	Between Moorreesburg and port of Saldanha	H2 storage volume: 114 t	Length: 73 km
Ammonia synthesis plant	Port of Saldanha – Landside facility within the SEZ	ASU annual N2 production: 233,000 tpa NH3 annual production: 280 000 tpa	5.25 ha (incl. ASU and ancillary facilities)
Battery	Port of Saldanha – Landside facility within the SEZ (adjacent to ammonia synthesis unit)	25MW Li-ion battery storage capacity: 8 hrs	1.5 ha
Ammonia storage	Port of Saldanha – Landside facility within the SEZ (adjacent to ammonia synthesis unit)	Single containment tank Tank volume: 30,000 cbm	1.8 ha
Ammonia pipeline	Between ammonia storage and the berth (GMQ)	Flow rate: 400l/s Requires a return line	Length: 1.25 km
Berth solution	Port of Saldanha – General Maintenance Quay (GMQ)	Bunker barge to fuel at GMQ Assumed loading rate: 400 cbm/hr	Existing infrastructure
Bunker vessel	Port of Saldanha (servicing both Saldanha and Cape Town)	Small size LPG carrier Carrying capacity: 9,145 cbm	n/a

Source: World Bank.

5.2.8. Risk spotlight: Safety and environment

Besides the development of physical infrastructure, detailed considerations must be given to safety, environmental and social aspects. However, also political, regulatory, and procedural implications should be carefully studied. They should be considered early in the planning process to mitigate negative impacts on infrastructure users and the environment.

Chemical and physical properties

The chemical and physical properties of ammonia and hydrogen warrant attention for project feasibility. Here, mainly the toxicity of ammonia and the flammability and explosion risk of hydrogen present key risks.

Table 5.22. Chemical and physical properties relating to hydrogen and ammonia safety

Property	Hydrogen	Ammonia
General properties	Small molecules allow easy penetration. ~14 times lighter than air	Lighter than air but can create a vapor heavier when released into the air. Liquefied ammonia can cause caustic irritation and severe burns on contact with the skin.
Flammability	Flammable, little energy needed for ignition. Flammability range between 4 and 75 volume percent in ambient air	Not very flammable
Ignition energy	0.02 mJ	680 mJ
Explosion risk	Possible if hydrogen accumulates in confined spaces.	Only possible in poorly ventilated spaces.
Toxicity	Not toxic	Toxic LBW value (1-hour) = 1.495 ppm
Odor	Odorless	Strong, pungent odor
Color	Colorless	Colorless

Source: World Bank.

Ammonia as a marine fuel

The choice of ammonia as a candidate ship fuel to decarbonize international shipping is based on techno-economic reasoning. On a lifecycle emissions basis, green ammonia could result in zero greenhouse gas emissions when used as a combustion fuel in ship engines. While having no carbon dioxide emissions, a valid concern is the management of nitrous oxide (N2O) emissions, which can result from the incomplete combustion of ammonia in internal combustion engines. If released into the atmosphere, nitrous oxides could partially offset greenhouse gas savings due to its high global warming potential (GWP).[18] Like methane slip, a concern in LNG-powered marine engines, the operational management and monitoring of potential N2O emissions from marine engines is therefore critical.

[18] The metric Global Warming Potential (GWP) allows comparisons of the global warming impacts of different gases. Carbon dioxide is the reference gas with a GWP=1. Methane has a GWP of 27-30 over 100 years, while nitrous oxide has a GWP 273 times that of carbon dioxide for a 100-year timescale (EPA 2024).

It is paramount to highlight that the adoption of rules for onboard safety, bunkering and emissions management as core issues must be resolved before ammonia can be safely introduced as a marine fuel (European Maritime Safety Agency 2023). Early studies into the potential risks of ammonia as a ship fuel, however, do not present showstoppers to the large-scale adoption of ammonia as a marine fuel (Mærsk Mc-Kinney Møller Center 2023). For crews, occupational health risks are found to be able to be kept within tolerable limits if the maritime industry works towards developing the necessary technical and human-element related safeguards (Lloyds Register 2023). Having said this, a mass uptake of ammonia as a bunker fuel increases the number of transfers, transactions and points of human interaction. Inevitably, this increases risk and therefore calls for diligent planning, effective prevention measures and risk management practices to sustain its social acceptance and environmental integrity.

Ammonia handling in ports

A key benefit of using ammonia as a hydrogen carrier and bunker fuel is that the gas is already handled and stored in ports. Around 20 million tons of ammonia are moved by ships and safely handled at approximately 150 port terminals globally (IEA 2023) every year, including South Africa. Hence, regulations and protocols for safe storage and handling are already in place today in many locations.

However, transferring ammonia in bunkering operations is novel and, while in preparation, regulations have not been adopted yet. However, there are numerous studies for bunkering in specific port locations, such as Oslo (DNV 2021b), Amsterdam (DNV 2021a) and Singapore (GCMD 2023). All three are city ports, and the latter two are amongst the world's biggest bunkering hubs.

Det Norske Veritas (2021b) and (2021a) undertook a Quantitative Risk Assessment (QRA) to determine location-specific individual risk (LSIR) contours for various scenarios. The Amsterdam study specifically aimed to determine the external safety distances for ship-to-ship (STS) bunkering of alternative fuels and to define the focus areas for three hazards: fire, explosion, and toxic cloud. A QRA determined location-specific risk distances for all fuels. Although location specific, the analysis provides a good indication of potential risk zones for ammonia bunkering, which was applied in the case study and indicate potential hazardous zones surrounding the example loading berth (Figure 5.21.).

Table 5.23. Safety distances for bunkering refrigerated ammonia

Risk Element	Flow rate = 400 cbm/h	Flow rate = 1000 cbm/h
	Distance from bunker hose (m)	
10^{-6}/year LSIR contour	255	427
10^{-5}/year LSIR contour	153	246
Toxic cloud focus area boundary	1,446	2,624

Source: DNV (2021b).

Figure 5.21. Indicative LSIR contours around candidate bunker vessel loading berth

Source: Google Earth, edits by World Bank.

The 10^{-5}/year and 10^{-6}/year LSIR contours are quite contained, not extending far from the loading berth. These contours indicate the regions where less vulnerable objects can fall, with all vulnerable objects, such as residential areas lying outside these zones, reducing the risk. As no high-risk objects appear to fall within the indicated LSIR contours, the selected bunkering location could be considered further.

Furthermore, the toxic cloud focus area indicated by the blue circle relates to the area where the toxic concentrations of ammonia can be 2.53 times the life-threatening value,[19] for a 1-hour exposure. The potential toxic cloud area is an essential consideration for ammonia handling, and the extent of this hazard zone is an important detail consideration for project development, especially considering the proximity of the port facility to the town of Saldanha.

Environmental and Social

The impact of large-scale infrastructure on the surrounding environment and local communities is a key consideration. For the location of the renewable generation, the selection was informed by socio-environmental screening and provided similar inputs into the site selection. At more advanced stages of

[19] Determined as the air concentration above which life-threatening conditions can occur indoor.

project development, the mitigation of environmental and social impacts is essential and require detailed investigation. This primarily relates to the renewable energy generation plant (e.g., noise and visual pollution, risk to birdlife, and disruption of agricultural activity), routing of the hydrogen pipeline and ammonia handling (production, storage, and bunkering operations design to mitigate safety and environmental hazards).

Regulation and procedural time requirements

In South Africa, the National Environmental Management Act No 107 of 2008 (NEMA) lists activities which upon their commenting require a formal Environmental Impact Assessment (EIA) and a positive Record of Decision before construction can begin. In a large-scale multi-component project, such as this case study, multiple activities will inevitably trigger EIA requirements, including the need for specialist studies for environmental compliance. Hence, EIA authorization would be expected to take around two years. In addition to environmental legislation, typical authorizations for renewable energy projects in South Africa include:

- Water use allocation/confirmation and license application
- Atmospheric emission license
- Heritage authority consent
- Land and resource use rights
- Conservation of agricultural resources act consent
- Grid connection
- Generation license
- Land use planning
- Building plan approval
- Civil aviation commission authorization

5.2.9. Logistics

Project logistics will require a plethora of project cargo handling, which come with individual challenges. For this specific case study, it will mainly relate to wind turbines (e.g., blades, masts, gearbox, generator, hub, shaft), solar technology (panels, structures, invertors, etc.), construction material (e.g., rebar, steel), process plant reactors and piping, as well as electrolyzer modules. The Saldanha multi-purpose terminal benefits from experience, and most importantly, space to handle and store project cargo for renewable projects, reducing cost for long-haul transport.

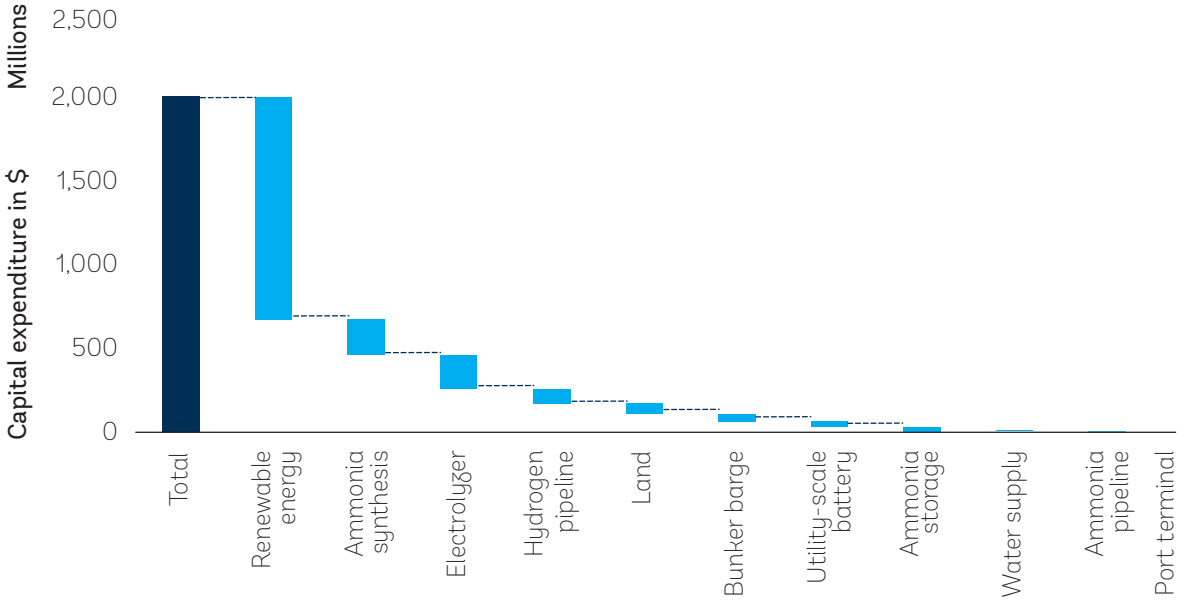
Figure 5.22. Unloading and storage of wind turbine blades at Saldanha

Source: Google Earth.

5.2.10. Cost estimate

The technical development determined the infrastructure requirements based on which a cost estimate including capital (CAPEX) and operational expenditures (OPEX) are calculated.[20] By far, the largest cost element is attributed to the renewable energy plant, followed by electrolyzers and ammonia synthesis.

Figure 5.23. Cost estimate for the case study project



Source: World Bank.

[20] Assumptions: All costing in 2018 US$ excluding cost of importing renewables components and excluding inflation (Roos 2021, IEA 2019, Guidehouse 2021, NREL 2022)

Table 5.24. Estimate of capital and operational expenditure

System element	CAPEX	OPEX		Comment
	$	$/yr	% of CAPEX	
Renewable energy generation plant	1,330,000,000	38,041,000	2.9%	Wind and SAT PV
Water supply	7,083,000	267,000	3.8%	Desalination plant, storage, and pipeline to electrolyzer
Electrolysis	202,119,000	4,044,000	2.0%	Alkaline electrolyzer
Hydrogen pipeline	91,874,000	7,351,000	8.0%	Hydrogen transport and storage
Ammonia synthesis plant	215,603,000	8,625,000	4.0%	ASU and Haber-Bosch reactor
Battery	37,824,000	946,000	2.5%	Charged from grid connection
Ammonia storage	22,955,000	919,000	4.0%	30,000 t NH3 tank
Ammonia pipeline	781,000	63,000	8.1%	High-pressure distribution pipeline
Berth solution	-	2,160,000	-	Common user
Bunker vessel	43,358,000	3,104,000	7.2%	Small-size LPG carrier
Land	55,470,000	555,000	1.0%	Could be leased
Total	**2,007,067,000**	**66,075,000**	**3.3%**	-

Source: World Bank.

5.3. Financial analysis

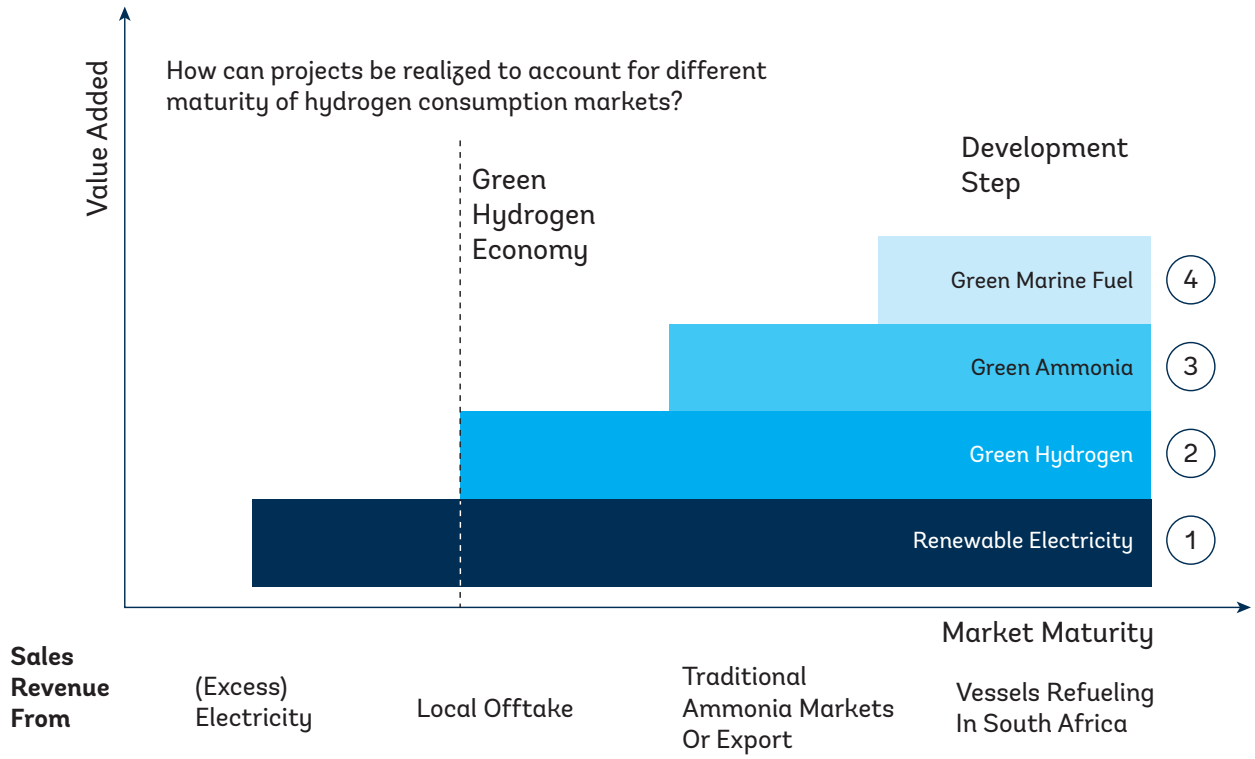
5.3.1. Approach

The objective of the case study is to determine at what price green ammonia can be supplied to ships in South Africa. It comprises a full value chain analysis, from producing renewable energy to refueling ships calling at the country's ports. At the same time, the appreciation of an end-to-end supply of ammonia as a bunker fuel is very complex in view of the infrastructure components and process steps. Hence, the financial analysis is designed to evaluate a step-by-step scenario approach in terms of output.

In practice, this means to divide the whole system into four additive scenarios, each component building upon the preceding process step. Namely, (i) producing renewables, (ii) producing green hydrogen, (iii) producing green ammonia, and (iv) supplying green ammonia to ships.

Consequently, each scenario adds additional CAPEX and OPEX, but is also funded by additional revenue streams as value is added to the product. This includes the use of hydrogen as industrial feedstock, the conversion of hydrogen into ammonia as a viable energy carrier and stand-alone product, and value-adding services through the supply of fuel to ships. There is a strong argument for a phased-in approach of certain steps for risk management purposes, too.

Figure 5.24. Stepwise development scenarios



Source: World Bank.

Table 5.25. Summary and results of the financial analysis

Scenario A	**Generates renewable electricity only,** through wind and solar at the identified sites. The scenario assists to evaluate an investment case into renewable electricity at the sites when electricity is sold at a competitive market tariff.
Scenario B	**Manufactures green hydrogen from renewable electricity,** which is assumed to be used by local off takers in the port's industrial cluster, such as in green steelmaking. Hydrogen is sold at a highly competitive price of $2/kg. Excess electricity, resulting from the renewable plant oversizing, is sold at cost, which makes this scenario conservative.
Scenario C	**Converts green hydrogen into green ammonia** as an energy carrier and is assumed to be sold to the (international) market at a sales price based on the three-year high for grey ammonia in Africa. Typical buyers of ammonia are the fertilizer industry.
Scenario D	**Targets the marine fuel market** by selling green ammonia to ships. To study the cost gap between today's most common ship fuel, very low sulfur fuel oil (VLSFO), the sales price for ammonia is set at the energy equivalent price of VLSFO. Around two tons of ammonia match the energy equivalent of one ton of conventional marine fuel. The sales price of two tons of VLSFO happens to be lower than the sales price for grey ammonia. This circumstance is further investigated in a sensitivity analysis.

	Scenario A	Scenario B	Scenario C	Scenario D
CAPEX and OPEX				
Renewable energy assets	✓	✓	✓	✓
Electrolyzers and desalination unit		✓	✓	✓
Ammonia production			✓	✓
Export infrastructure in the port			✓	✓
Bunker vessel				✓

		Scenario A	Scenario B	Scenario C	Scenario D
Production outputs (actuals)					
Electricity	MWh	3,727,294	3,727,294	3,727,294	3,727,294
Hydrogen	Metric tons	not applicable	55,037	55,037	55,037
Ammonia	Metric tons	not applicable	not applicable	280,270	280,270
Financial results					
NPV (in 2025, at 6% WACC)	$	242	(692)	219	(1,672)
Payback period (from start of operation)	Years	12	19	13	not applicable
Internal Rate of Return (IRR)	%	7.2	2.7	6.8	not applicable

Source: World Bank.

5.3.2. Discussion of the financial analysis' results

The financial feasibility differs considerably in each scenario, based on the primary market the scenario penetrates. While a renewable energy project, i.e., selling green electricity into the grid, presents a viable undertaking in the South African context (Scenario A), the market for green hydrogen as a feedstock for industrial use is still at a nascent stage (Scenario B). Here, revenues from a competitive sales price are insufficient to recover the investment. However, subject to the off-taker's willingness, sellers could get them to pay a potentially higher price for green feedstock. Value addition, i.e., converting green hydrogen into ammonia, targets a different market, the ammonia market, and returns a financially feasible project case (Scenario C).

Certainly, the most immature market is truly green shipping fuels. This is equally reflected in the negative net present value (NPV) for Scenario D, where ammonia is not sold to the existing ammonia markets but must compete at a price equivalent to conventional bunker fuel. While this may not surprise given the awareness that low-carbon energy alternatives for ships are more expensive, the analysis presents a concrete production case to determine a real-world price differential.

Furthermore, the scenario results must be viewed in the context of underlying infrastructure selection for the case study. Additional optimizations regarding technical efficiencies could be explored, as well as the creation of spin-off synergies for socio-economic gains. In practice, this could be an even larger oversizing of the renewable power plant to add more generation power to the grid for energy security or oversizing the desalination plant for water security. However, such technical modifications can increase cost and must be carefully weighed against social and economic benefit. These considerations open an avenue for transitioning the project to meet additional objectives beyond the production of green shipping fuel.

5.3.3. Sensitivity analysis

The objective of the analysis is to determine the price at which ammonia can be supplied to ships in a concrete location in South Africa without bias towards current, historic, or future price assumption. Hence, the sensitivity analysis is set at project feasibility (where the net present value is zero) and varies inputs with respect to discount rate (weighted average cost of capital, WACC) and reduction in capital expenditure (i.e., CAPEX subsidy) to explore a financially viable sales prices within each of the four production scenarios.

Lower WACC and partial CAPEX subsidies naturally result in more competitive prices at which a product can be sold. The benefit of this perspective is particularly relevant to determine the price gap for products like green hydrogen and ammonia, which broadly suffer from a large price differential with conventional, fossil-based feedstocks, or fuels. The prices are benchmarked against the five-year historic average and three-year high to identify which financial condition the project could aim for to land in competitive price ranges. Concretely, this could be the sourcing of, for example, low-cost debt and equity or benefiting from grant schemes available to climate-related projects.

While renewable energy can be offered at competitive market prices across all conditions, green ammonia, as derivative becomes highly price competitive with fossil-based ammonia at WACC of six percent or below, or a combination of higher capital cost, but the use of subsidies. Green ammonia as bunker fuel does not reach competitiveness with conventional marine fuel though. Even under aggressive financing and investment conditions, the price differential with conventional bunker fuel is not overcome.

Nevertheless, at low capital cost, green ammonia could be offered at double the conventional fuel's price, while subsidies could reduce the cost differential to around 40 percent of the conventional fuel's price considering its three-year high, resulting in a price range between ≈1.4 to 4.3 times the cost of fossil-based marine fuel.

Table 5.26. Sensitivity analysis

NPV = 0			Capital investment subsidies					Historic benchmark	
WACC	Scenario	Unit	0%	10%	20%	30%	40%	5-year average	3-year max[21]
12%	A - RE	$ per kWh	0.072	0.066	0.061	0.055	0.049	0.069	-
	B – H2	$ per kg	5.21	4.72	4.24	3.75	3.27	-	-
	C – NH3	$ per ton	1,240	1,130	1,020	910	790	490	780
	D – NH3 (B)	$ per ton	1,280	1,170	1,050	940	820	230	300
10%	A - RE	$ per kWh	0.062	0.056	0.053	0.048	0.043	0.069	-
	B – H2	$ per kg	4.38	3.98	3.58	3.17	2.77	-	-
	C – NH3	$ per ton	1,050	960	870	770	680	490	780
	D – NH3 (B)	$ per ton	1,090	990	900	800	710	230	300
8%	A - RE	$ per kWh	0.053	0.049	0.045	0.041	0.037	0.069	-
	B – H2	$ per kg	3.63	3.30	2.97	2.65	2.32	-	-
	C – NH3	$ per ton	880	800	730	650	580	490	780
	D – NH3 (B)	$ per ton	910	830	760	680	600	230	300
6%	A - RE	$ per kWh	0.045	0.042	0.039	0.036	0.032	0.069	-
	B – H2	$ per kg	2.95	2.69	2.43	2.17	1.91	-	-
	C – NH3	$ per ton	720	660	600	540	480	490	780
	D – NH3 (B)	$ per ton	750	690	630	570	510	230	300
4%	A - RE	$ per kWh	0.038	0.035	0.033	0.031	0.028	0.069	-
	B – H2	$ per kg	2.34	2.15	1.95	1.75	1.55	-	-
	C – NH3	$ per ton	590	540	490	450	400	490	780
	D – NH3 (B)	$ per ton	610	560	520	470	420	230	300

Unit cost lies below the historic 5-year average

Unit cost lies below the historic 3-year max

Source: World Bank.

[21] 3-year max price refers to the average of each year's peak price

Table 5.27. Sensitivity analysis for the bunker scenario (Scenario D) and VLSFO equivalent price

NPV = 0			Subsidies to capital expenditure					Historic benchmark	
WACC	Scenario	Unit	0%	10%	20%	30%	40%	5-year average	3-year max
12%	D (VLSFO-eq)	$ per ton	2,860	2,610	2,350	2,100	1,830	510	670
10%	D (VLSFO-eq)	$ per ton	2,440	2,210	2,010	1,790	1,590	510	670
8%	D (VLSFO-eq)	$ per ton	2,030	1,850	1,700	1,520	1,340	510	670
6%	D (VLSFO-eq)	$ per ton	1,680	1,540	1,410	1,270	1,140	510	670
4%	D (VLSFO-eq)	$ per ton	1,360	1,250	1,160	1,050	940	510	670

Source: World Bank.

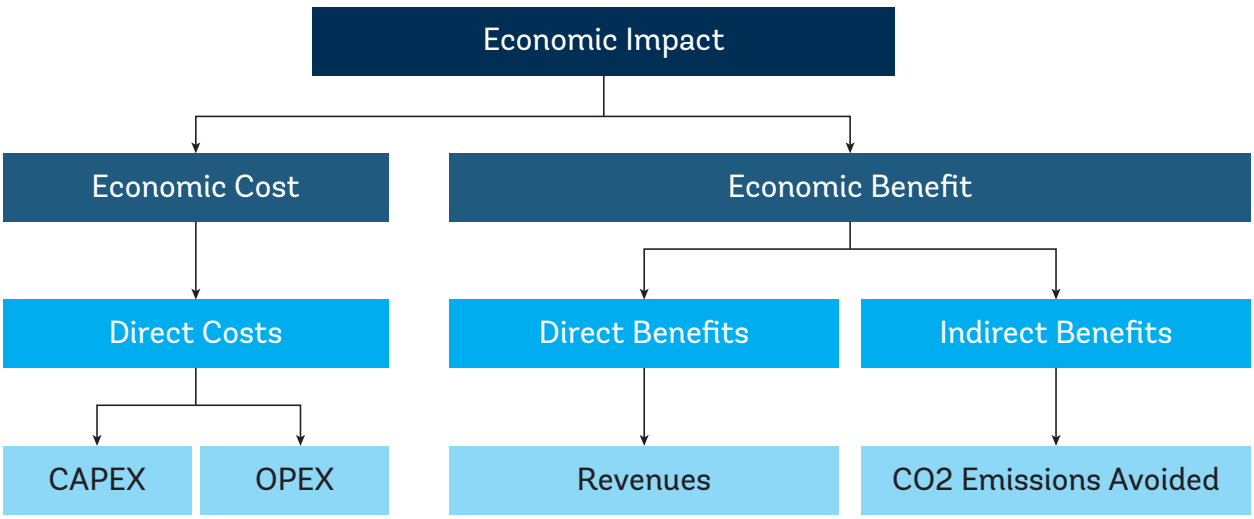
5.4. Economic analysis

The economic cost-benefit analysis (ECBA) complements the financial analysis with an overview of the economic effects of the study case. The premise of the analysis is to compare the economic effect of the implementation scenario (i.e., project case) with not implementing a possible project (i.e., no project case). The project case is the execution of Scenario D, the production and supply of green ammonia to ships as marine fuel. The no project case retains the use of conventional fossil-based marine fuels.

5.4.1. Approach

The economic impact of developing the project is calculated by determining the difference between project case and the no project case. The ECBA, therefore, focuses on the incremental economic effects expected to occur during the period 2028-2052, in line with the financial analysis.

Figure 5.25. Economic cost-benefit analysis



Source: World Bank.

Economic costs

The *direct economic* costs consist of the incremental CAPEX and OPEX resulting from the example project. Understanding the ECBA means understanding what is considered incremental and what is not. While it is true that the overwhelming share of direct costs (CAPEX, OPEX) associated with the project case is incremental, there are a few corrections required. These exceptions come in two forms: imported project cargo and labor costs.

The CAPEX and OPEX forecast includes the investment in materials, with a likelihood of the majority being imported. Consequently, an import duty – in South Africa, that is assumed to be 15 percent of the asset's value – will be imposed on the imported goods. This import duty is included in the economic costs for CAPEX. Moreover, the costs also include labor, specifically wages. These wages are subject to income tax at approximately 30 percent. Consequently, the incremental CAPEX and OPEX utilized in the ECBA are lower than those employed in the financial analysis.

Economic benefits

The *direct benefits* consist of the additional revenues that the project case generates. The *indirect benefits* comprise the prevented carbon dioxide (CO2) emissions, which would be combusted if the project would not be developed. The avoided CO2 emission from renewable electricity generation is included, too. The direct benefits are mainly driven by the additional revenues in the region. The revenue generated due to the sale of ammonia is additional; there will not likely be a regional cannibalism effect in this market. The revenue generated from the sale of electricity is only partly seen as additional. The surplus electricity sold to the market will cannibalize a portion of the existing power plant supply.

The *indirect benefits* are mainly driven by the prevention of CO2 emissions, coming from replacing fossil fuels (e.g., VLSFO) with green ammonia. In this analysis, it is assumed that emissions are avoided only because the vessel's engines do not burn fossil-based marine fuels. The avoidance of emissions by producing ammonia sold for export (occurs during the first seven years since bunker demand is below production level) and generating excess electricity sold to the grid is also appreciated.

The economic benefit of the avoided carbon emissions during the project's life cycle (in tons CO2) is monetized by adding a carbon price ($ per ton of CO2).[22]

5.4.2. Results

The economic cash flows derive from the economic cost, consisting of incremental CAPEX and OPEX as well as the direct and indirect, external economic benefits comprising of additional revenues and the avoidance of carbon emissions, under application of allocation and conversion factors as well as the social discount rate.

The project case returns total positive discounted economic cash flows, resulting in an economic net present value of around $155 million and generating an economic rate of return of 6.7 percent. Hence, the case study presents an economically viable investment with social and economic benefits to the country.

[22] For the purpose of the cost-benefit analysis, a carbon price of 100 $/t was assumed.

Table 5.28. Results of the economic analysis

Item	Unit	2025	2026	2027	2028	2030	2035	2040	2045	2050	2052
CAPEX	$ million	(746)	(683)	(696)	-	-	-	(19)	-	-	356
OPEX	$ million	-	-	-	(65)	(68)	(75)	(82)	(91)	(100)	(104)
Economic costs	$ million	(746)	(683)	(696)	(65)	(68)	(75)	(102)	(91)	(100)	252
Additional revenues	$ million	-	-	-	94	98	108	119	132	146	151
CO2 avoided Electricity		-	-	-	101	101	101	101	101	101	101
CO2 avoided Ammonia (marine fuel)	$ million	-	-	-	3	9	39	40	40	40	40
CO2 avoided Ammonia (export)		-	-	-	70	59	2	-	-	-	-
Economic benefits	$ million	-	-	-	268	267	250	261	273	287	292
Net economic cash flows	**$ million**	**(746)**	**(683)**	**(696)**	**203**	**199**	**175**	**159**	**182**	**187**	**544**
Economic net present value (ENPV)	**$ million**						**155**				

Source: World Bank.

5.5. Project risks

The case study outlines a large-scale infrastructure project which involves nascent technology and emerging markets for green hydrogen. A project of this scale comes with risks, which require further assessment and proper mitigation, especially for the downstream components:

- **Dependency risk:** The scale of the infrastructure development and the involved technology, e.g., electrolyzers, is not common for contractors and construction companies. There is a lack of experience to deal with green hydrogen projects. This circumstance is considered a medium risk and can be addressed by diligent procurement and early assessment of contractor capabilities.
- **Technology risk:** Some of the technology components have not been deployed at scale yet and could be considered not fully mature. This includes, for example, electrolyzer technology or the risk of

embrittlement in hydrogen transfer. This could cause complication during development. Of medium risk level, technology risk should be carefully addressed in detailed engineering. Exploring contingency scenarios can help to identify alternative development pathways should errors arise.

- **Commercial risk:** To be financially sustainable and recover invested capital, the project primarily requires steady revenues from a to date immature market for hydrogen-based marine fuels. The delay of regulatory mandates or incentives for the use of low carbon marine fuels therefore requires a customer who is prepared to pay a premium, since the product is not cost-competitive with conventional fuels just yet. While long-term offtake agreements mitigate this risk to a large extent, the possible default of a single off-taker presents a counterparty risk. Equally, if the project's offtake is dependent on financial subsidies, the limited duration or termination of such support could have a detrimental effect on the project profitability. Diversifying off-takers and exploring several hydrogen or ammonia markets can mitigate financial risk, which is viewed high.
- **Environmental, Safety and Permitting risk:** Permits and licenses play an important role during the development phase. If not in place, missing or unsuccessful permitting processes can cause project delays or even project stop. Besides permitting and licensing for commercial activities, the hazardous properties of hydrogen derivatives pose a risk during the project's operational life, too. Early engagement with relevant authorities, and quantitative risk assessment can mitigate environmental and human safety related risks. Reducing these risks is fundamental to sustain the social acceptance of such an investment and its effect on people and the environment.

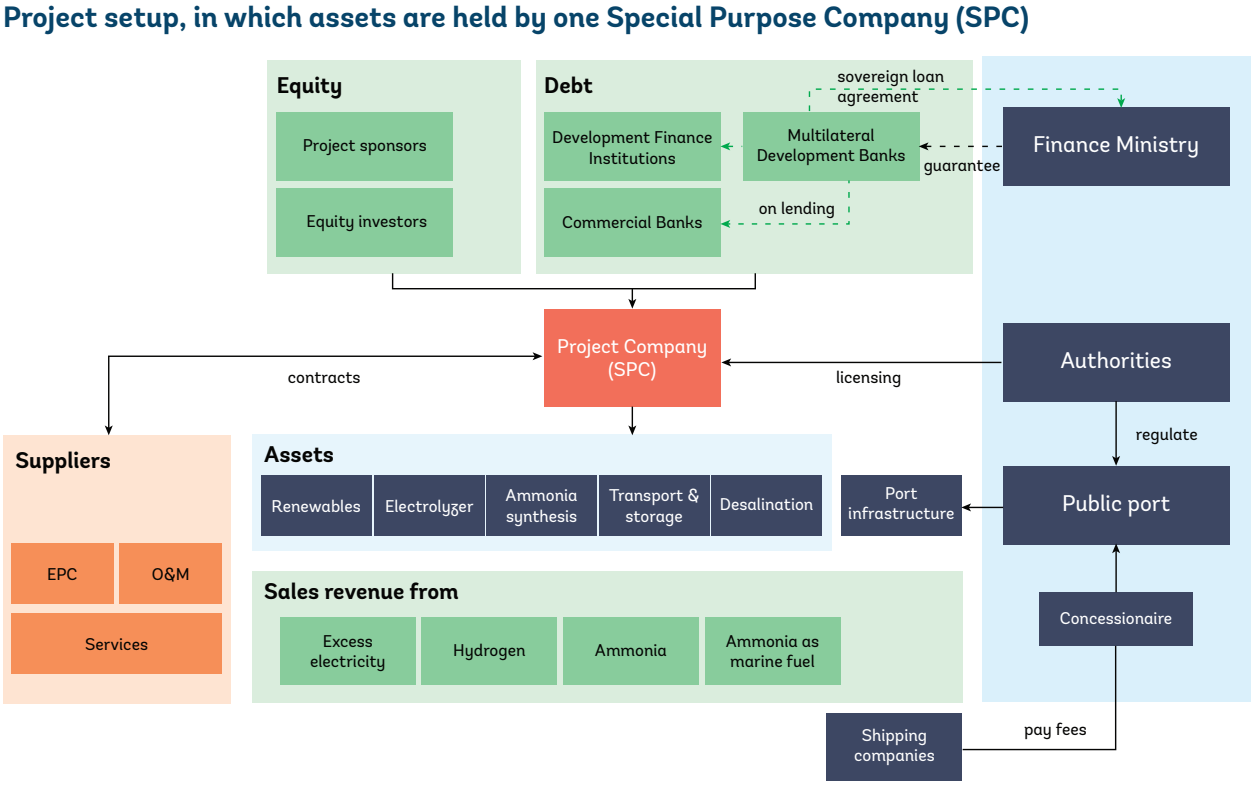
5.6. Project execution

5.6.1. Project structure

The project can be viewed as comprising of four main components, namely (i) renewable power generation plant, (ii) hydrogen production, (iii) ammonia production and (iv) port infrastructure. The final structure is influenced by variations in ownership and operational management of the main components.

If the core objective is to produce and sell green marine fuel, then as a result, one controlling company would be a reasonably assumed setup (Figure 5.26.). However, other options are possible. Individual project components could be further separated or ringfenced from one another.

Figure 5.26. Example project setup



Source: World Bank.

5.6.2. Finance structure

A green marine fuel project can be viewed as a green hydrogen project given that it produces an energy carrier derived from hydrogen. One major consideration for green hydrogen projects will be to structure an acceptable financial risk profile by allocating risks to those parties best able to accommodate them. These include project sponsors, insurers, financiers, and government.

In the port industry, Public-Private Partnerships (PPPs) are often used for major projects, while a fully private ownership set-up is more common in the renewables and hydrogen value chain. PPPs are defined as to create or add public assets to meet common infrastructure needs for the economy and its users, where the private party bears significant risk and management responsibility. This raises the question of the appropriate level of government involvement.

Green hydrogen development is in its early stages, and financial support, especially in budget-constrained economies, is expected to be limited. While green ammonia production creates an economic opportunity for the country, this activity can be considered private. As a result, projects like these could be financed through either project finance or balance sheet finance, whereby the former appears to be the most common for hydrogen projects these days.

While government may still aim to invest in the green hydrogen value chain to support decarbonization efforts and secure green energy, it may not be a major off taker of the product itself. However, a possible involvement of government-owned entities is the purchase of excess renewable electricity from a private independent power producer. Furthermore, where new port infrastructure is required, this would traditionally fall into the realm of the public port landlord. Beyond this, the government can play an important role, in for example mobilizing lower cost capital, and as a broker of potential financial support.

5.6.3. Financing options

The main applicable financing options are balance sheet financing or project finance.

Balance sheet financing

A company takes out financing based on the health of its balance sheet evaluated by for example, lenders. This form of financing is also known as corporate finance. Banks often rely on the credit ratings provided by rating agencies. Corporate bank loans or bonds are secured against assets or operating performance of the company, assessing its short- and long-term debt to evaluate the counterpart's debt capacity. As such, the company's risk profile plays an important role in the finance due diligence by the lenders. Also, working capital loans are often structured as balance sheet financing. In general, the cost of capital for corporate financing can be lower, as opposed to project finance, if the underlying corporate borrower shows a strong balance sheet.

Project finance

In project finance arrangements, the project itself is subject to credit assessment as opposed to the balance sheet of the parent company, normally with limited or no recourse to the latter. Private investors often support project finance as the project company itself has no financial track record. The Special Purpose Company (SPC) is initially not considered valuable unless it has received development or land ownership rights to further the project. Project finance normally commences only after due diligence by the sponsors and financial investors.

While operating cash flows present the primary collateral against which financing is secured, secondary security is provided by the SPC's assets, rights, and interests during operation. The underlying projected cash flow of the project's operations must suffice to cover operational expenses, service debt and repay equity. Project finance is often used for large-scale asset finance such as infrastructure investments. In terms of risk, lenders are less exposed to the performance of the parent company, while the project sponsor can protect his own balance sheet (for example, if there is no recourse to the parent company by way of a guarantee) should the project not perform well.

5.6.4. Project structure summary

As outlined, a large-scale project, such as the Saldanha case study could be delineated into four main additive components: Renewable energy, hydrogen, ammonia and the supply of marine fuel. However, if the goal is to produce marine fuel, then a straightforward approach with only one controlling company, a special purpose company, can be an adequate setup. In the context of the production of a nascent product, such setup primarily mitigates risk for the project developer, too. The income of the SPC is derived from excess electricity sales, hydrogen sales to local off-takers, ammonia sales to local and foreign markets, and the supply of ammonia to the regional bunker market.

5.7. Delivery schedule

To deliver a large-scale infrastructure project of this size, four key workstreams must work together to meet an ambitious, but achievable timeline until fuel production can start. However, all workstreams directly or indirectly hinge on a successful financial close, which is assumed to take approximately two years. In practice, key dependencies are:

- **Technical:** To substantiate bankability, a feasibility study including frontend engineering design (FEED) must be completed to control project expenses and issue procurement for engineering, procurement, and construction (EPC) contracts.
- **Financial:** Financial feasibility must be confirmed based on the chosen business model and structure, assisting with offtake agreements, and securing finance from lenders and investors.
- **Environmental:** Financial close is dependent on environmental approval including several environmental impact assessments (EIA), associated specialist studies and stakeholder engagements.
- **Regulatory and legal:** Establishment of the special purpose entity including all relevant permits and documentation including but not limited to air emission licenses, land use agreements, port use and bunkering licensing. Consideration to timelines relating to certification under overseas rules, such as RED III must be given in respect to the renewable power generation plant.[23]

Following financial close, an EPC contractor can commence with detailed design, procurement, and construction, which would be expected to take around three to four years until completion and operations start. A phased-in approach of value adding processes such as ammonia production and supply of the latter as marine fuel could advertently defer those components.

5.8. Conclusions of the case study

The case study presented a full value chain assessment of a green hydrogen project, aiming to produce hydrogen-based marine fuel in a South African port location. A bunker demand assessment determined that the pilot project could produce around 280,000 tons of green ammonia per annum, requiring a feedstock of around 50,000 tons of green hydrogen.

Green ammonia was identified as the preferred zero-carbon bunker fuel due to its truly zero-carbon production characteristics and flexibility as a hydrogen carrier to penetrate different markets and applications. Green methanol, however, remains a contender, but depends on the successful identification of a sustainable carbon source, and can be reconsidered in further analysis.

The technical assessment in the case study concluded that renewable power generation, at a split of 40/60 between wind and solar, could be co-located with the electrolyzers at an inland site around 70 km away from the port. Hydrogen produced at the inland site would then be piped to Saldanha, which turned out the most cost-effective option as opposed to building a power transmission line and wheeling electricity to the port. To take advantage of fiscal incentives, ammonia production is located within the limits of the special economic zone (SEZ).

[23] Production start no more than 36 months before electrolysis and derivate production start under RED III.

The intermittency of renewable power poses two challenge to green ammonia production. Firstly, to reduce the cycling of the ammonia synthesis plant. Secondly, the continuous supply of renewable electricity to comply with certification rules, such as RED III in the European Union, as South Africa's electricity generation is mainly based on coal-based power. A utility-scale battery, which is charged during hours of solar and wind can supply around eight hours of steady power to address these two issues. The port of Saldanha would serve as a loading hub for a bunker vessel, which can supply green ammonia to vessels calling the port of Saldanha and the port of Cape Town nearby. If ammonia was exported as a commodity to overseas markets, the port can serve as an export hub.

The estimated capital expenditure for the project is around $2 billion, with most of the expenditure relating to renewable energy production (around 65 percent), followed by electrolyzers (around 10 percent) and ammonia synthesis (around 10 percent). In the case of Saldanha, existing port infrastructure could largely be used to load a bunker barge or an ammonia carrier. However, in greenfield projects, investments into port infrastructure could be substantial.

In the case study, green ammonia as a marine fuel could be produced at a price range between ≈1.4 to 4.3 times the cost of conventional marine fuels, depending on the financing conditions (weighted average cost of capital or subsidies). Other target markets, such as the sale of green hydrogen and green ammonia, would either require lower capital costs, subsidies, or a market prepared to pay a premium for a green product. In view of recent price hikes in, for example, conventional ammonia markets, green ammonia could even become cost-competitive on the African continent under less aggressive financing terms. Local industrial off-takers, such as steelmaking, can present special markets. It is worthwhile to further explore and understand the financially viable production cost for green hydrogen as feedstock. The study also explored the potential benefits of disaggregating project components. From an economic perspective, the socio-economic benefits of producing hydrogen-based marine fuel outweigh the cost, through additional revenue and the avoidance of carbon emissions.

The example project presents a major infrastructure development, which requires strong private sector support. The main commercial risks are (i) the lack of demand for hydrogen-based marine fuels, and (ii) the risk of banking on a product which is not price-competitive with its fossil-based alternative just yet. Given the scale of investments, a possible project could be de-risked through the gradual development of value-adding process steps, as different markets mature.

Another key risk, which require further, detailed study are the safety and environmental implications of handling hydrogen-derivatives at Saldanha. While early studies indicate that safety challenges in the transfer of ammonia can be overcome, location specific study and consultation is needed to identify risks and establish effective safeguards.

As such projects comes with significant risk to the commercial project sponsor, the public sector has a key role to play. Partnership between the government and private sector is critical to manage risk when developing complex, nascent infrastructure projects, and helps to unlock co-benefits for the local community. Here, support regarding possible common-user infrastructure needs, regulatory hurdles and legal considerations must be addressed in sync for successful project delivery. Saldanha present a special case, where several demand sources for green hydrogen aggregate: local heavy industry, export, and supply of marine fuel. This offers a distinct opportunity for possible risk-sharing on the common-user infrastructure side and would benefit from further study.

Please note: *The case study presents an options case, where the technical or commercial setup should neither be viewed prescriptive, nor conclusive. Further detailed analysis, especially regarding environmental, safety and social implications is needed. However, the study can serve as a reference point for public and private sector stakeholders, including, but not limited to, discussions with development partners, to support the advancement of large-scale green hydrogen infrastructure projects in South Africa.*

Harboring hydrogen: Ports as green energy hubs

Ports are no strangers to massive trade; they are critical to global supply chains. So why not for hydrogen as well? They serve three pivotal roles for the energy economy. Governments are counting them into national strategies and incentives are being given to make them fiscally attractive for investors. Plus, they're encouraging the growth of green corridors.

06

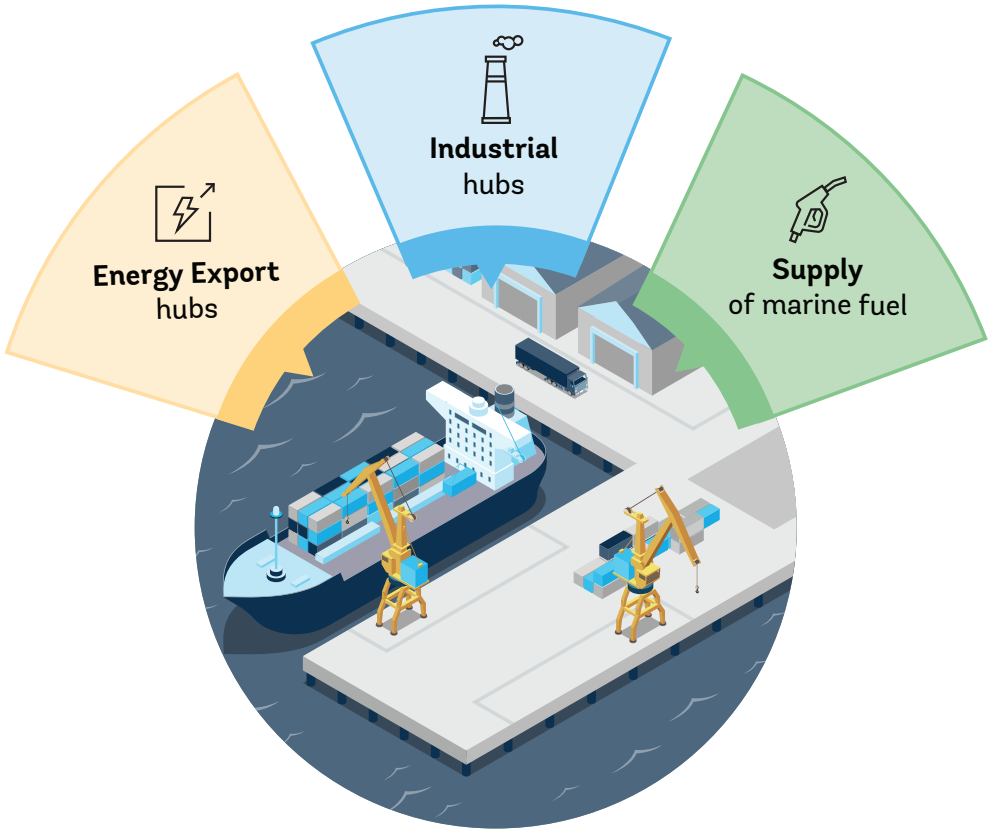
Seaports handle around 80 percent of global trade (UNCTAD 2023), and are critical nodes in global supply chains. For the uptake of hydrogen, the International Energy Agency (IEA 2019) identifies industrial ports as the "nerve centers" to scale up clean hydrogen use.

6.1. Three roles for ports

Ports today serve the conventional energy economies, for which three functional dimensions are traditionally identified in the literature: energy transport platforms, energy transformation platforms and energy generation platform (Notteboom, Pallis and Rodrigue 2022). Similarly, ports can play three critical roles in South Africa's green hydrogen economy: export hubs, industrial hubs for consumption and production of hydrogen, and the supply of hydrogen-based marine fuel.

Figure 6.1. Three roles for ports in the hydrogen economy

Different roles for ports



Source: World Bank.

Energy export hubs

Today, mineral extraction is a key industry for South Africa. In this energy field, ports are already adjudged a catalytic role for economic growth (DMRE 2022), leveraging on a port's role as transport platform. For global hydrogen trade, ports will act as transport platforms, too, since 45 percent of the global hydrogen trade is expected to be seaborne (De Maigret, Macchi and Herib 2022).

Industrial hubs for consumption and production

For the hydrogen economy, the port as a transformation platform will be of great relevance. Based on the sectors which are expected to utilize hydrogen as alternative feedstock, it was found that, for example, in Europe, 50 percent of total hydrogen demand will occur within port areas (Clean Hydrogen Partnership 2023).

The production of green hydrogen within or around ports can also lead to renewable energy generation becoming part of the immediate or extended port ecosystem as the case study shows (Chapter 5). This is also exhibited by the global mapping of low-cost hydrogen production and export centers as a function of solar and wind resources (IEA 2019), along coastlines, and ultimately nearby ports.

Supply of hydrogen-based marine fuel

Likewise, ships are calling or passing by ports and will need to refuel. Verschuur et al. (2024) studied the optimal supply of green ammonia to decarbonize shipping. The modelling forecasts a regionalization of bunker supply, entailing a few large supply clusters that will serve regional demand centers, with limited long-distance shipping of green ammonia as a ship fuel. In a cost-efficient model, green ammonia production is predicted to lie within 40 degrees latitudes North/South, in which South Africa is situated. In the study, South Africa ranks among the top 10 countries that, as a group, is projected to demand and supply up to 58 percent of the global ammonia marine fuel market.[24]

South Africa ranks among the top 10 countries that, as a group, is projected to supply up to 58 percent of the global ammonia marine fuel market.

This group of countries has in common that it depends on maritime transport over long distances. For South Africa this means, that ships would need to refuel at its ports and underpins its role as a production and bunker trading hub for ammonia. The demand forecast for hydrogen-based shipping fuel presented in Chapter 4 and the case study for the port of Saldanha, described in Chapter 5, confirm the relevance and the possible structure of a green bunkering market in South Africa.

Attention for the special role of ports

Internationally, the Clean Energy Ministerial, a group of countries accounting for around 80 percent of global greenhouse gases, launched the Clean Energy Marine Hubs (CEM Hubs) initiative, which aims to establish maritime hubs for low-carbon fuels.[25] The initiative was officially launched in July 2023 to address the readiness of ports to handle the more than 50 percent of all traded low-carbon fuels as per projections.

The CEM Hubs initiative's two main objectives are (i) providing an international platform that supports government and decision-makers with a pathway and criteria to transform ports into CEM Hubs, and (ii) to educate and engage stakeholders across the energy-maritime value chain on pace, timelines, and priority

[24] The group of countries comprise: United States, Singapore, Japan, South Korea, China, Brazil, India, Australia, South Africa, and Malaysia.
[25] The Clean Energy Ministerial (CEM) has 29 member governments, including South Africa. As of January 2024, the CEM Hubs initiative's members are: Brazil, Canada, Norway, Panama, Uruguay, the United Arab Emirates and two industry associations, namely the International Association of Ports and Harbors (IAPH), which represents some 180 ports and 140 port-related organizations and businesses, as well as the International Chamber of Shipping (ICS), the global trade organization for ship owners and operators, representing over 80 percent of the world merchant fleet.

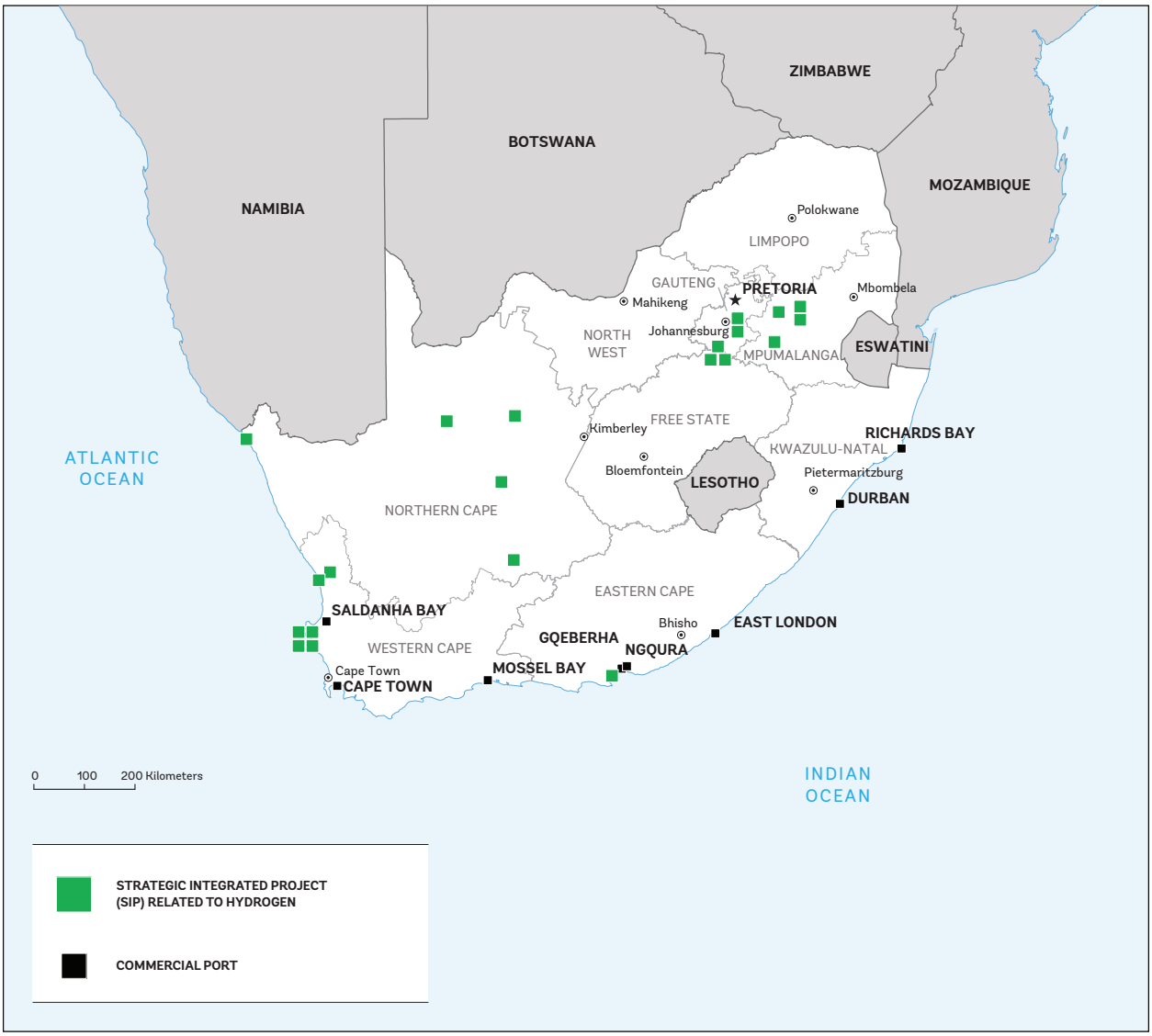
of fuels for multiple industries, including shipping. The International Association of Ports and Harbors (IAPH) notes that, in contrast to previous transformative changes in the shipping industry, such as containerization, the global energy transition has set global targets and timelines through the UNFCCC process as well as sectoral commitments, such as through the IMO (International Association of Ports and Harbours 2023). Hence, global coordination and planning amongst countries is required, to unlock funding and financing to build out infrastructure at all ends of the maritime supply chain.

6.2. National priorities and port authorities

In South Africa, the Presidential Infrastructure Coordinating Commission (PICC) is identifying a growing number of Strategic Integrated Projects (SIPs) through the Green Hydrogen National Program (GHNP). SIPs are projects that hold significant economic or social importance, contribute to national strategies or policies, or have a certain monetary value. These projects benefit from an expedited approval process and shorter timeframes for delivery, as outlined in the Infrastructure Development Act of 2014. Currently, there are 21 hydrogen-related projects registered with the Department of Public Works and Infrastructure South Africa (ISA).

The current list of registered hydrogen-related SIPs also confirms that ports are strategic to developing the country's hydrogen economy as shown in Figure 6.2., where many projects are either at or nearby a seaport. Since private companies apply for SIP status, one can conclude that industry sees great value in locating their planned infrastructure investment at a port location. Private developers also recognize the versatile options regarding final hydrogen-based products. While many projects aim to produce pure hydrogen or ammonia, some identified carbon sources to produce methanol sustainably.

Figure 6.2. Strategic Integrated Projects (SIPs) relating to green hydrogen

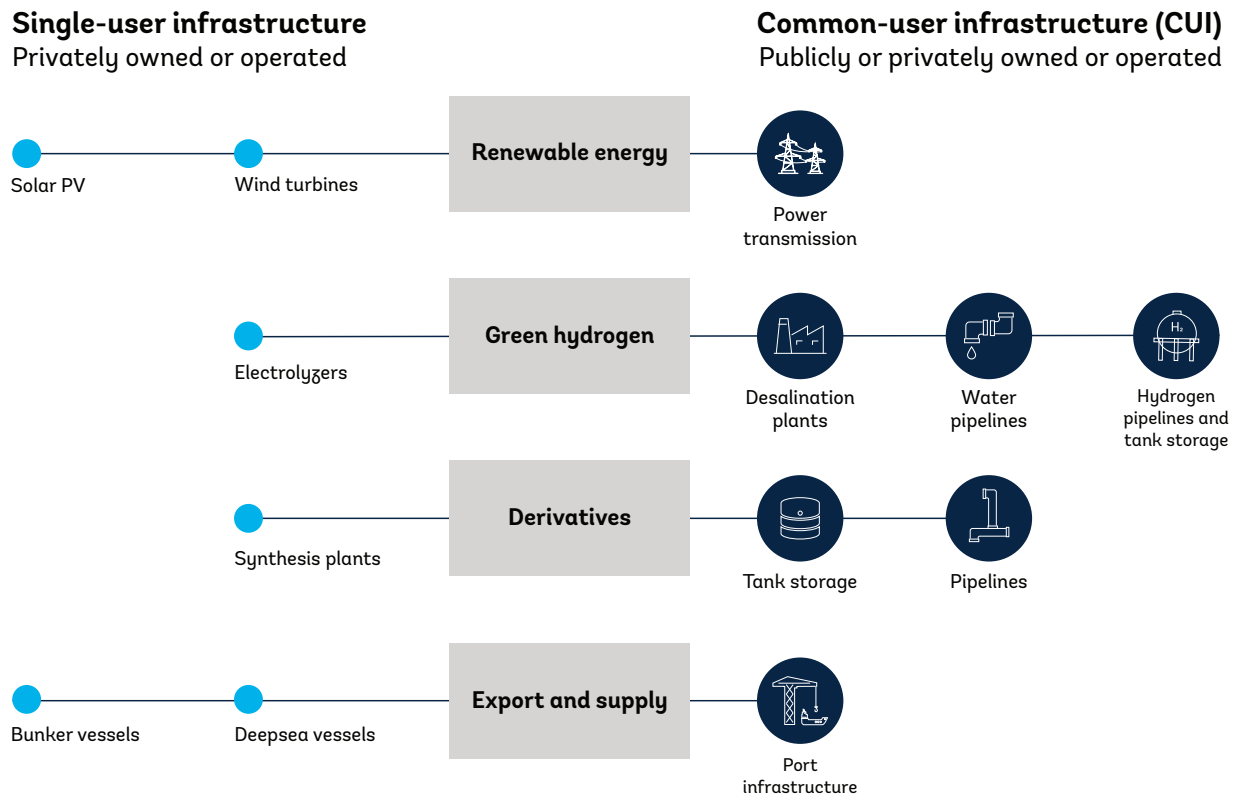


Source: The Presidency Republic of South Africa (2023). For accessibility purposes, the locations of ports and projects are approximate only.

Port authorities have an advanced, complex, role to play (Clean Hydrogen Partnership 2023): As a landlord, to provide land leases for production plants, terminals, pipelines, and bunker facilities. As a community builder and enabler, to promote and attract hydrogen investments in the port. And as an investor, to create, for example, enabling common-user infrastructure, Figure 6.3. closely partnering with the private sector.

South Africa's national port operator, Transnet National Ports Authority (TNPA), is taking important initial steps to respond to a growing number of private sector pipeline projects. Through a request for information (RFI), the port operator is looking to understand, amongst others, the market demand, and requirements for establishing common-user infrastructure and terminal facilities to facilitate the hydrogen economy (TNPA 2023).

Figure 6.3. Single-user and common-user infrastructure options in green fuels production



Source: World Bank.

6.3. Incentives and cost reduction

To bring down the production cost of green hydrogen, ports can offer a distinct, favorable environment through local fiscal incentives and the pooling of activities.

Special Economic Zones (SEZ) are government designated areas that offer business more favorable conditions regarding taxation, regulation, and possibly infrastructure support, compared to the rest of the economy. Globally, around five and a half thousand economic zones are established in more than 140 countries, many of which adjacent to seaports (UNCTAD 2019).

In South Africa, 11 SEZs or Industrial Development Zones (IDZs) are licensed, of which four are located directly at an official commercial seaport: Saldanha, Coega, East London, and Richards Bay. Through the establishment of SEZs by the Special Economic Zones Act 16 of 2014, they offer incentives like (i) the reduction of the corporate tax rate from 28 to 15 percent, (ii) accelerated depreciation allowances, (iii) VAT and customs relief as well as (iv) employment tax incentives. Many countries are looking to utilize SEZs for green hydrogen related projects today and so has South Africa, by giving express reference to SEZ incentives in its Green Hydrogen Commercialization Strategy (DTIC 2023).

The establishment of ports as hydrogen hubs are viewed a distinct hydrogen business model, can offer risk mitigation, and are even expected to lower the cost of hydrogen production (Lee and Saygin 2023). Hydrogen hubs also play an important role for hydrogen production cost. First-mover projects, which carry even higher risk, can benefit if they leverage project location.

The World Bank currently supports the government of Brazil in a $155 million facility to develop the Ceará Hydrogen Hub (Vilar Ferrenbach 2023). The port of Pecém is a deepwater in one of Brazil's economically weakest states, Ceará. The port is supposed to catalyze the state's renewable endowment and transform it into a hydrogen-derivative export hub. Despite private sector interest, investment risk still remains elevated and has inhibited commercial investments at the scale at which would be required. To address this, the government of Brazil, with support from the World Bank and national development banks, is looking to upgrade port and logistics infrastructure to capitalize on a first-mover advantage and de-risk private investments. The port also features an SEZ, ZPE Ceará, which houses amongst others, steelmaking, and industrial manufacturing.

6.4. Green corridors

The concept of green corridors has emerged recently. While there is no uniform definition, green corridors are considered maritime shipping routes around or between one or several ports (e.g., point to point, networks), on which vessels operate with a lower carbon footprint. To achieve this, several measures can be employed. These can range from energy efficiency gains through voyage optimization (from e.g., better ship-to-shore communication) or the use of zero-carbon fuels, such as green hydrogen-based marine fuels. The objective of establishing green shipping corridors is to accelerate the decarbonization of shipping on selected shipping routes, which offer favorable conditions to practice generally more expensive, low-carbon, operations.

The Global Maritime Forum is currently counting an increasing number of green corridor initiatives, totaling 44 possible routes globally. In terms of geographies, most initiatives are still found in Europe, while recently, corridor concepts in the Southern Hemisphere and Asia have increased (Boyland, Talalasova and Rosenberg 2023). To facilitate the implementation green corridors, the Mærsk Mc-Kinney Møller Center for Zero Carbon Shipping together with McKinsey & Company have developed a blueprint for the feasibility of green corridors (Mærsk Mc-Kinney Møller Center for Zero Carbon Shipping 2022).

Since these first-mover green shipping routes are viewed to be a joint effort between private and public stakeholders, this blueprint attributes a special role to government as well. Besides data collection and permitting on possible port level, the proposed role of national government is heavily centered around making the corridor economically feasible. This would entail financial support to bring down the cost differential of low carbon fuel options used on a possible corridor, but also financial instruments towards capital investments, such as in fuel production. The stock take of green corridor initiatives also concludes that governments should focus on support to implementation, including financial support.

Having said that, fiscal limitations are a key barrier. Considering the fiscal constraints of Emerging markets and developing countries (EMDCs), financial support from government to improve the economic viability of an early moving corridor concept, appears challenging. However, corridors proposed between EMDCs and those countries where such external funding is available, could benefit from using the green corridor to gain access to funding. Green corridors would not only connect two ports, but would connect financial support, available on the one end of the corridor, with financial needs on the other end of the corridor. At present, the only green corridor project linking South Africa, is a proposed South Africa to Europe iron ore corridor. This project could capture the diverse green hydrogen use cases in the port of Saldanha and take advantage of possible financial and policy support in the European Union.

Financing and funding green hydrogen

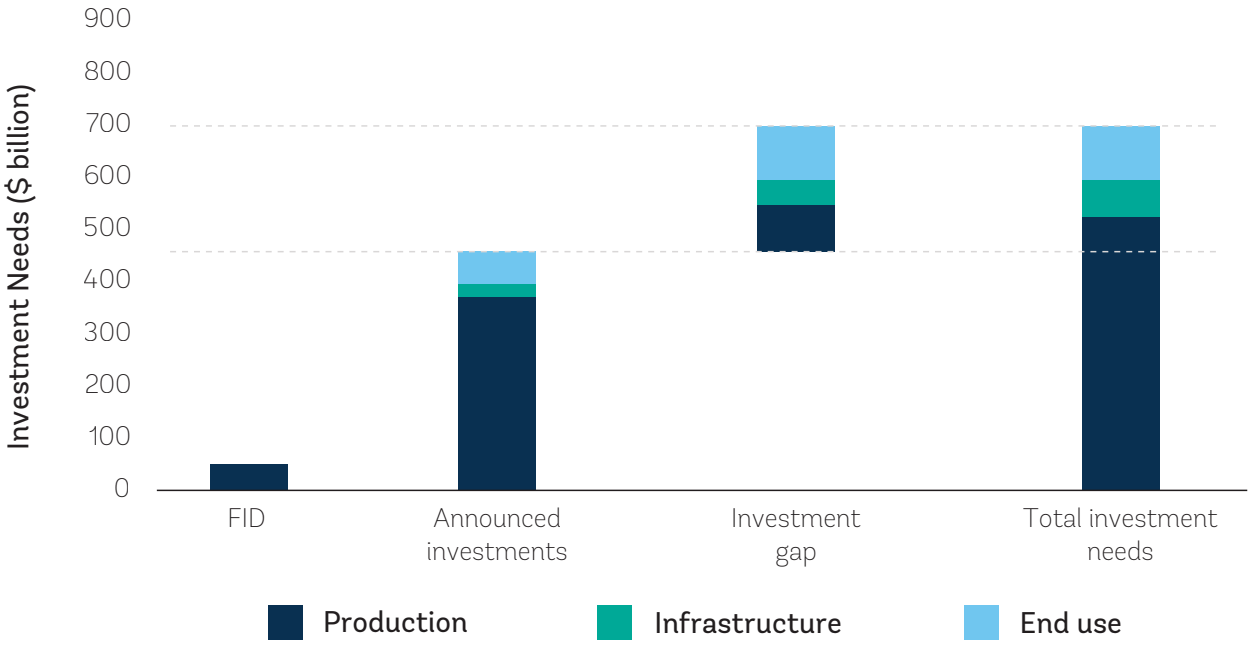
The manufacture of green shipping fuels and big green hydrogen projects face similar challenges. Currently, there's a gap in funding for these projects. This includes challenges in getting money from outside sources and the fact that green fuels are more expensive than traditional energy sources. Because emerging market and developing countries (EMDCs) have limited funds, investments in green hydrogen, like making green hydrogen-based marine fuels, need to be carefully managed to reduce risks.

07

7.1. Challenge

Emerging markets and developing countries (EMDCs) are projected to supply 25 to 50 percent of global clean hydrogen (World Bank 2024). The investment needs for clean hydrogen in EMDCs until 2030 is estimated at $700 billion of which the announced investment project pipeline currently makes up around $450 billion. However, only less than $50 billion of investment needs have reached financial close, committing the full investment amount (see Figure 7.1).

Figure 7.1. Investment needs in clean hydrogen until 2030

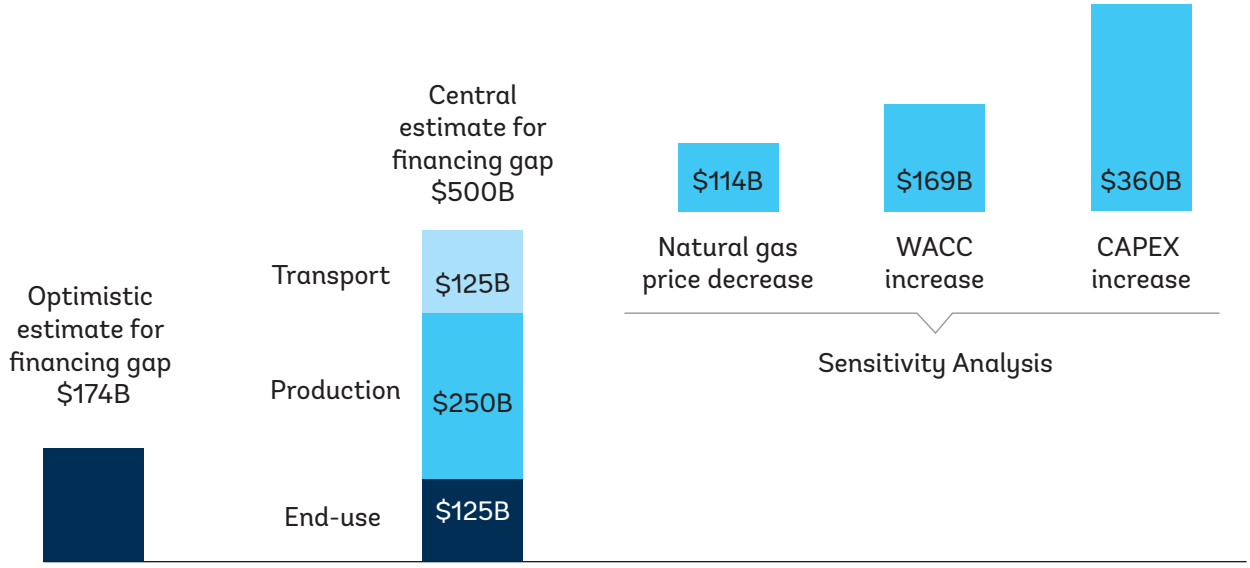


Source: ESMAP (2024).

For the competitiveness of clean hydrogen, analysis indicates that the gap between product value and production cost amounts to $10-40 billion per year between today and 2030 globally (ESMAP 2024). This shortfall could be viewed as a gap which would need to be filled by some form of financial support to make clean hydrogen competitive with conventional energy sources. Consequently, if such support was mobilized, this would reduce the global average levelized cost of hydrogen (LCOH) from $5 to $2/kg in an optimistic case.

The investment needs for clean hydrogen in EMDCs until 2030 is estimated at $700 billion, of which the announced investment project pipeline currently makes up around $450 billion. However, only less than $50 billion of investment needs have reached financial close, committing the full investment amount.

Figure 7.2. Financing gap under different assumptions



Source: ESMAP (2024).

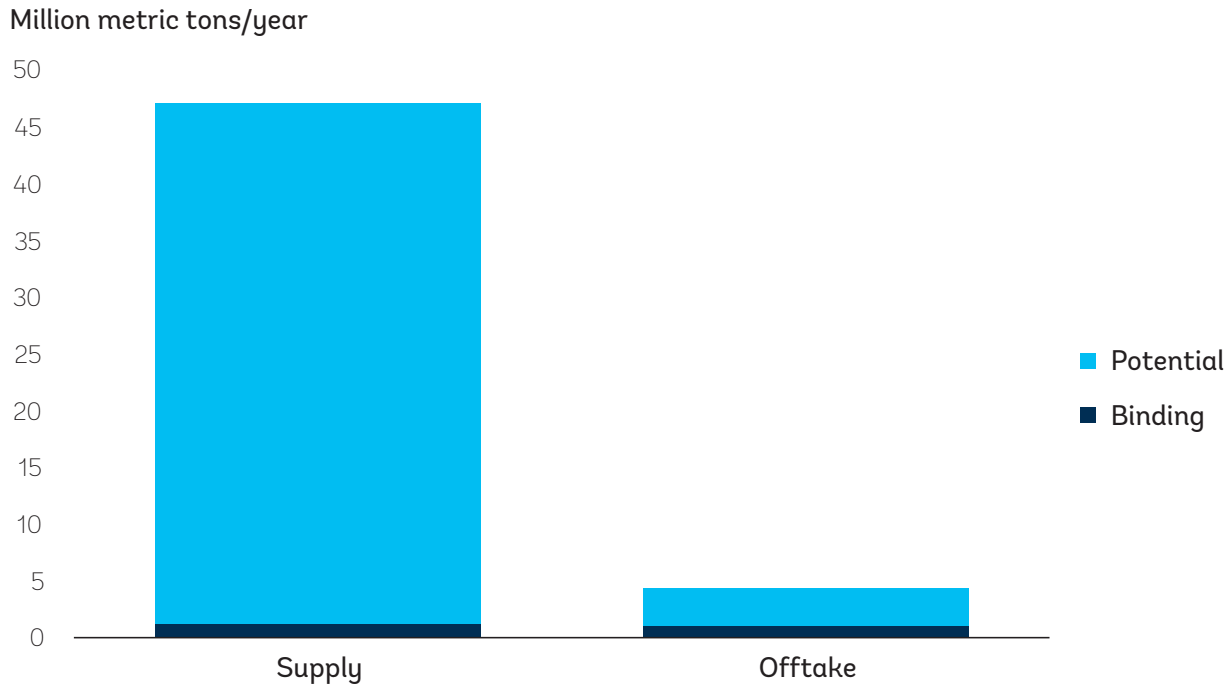
In EMDCs, fiscal space has generally been shrinking since the global financial crisis (World Bank 2018, Ayhan Kose, et al. 2022). Mobilizing public financial support to subsidize clean hydrogen is a likely constraint. The International Monetary Fund (IMF) highlights that countries with less fiscal space will also need to prioritize spending. Additional debt in these economies will be particularly challenging given their inherently higher debt levels. Rising interest costs put additional stress on debt capacity (IMF 2023). In EMDCs, the private sector therefore has an even greater role to finance green hydrogen investments.

The issue with missing offtake agreements

For green hydrogen projects to reach final investment decision today, developers and investors must find buyers who are prepared to pay a premium for green hydrogen or who operate in a market where today's production prices can match the price of conventional feedstock. In project finance setups, financial investors require their risk to be mitigated through long-term offtake agreements. Offtake agreements can come from the private sector or are promoted through governments, coupling subsidies for hydrogen imports with private or public sector final consumption.

Bloomberg highlights that only 10 percent of planned hydrogen projects globally has secured offtake (Liebreich 2023). This issue is exacerbated by the fact that 80 percent of those 149 agreements are only memorandums of understanding (MoUs) (or remain unspecified). For hydrogen projects, the median length of offtake agreements is around 15 years, however, could range from seven to 30 years (Liebreich 2023). Securing long-term cashflows to recover investments at a reasonable return appears to be a huge challenge and jeopardizes the bankability of announced projects.

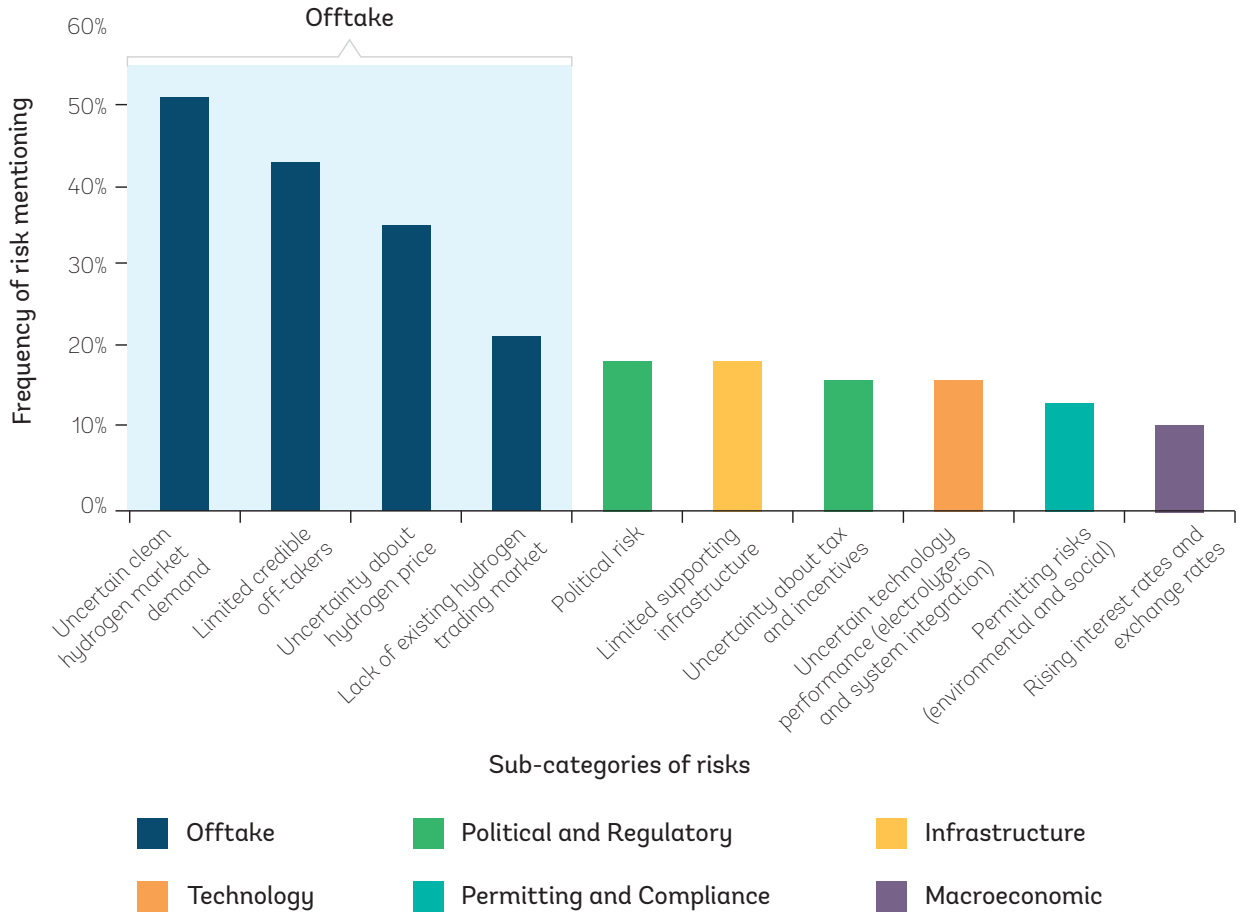
Figure 7.3. Hydrogen supply and offtake by 2030



Source: Liebreich (2023). Note: Binding supply refers to projects that reached final investment decision.

In Chapter 5, the case study identified two significant commercial project risks, relating to (i) the lack of demand for green bunker fuels stimulated by regulation, and (ii) the financial risk of banking a product, which is not price-competitive with its fossil-based alternative, i.e., the realized sales price. This risk assessment aligns with findings from a global market sounding amongst financiers, project developers and other key stakeholders, where the top four risks relate to firm offtake (Figure 7.4.) (Lee and Saygin 2023). In turn, survey respondents indicated that if risks around the uncertainty of market demand and hydrogen price were mitigated, such projects would be able to successfully secure financing.

Figure 7.4. Key risks for green hydrogen projects



Source: ESMAP (2024).

For the maritime sector, overcoming the barriers to successful project delivery is critical to meet its own greenhouse gas reduction targets. Shipping will demand around five million tons of hydrogen by 2030 globally to sustain its climate change ambition levels (compare Chapter 3). Further, a smooth and equitable maritime energy transition for countries can only be achieved if a critical mass of first-mover projects reaches financial close soon.

A smooth and equitable maritime energy transition for countries can only be achieved if a critical mass of first-mover projects reaches financial close soon.

7.2. Financing and funding sources

Multiple sources of financing and funding are needed to for capital-intensive green hydrogen projects. Some of which are traditionally used in project finance, while others are novel funding or financing mechanisms specific to this nascent hydrogen investing landscape.

Private finance

Developers may secure their own financing sources through equity and debt financing, such as loans and bonds. Regional institutional investors and fund managers have experience in the South African context, mobilizing equity and debt for renewable energy projects. Notably, private fund managers are working jointly with the government of Namibia to establish a fund, SDG Namibia One, to mobilize finance for hydrogen and to empower the country to make strategic decisions for projects that support socio-economic development (Climate Fund Managers B.V. (CFM) 2022). In South Africa, the Dutch and Danish government, together with national institutional investors, intend to setup a $1 billion fund for hydrogen initiatives (Reuters 2023). In the case of project finance, project bonds are also an attractive alternative to bank loans, especially in view of more stringent compliance and liquidity requirements of lenders (Mawkin, Young and Torres Caro 2022). Given the large capital requirements, the institutional project bond market is largely focused on the power industry (around 40 percent). It has grown significantly post the financial crisis, becoming an important source for the vast investment needs of the energy transition (Global Investment Banking 2018).

Commercial banks

Commercial bank loans are considered debt financing. Domestic or international banks typically provide medium to long-term amortizing loans secured against the financed assets, but also provide working capital financing. Lending policies of commercial banks increasingly focus on the carbon intensity of projects and borrowers, shifting away from fossil fuel projects. Commercial banks can also be used as vehicle to deploy lower-cost capital provided by development finance institutions (DFIs), usually backed by a sovereign guarantee.

Export credit agencies

Export credit agencies (ECAs) act as intermediaries between exporters and government to promote exports through the provision of risk mitigation instruments and loans. The objective is to offer lower financing cost normally in markets, where an inherently higher foreign investment risk demands higher risk premiums. Hence, ECA guarantee cover lowers interest rates for loans and insures equity investments against political risk.

Venture capital

Venture capital providers provide companies with capital in exchange for equity. Often, venture capitalists pool and manage money from family offices, or institutional investors such as mutual funds, pensions funds and insurance companies. A similar shift – as happened with the rapid uptake of solar and wind energy after 2010 – is anticipated by the venture capital industry to happen with hydrogen as most recently deal activity doubled (Kraan and van Haren 2022). However, venture finance providers normally invest in early-stage company's balance sheet and not in project finance as they are more concerned with the corporate value. As a result, venture capital is less relevant for financing large-scale, ringfenced projects.

Multilateral development banks

Also known as international financial institutions (IFIs), multilateral development banks (MDBs) offer financial assistance to developing countries to support socio-economic development. The five major international MDBs are the World Bank, African Development Bank (AfDB), Asian Development Bank (ADB), the European Bank of Reconstruction and Development (EBRD), and the Inter-American Development Bank (IDB).

Traditionally, MDBs provide infrastructure financing and general budget support through concessional, non-concessional, and blended finance to government, state-owned enterprises (SOEs), but also to the private sector. Further, they provide risk insurance and guarantees to lower credit risk and attract private capital. In financing hydrogen, MDBs play a strategic role, supporting governments in their aspiration to attract private investments through de-risking, lowering capital cost and aiding in developing enabling policy frameworks. To date, all MDBs have only committed less than $5 billion in financing for hydrogen per year. Especially for clean hydrogen, where financial resources and assistance exceeds the capacity of a single MDB, platform approaches, where multiple MDBs cooperate could be sensible financing mechanisms to address lending capacity limitations and promote risk-sharing (ESMAP 2024).

Practical examples of MDB engagement in hydrogen are, for example, the provision of a $1.5 billion policy loan by the World Bank to the government of India. The program is supposed to support the India's National Green Hydrogen Mission, by addressing viability funding gaps, reducing off-taker risk, and boosting grid integration of renewables (World Bank 2023a). In Chile, the World Bank worked with the country's national development agency, CORFO, to set up an innovative financing facility for hydrogen. A first $150 million credit line by the World Bank can be on-lend to commercial banks to blend finance, reducing the cost of capital for private hydrogen investments (World Bank 2023b).

Government and supranational initiatives

Multiple governments are developing and implementing policies to promote the decarbonization of their industries using green hydrogen or establish export-oriented hydrogen economies. Globally, the lack of clarity on support schemes has for a long time contributed to missing traction in hydrogen investments. However, recently launched initiatives for example in the EU, are likely to have a direct impact on South Africa as a hydrogen trading partner in the near term. To achieve the decarbonization ambition under the EU's Fit-for-55 package, including the EU's hydrogen strategy (European Commission 2020) and REPowerEU (European Commission 2022), the EU is looking to produce 10 million tons of green hydrogen domestically and import another 10 million tons by 2030.

Finance is of the core concerns and several mechanisms support green hydrogen not only in member states, but also in non-EU countries. The European hydrogen bank (European Commission 2023a) aims to cover and lower the cost gap between renewable hydrogen and fossil fuels for early projects through an auction system for renewable hydrogen production supporting producers through a fixed price payment per kilogram of hydrogen produced for a maximum of 10 years of operation. The first pilot auctions were opened in November 2023 and are backed by €800 million emissions trading revenues, administered through the Innovation Fund (European Commission 2023b). Applications for grants from the Hydrogen Bank are open to both domestic and foreign green hydrogen producers (exporters). Germany set up H2Global, a government-backed "double-auction" platform, which aims to bridge the difference in price at which green hydrogen is traded on the global market and the lower price at which it can be sold to European off takers. In the first auction, the lowest international bidder is granted a long-term contract. In the second auction, the hydrogen delivered into the EU, will be auctioned to the highest bidder (Federal Ministry for Economic Affairs and Climate 2024). The first auction round is endowed with €900 million in funding to close the price gap for green hydrogen and derivatives, acting like a contract for difference (CfD) mechanism.

In 2023, it was announced that H2Global will become part of the European Hydrogen Bank and will open hydrogen tenders for all European countries (European Commission 2023c). H2Global has recently expanded its reach to possibly include €400 million partial funding from the Australian government. From the contribution jointly funded by the German and Australian state, Australian producers would be able to bid for exports to Europe. If successful, the lowest supply-side offer would be matched with the highest bidder on the demand side in Europe and the government funding will come up for the price gap (Macdonald-Smith 2024).

The Republic of Korea and Japan are also looking at support schemes, which are however not as advanced yet. Clear guidelines for the import of ammonia, expected for both countries in 2024, are bound to trigger import contracts for hydrogen derivatives, such as ammonia, under yet-to-be-launched CfD schemes (Garg and Lim 2023). However, Japan has recently firmed its intention to allocate $21 billion to clean hydrogen CfD mechanisms (Collins 2023). In the United States, the inflation reduction act (IRA) is looking to provide tax credits for domestically produced clean hydrogen. While not aimed to support imports, i.e., production in countries overseas, green hydrogen produced in the United States could potentially compete with developing countries, when targeting, for example, exports to Europe (Tatarenko, et al. 2023).

Carbon border adjustment mechanism (CBAM)

The European Union's emissions trading scheme (ETS) is one of the key measures to cost-effectively reduce greenhouse gas and meet the Union's ambition of 55 percent reductions by 2030. Now, the ETS covers over 10,000 installations in power generation, manufacturing, and aviation emissions. Since 2024, emissions from international shipping are also subjected to carbon pricing in the EU. To prevent carbon leakage through the relocation of production outside the European Union, the carbon border adjustment mechanism shall ensure that goods produced outside member states have the same cost as if they were produced within the EU.

Therefore, importers, from 2026 onwards, must surrender allowances for imported products. While CBAM only covers goods from selected sectors, hydrogen, ammonia, and green steel are regulated by the mechanism. Products produced with less carbon emissions can hence become more competitive than products with higher embedded emissions from production.

CBAM can add significant cost to products and is expected to create a strong incentive for low-emissions production routes. For example, green ammonia is expected to have competitive advantage of up to EUR 110 per ton of ammonia[26] (International PtX Hub 2023), the production route most promising in South Africa. Today, the country's close ties with the European Union has put it at 10th place in a ranking of top CBAM supporters amongst the 50 largest economies. South Africa currently exports CBAM applicable products worth $423 million (Sabyrbekov and Overland 2024). Green hydrogen and derivative production, to which CBAM applies, would increase this figure significantly and can gain a competitive edge over other economies. There remains criticism however that under the current ruleset, not all hydrogen carriers, like methanol and e-kerosene are covered (Hydrogen Europe 2022).

[26] At ETS price of EUR 150 ton/CO2 in 2030.

Revenues from global maritime emissions pricing mechanism

At the International Maritime Organization (IMO), member states have agreed to develop new climate measures for international shipping. A global emissions pricing mechanism covering ship emissions shall help the sector to decarbonize to reach net zero emissions by around 2050. Many member states of the IMO believe that carbon pricing can not only reduce greenhouse gas emissions but also make the energy transition in shipping more effective and equitable through the recycling of collected revenue – a by-product of an emission pricing scheme.

The money could be distributed to support the decarbonization efforts in the shipping industry, invest in green fuel production, develop maritime infrastructure, improve port efficiency, and help nations and industries mitigate and adapt to climate change. The state of discussions involves considering the socio-economic impacts of the policy measures, such as their effects on food security and their ability to achieve the agreed GHG reduction targets. IMO member states envision an entry into effect from 2027. It would be the first carbon pricing instrument to cover all greenhouse gas emissions of an entire sector globally. For high potential green hydrogen producers, such as South Africa, this opens a window of opportunity to benefit from such a measure as it could stimulate green hydrogen economies if revenues are recycled as capital, subsidies, or incentives.

7.3. Risk spotlight: Fiscal considerations

South Africa's Green Hydrogen Commercialization Strategy (GHCS) considers options on how to finance green hydrogen investments. The study estimates around ZAR 319 billion ($16.8 billion) in financing needs in the five-year period, 2023 to 2027 alone (The Presidency, Republic of South Africa 2023). This estimate covers the current private sector investment pipeline and 50 percent is considered public investment in ports. The GHCS outlines options how to mobilize private finance, but also discusses how national budgetary processes can unlock levers to accelerate the development of green hydrogen.

For private capital mobilization, the GHCS considers options such as the strengthening partnerships with countries in the international partners group (IPG) and outside, the provision of risk-mitigation instruments and blended concessional finance. Tapping into grants or subsidies for pilot projects is also considered. National departments, like the Department of Trade, Industry and Competition, and the Industrial Development Corporation (IDC), can explore how redirecting income from carbon taxes, special economic zones, and income tax provisions or deductions can support green hydrogen, all of which are levers which other countries are considering (DTIC 2023).

South Africa's own fiscal space is limited. Hence, government spending in green hydrogen must be viewed against this background and is to be considered cautiously. Managing fiscal space, i.e., the availability of budgetary resources to pursuit strategic growth opportunities is an essential instrument of macroeconomic risk management and should be appreciated in the context of the green hydrogen economy, too (World Bank 2013).

Despite having recently recovered from the pandemic well, South Africa's fiscal deficit remains high. Rising debt service costs, cash transfers to state-owned enterprises, amongst others, put pressure on the available resources and subdue the fiscal outlook. According to the World Bank (2023d), the fiscal deficit is expected to grow to 6.2 percent of GDP this fiscal year, and only fall to 4.8 percent in 2026/27. It is anticipated that South Africa's spending will be tilted towards current expenditure, which limits capital spending to 2.3 percent of

GDP, leaving insufficient room to support infrastructure investment and growth. Nevertheless, the country's commercial banking sector reports strong capital and liquidity buffers, and the structure of public debt is favorable, with average maturity of public bonds of around 12 years, being one of the longest among middle-income countries.

One way to prudently manage fiscal room is to focus on private capital mobilization and external support. Nevertheless, the inherent investment risk of a nascent industry persists for the private sector and can be seen in the few numbers of projects reaching financial close globally as described earlier. The government can prioritize the choice of instruments to help reduce private sector risk and mobilize lending from private financial investors. Given the capital-intensity, financial risk can be mitigated through blended finance solutions, for instance. Credit enhancement guarantees and non-financial risk insurance can further mobilize private capital, leveraging the credit rating of development banks (most major development banks are triple-A credit-rated), reducing risk and increasing the risk-adjusted return that private investment would be expecting. In the context of financing the green hydrogen economy, the choice of instruments is important. Firstly, to manage government's fiscal space, preventing the redirection of resources from existing spending needs and secondly, to efficiently use development finance resources, which are limited, too.

To manage fiscal space, the government can furthermore support the private sector to work towards a sound pipeline of investment projects, which qualify for external support from international partners. In areas, where there is overlap with the public sector, such as in common-user infrastructure, land-leasing, zoning of SEZs, and environmental assessments, government can unblock project barriers. The realization of co-benefits for local communities, for example for water resilience, is another possible area of government involvement, which can help sustain the social operating license of substantial infrastructure developments for green hydrogen.

Looking ahead for policymakers

The marine fuel market has the potential to implement South Africa's hydrogen ambitions, with the maritime value chain playing a crucial role. Nevertheless, each new opportunity is accompanied by its unique set of challenges, ranging from global to country-specific issues. Policymakers and industry stakeholders in South Africa must consider what steps they can take to capitalize on this opportunity and establish a market for green marine fuels.

08

The report explored how the maritime sector can help unlock the potential of South Africa's hydrogen economy using marine fuel as a promising downstream opportunity for green hydrogen production. This opportunity is framed within South Africa's ambition to become a leader in green hydrogen (Chapter 2) and the hydrogen-centered decarbonization pathway for deep-sea shipping (Chapter 3). The analysis then quantified the potential market size for hydrogen-based marine fuel in the country's eight commercial ports and from by-passing ship traffic (Chapter 4). To understand what it takes to develop production and supply of green marine fuel, the study investigated a specific case for a possible project at the port of Saldanha (Chapter 5). Chapter 6 explored the important role of ports in the hydrogen economy, and Chapter 7 discusses how clean hydrogen can be financed, which is the main barrier for hydrogen globally.

The report reflected on four risks, which will require continuous effort and monitoring to create a successful green marine fuel market in South Africa: Electricity crisis (Chapter 2.3), maritime sector challenges (Chapter 3.3), safety & environment (Chapter 5.2.8), and fiscal considerations (Chapter 7.3).

The report concludes on five main messages and actionable policy recommendations:

South Africa strategically advances a national green hydrogen economy, and the maritime sector plays a crucial role.

Through the Green Hydrogen Commercialization Strategy (GHCS), South Africa is moving towards establishing a hydrogen economy, which can realize development co-benefits for the country. To unlock these additional benefits, critical reforms in the power sector are already under way (Chapter 2.3). The ultimate success of this new industry, however, also hinges on the maritime sector, where an enabling regulatory environment can support South Africa's green hydrogen ambition. At the same time, the shipping industry is dependent on the successful implementation of South Africa's hydrogen economy to secure green fuel supply. Policymakers may therefore consider to:

- Advance objectives of the Green Hydrogen Commercialization Strategy (GHCS) to kickstart green hydrogen production.
- Facilitate investments in renewable electricity and power grid expansion to realize co-benefits.
- Develop a regulatory framework for hydrogen derivatives in the marine environment to preserve environmental integrity and social acceptance.

International shipping can be a catalyst to commercialize South Africa's green hydrogen economy, unlocking an up to 2-million-tons hydrogen market.

Hydrogen-based marine fuels present a unique downstream opportunity for the South African green hydrogen economy offering a stable demand source. The report finds that in a base case, up to 56,000 tons of annual hydrogen demand could come from the supply of marine fuel, as early as 2030. By 2050, this demand could rise to over 0.5 million tons per year. If capturing the demand from ships passing by the Cape, South Africa could supply international fleets with up to 182,000 tons by 2030 and a substantial 2 million tons of green hydrogen by 2050. To commercialize the sale and supply of green marine fuel in South Africa's ports, policymakers may therefore consider to:

- Promote the adoption of global rules for the uptake of green hydrogen-based marine fuels at the International Maritime Organization (IMO) to unlock demand.
- Develop a clear and transparent licensing system for marine fuel suppliers to foster competition.
- Identify and conduct pilot initiatives to build investor confidence and mitigate operational risk.

Green shipping fuel production is technologically feasible, but the business case faces challenges, not only in South Africa.

Green hydrogen-based marine fuels are more expensive than the conventional oil-based fuels. Depending on the financing conditions,[27] the case study (Chapter 5) shows that green ammonia could be manufactured within a price range between ≈1.4 to 4.3 times the cost of conventional marine fuels. This cost differential is not unique to South Africa but highlights the need for global regulation to be the key demand driver for this hydrogen use case. Access to lower cost of capital is important where fiscal space is constrained to deploy subsidies or development priorities would compete. To overcome cost-competitiveness barriers, policymakers may therefore:

- Consider public spending carefully to safeguard fiscal space.
- Develop global mandates at the IMO to leverage the policy-driven nature of the marine fuel market.
- Support the development of a sound project pipeline to unlock external support for projects.

Mobilizing private capital is key to successfully develop green marine fuel projects at scale.

The capital-intensity of large-scale green hydrogen projects exceeds the capacity of single governments and many financial investors. Platform approaches, where multiple (development) banks cooperate, can help to blend capital from commercial or concessional sources to accommodate different risk profiles and spread investment risk. To mobilize private capital, policymakers may consider to:

- Identify suitable financial instruments to de-risk first-mover projects.
- Partner with group of development finance institutions to scale impact.
- Promote novel funding sources such as global ship emissions pricing to support private investments.[28]

Ports in South Africa can develop into enabling hydrogen hubs.

Efficient ports are critical for the hydrogen economy to be successful and will act as hydrogen production-, export-, and supply hubs. A commercial and regulatory enabling environment is centered around ports, while common-user infrastructure can increase competitiveness, mitigate risk, and realize co-benefits for local communities. To develop ports into hydrogen hubs, policymakers may therefore consider to:

- Assess common-user infrastructure needs at the port, municipal and regional levels to reduce cost and spread risk.
- Consider expanded private sector participation and local incentive structures to attract investment.
- Address current inefficiencies in the port sector to mitigate operational risk.

[27] Based on a weighted average cost of capital (WACC) range of 4-12 percent and a reduction in capital expenditure of between 0-40 percent, compared against a three-year max average price of VLSFO, and considering a delivered price (fuel supplied).
[28] Also see World Bank publication on how revenues from a global emissions pricing scheme could be used (Dominioni, et al. 2023)

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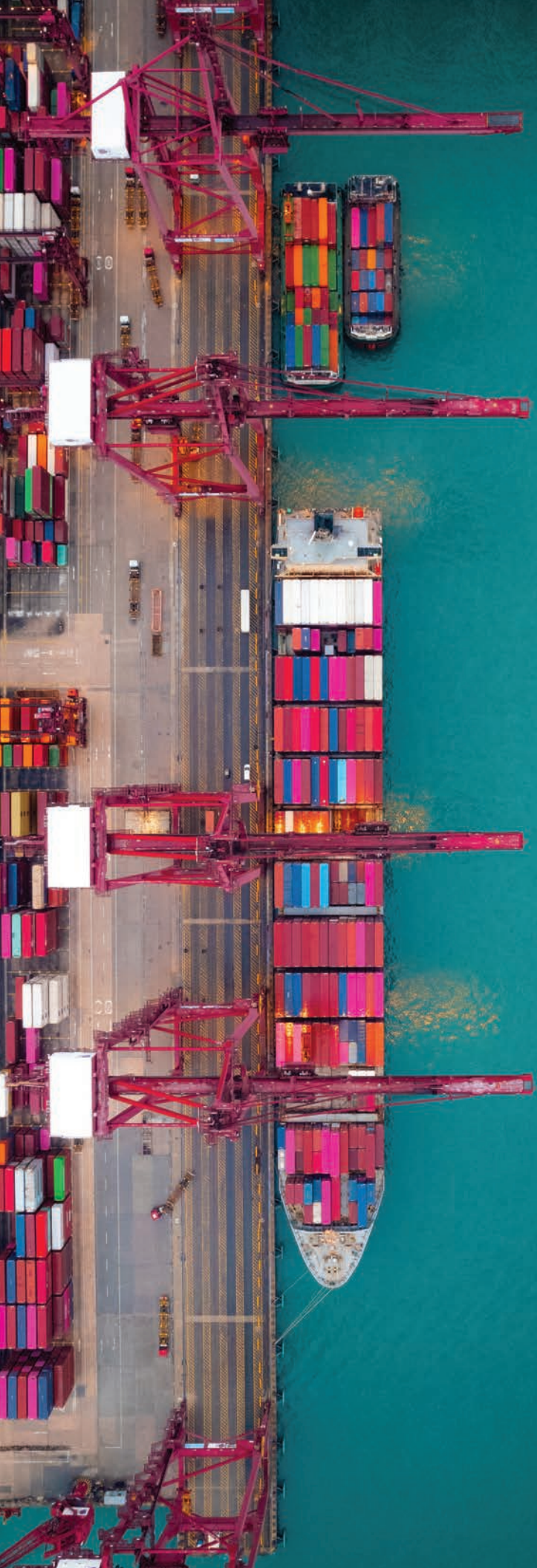

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